

Asymptotically Optimal Mixed-State Cloning

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Abstract

We study approximate quantum mixed-state cloning in the asymptotic regime. The input consists of n copies of a d -dimensional state, and the output contains m_n systems, with fixed d and $m_n/n \rightarrow \gamma > 1$. For a prescribed simple full-rank spectrum $p_1 > \cdots > p_d > 0$, we determine the asymptotically optimal fidelity on its unitary orbit:

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} \frac{\sqrt{\gamma} + \sqrt{q_{ij}(\gamma - 1 + q_{ij})}}{\gamma + q_{ij}}, \quad q_{ij} = \frac{p_j}{p_i}.$$

For unknown spectrum, we construct a universal cloner whose asymptotic fidelity is

$$F_{\text{univ}}(\gamma, p) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2} F_{\text{orb}}(\gamma, p),$$

uniformly on compact subsets of the simple full-rank chamber. This cloner is minimax optimal on nonempty compact spectral sets equal to the closure of their interiors within the affine hyperplane $\sum_i p_i = 1$. Separately, for flat projector states $\rho = P/r$, with P an orthogonal projector of rank $1 < r < d$, we prove that the optimal asymptotic cloning fidelity is $\gamma^{-r(d-r)/2}$. We also determine the exact asymptotic cloning fidelity of the purify-clone-trace (PCT) protocol at simple full-rank spectra and bound it for projector states. Our cloners strictly outperform PCT at every simple full-rank spectrum and for projector states.

The cloner constructions combine Schur–Weyl label processing with a Cartan-sector covariant channel extending Werner’s pure-state cloner. For the converse, local asymptotic normality reduces the problem to classical Gaussian dilation and quantum Gaussian amplification. The projector result follows from exact Schur-sector compression and a direct finite-dimensional converse.

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1 Introduction

The no-cloning theorem rules out perfect copying of an arbitrary unknown quantum state [WZ82, Die82]. Approximate cloning asks how well a channel can transform n identical inputs into an output resembling m ideal copies. Here we measure fidelity with the entire product target $\rho^{\otimes m}$. This global criterion is stronger than reproducing one-site marginals and is distinct from broadcasting, which permits correlations among the outputs [BCFJS96]. For mixed states, the eigenvalues and eigenbasis may both be unknown, and these two kinds of information behave differently under repeated sampling. Earlier work on pure-state cloning, mixed-state bounds, marginal criteria, and Gaussian models is discussed in Section 1.3.

We call a full-rank spectrum *simple* if its eigenvalues are distinct. In the known-spectrum problem, the eigenvalues are supplied and the state ranges over one unitary orbit; cloning must reproduce only the unknown eigenbasis. In the unknown-spectrum problem, a single spectrum-independent sequence of channels must handle both eigenvalues and eigenbasis, uniformly over compact subsets of the chamber of simple full-rank spectra. This is not merely the first problem preceded by spectrum estimation: the measured spectrum retains fluctuations of order $n^{-1/2}$, which must themselves be converted from the n -copy to the m_n -copy scale.

We solve both problems in fixed dimension when $m_n/n \rightarrow \gamma > 1$. For known spectrum, we determine the optimal global fidelity on each simple full-rank orbit. For unknown spectrum, we construct one cloner, determine its asymptotic fidelity uniformly on compact subsets of this chamber, and determine the minimax value over the full-dimensional compact spectral sets specified in Theorem 1.2. To our knowledge, no sharp solution under global product fidelity was previously available for these non-degenerate full-rank models at fixed gain. We also solve cloning of flat rank- r projector states $\rho_P = P/r$, with P any rank- r orthogonal projector, by a direct finite-dimensional argument. This gives an exactly solvable rank-deficient family without requiring a general treatment of fixed-rank strata. Section 5 compares these results with purify-clone-trace (PCT) at fixed gain and in the high-fidelity sample-complexity regime.

Two complementary decompositions drive the solution. Schur–Weyl duality separates the spectral information in the Young label from the eigenbasis information in the conditional irreducible state [KW01]. Our cloners process the label classically and apply a multiplicity-one Cartan-sector channel to the irreducible state [PRV67, FH91]. Two-way local asymptotic normality (LAN) gives the corresponding separation into a classical Gaussian shift and displaced thermal oscillator modes. A cloner exceeding our orbit or compact-family minimax value would induce a channel exceeding the corresponding Gaussian dilation or amplification optimum. We prove these bounds by a unified flat-prior fidelity argument; the quantum optimum is attained by the least-noise amplifier. Projector states instead admit exact Schur-sector compression and a direct finite-dimensional converse.

1.1 Results

Fix an integer $d \geq 2$ and positive integers m_n such that

$$\gamma_n := \frac{m_n}{n} \longrightarrow \gamma > 1, \quad r_d := \binom{d}{2}. \quad (1)$$

In particular, $m_n > n$ for all sufficiently large n .¹ Write $\mathbf{A}_d := \{x \in \mathbb{R}^d : \sum_i x_i = 1\}$. Let

$$\Delta_d^\circ := \{p \in \mathbf{A}_d : p_1 > \dots > p_d > 0\}, \quad \rho_p = \text{diag}(p_1, \dots, p_d), \quad \rho_{p,g} = g\rho_p g^*. \quad (2)$$

For positive trace-class operators A, C , not necessarily normalized, we write

$$F(A, C) = \|\sqrt{A}\sqrt{C}\|_1. \quad (3)$$

For states, this is the root (unsquared) Uhlmann fidelity. For nonnegative functions or measures, the same notation denotes their Hellinger affinity; in particular, $F(P, Q) = \sum_x \sqrt{P(x)Q(x)}$ on a countable space. The target in every statement below is the entire product state $\rho_{p,g}^{\otimes m_n}$. For $p \in \Delta_d^\circ$, define

$$q_{ij} = \frac{p_j}{p_i}, \quad f_\gamma(q) = \frac{\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)}}{\gamma + q}, \quad (4)$$

and

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} f_\gamma(q_{ij}), \quad F_{\text{cl}}(\gamma, d) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{(d-1)/2}, \quad F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d) F_{\text{orb}}(\gamma, p). \quad (5)$$

The factor F_{orb} measures the cost of amplifying the unknown eigenbasis, whereas F_{cl} measures the additional cost of dilating the unknown spectral fluctuation.

Known spectrum Fixing p isolates the genuinely noncommutative part of the problem. The target distribution of Young labels is then known and can be sampled exactly at size m_n ; only the conditional representation sectors, which encode the eigenbasis, need to be cloned.

Theorem 1.1 (Known-spectrum optimum). *For every $p \in \Delta_d^\circ$,*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{g \in \text{U}(d)} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = F_{\text{orb}}(\gamma, p), \quad (6)$$

where the supremum is over all quantum channels from n to m_n qudits. The explicit sequence of channels $\mathcal{C}_n^p$ defined in [Section 2.3](#) asymptotically attains the displayed value.

Thus every pair $i < j$ contributes one displaced-thermal amplification factor $f_\gamma(p_j/p_i)$. The Schur–Cartan cloner attains their product, while fixed-spectrum LAN shows that any better finite-dimensional cloner would violate the Gaussian amplification optimum proved here.

Unknown spectrum If p is not supplied, the target Young-label distribution can no longer be sampled directly. The measured input label determines the spectrum only at the $n^{-1/2}$ scale, and converting n copies into $m_n \sim \gamma n$ copies requires dilating this classical fluctuation. This produces the extra, spectrum-independent factor $F_{\text{cl}}(\gamma, d)$.

Theorem 1.2 (Unknown-spectrum cloner fidelity and compact-set optimum). *There exists a sequence of spectrum-independent, $\text{U}(d)$ -covariant channels $\mathcal{C}_n^{\text{univ}}$ with the following properties.*

(a) *For every compact $K \subseteq \Delta_d^\circ$,*

$$\lim_{n \rightarrow \infty} \sup_{\substack{p \in K \\ g \in \text{U}(d)}} \left| F(\mathcal{C}_n^{\text{univ}}(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) - F_{\text{univ}}(\gamma, p) \right| = 0. \quad (7)$$

¹At the finitely many exceptional indices with $m_n \leq n$, both cloners are defined as the exact partial trace channel (and the identity when $m_n = n$). This convention does not affect any limit, and all construction formulas below concern $m_n > n$.

(b) Let $K \subseteq \Delta_d^\circ$ be nonempty and satisfy $K = \overline{\text{int}_{\mathbf{A}_d} K}^{\mathbf{A}_d}$. Then

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{\substack{p \in K \\ g \in \text{U}(d)}} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = \inf_{p \in K} F_{\text{univ}}(\gamma, p), \quad (8)$$

where the supremum is over all quantum channels from n to m_n qudits. The same sequence $\mathcal{C}_n^{\text{univ}}$, defined in [Section 2.4](#), asymptotically attains the displayed value.

Part (a) identifies the fidelity attained by the specified spectrum-independent cloner; it is not a pointwise optimum when p is known. In that case [Theorem 1.1](#) gives the larger value $F_{\text{orb}}(\gamma, p)$.

Theorem 1.3 (Cloning projector states). *For fixed $1 \leq r < d$, let $\text{Gr}(r, d)$ denote the set of rank- r orthogonal projectors on $\mathbb{C}^d$. Then*

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{P \in \text{Gr}(r, d)} F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) = \gamma^{-r(d-r)/2}. \quad (9)$$

Equivalently, the squared-fidelity value is $\gamma^{-r(d-r)}$. For $r = d$, the family contains only I_d/d , and the value is one.

This projector-state result is proved separately in [Section 4](#). It uses the same Cartan-sector channel after an exact compression to the positive support, but neither the full-rank LAN converse nor a general rank-deficient LAN theorem.

The common architecture of $\mathcal{C}_n^p$ and $\mathcal{C}_n^{\text{univ}}$ is summarized in [Fig. 1](#). After the Schur transform, the Young label μ carries the spectral information, while the conditional irrep state carries the unknown eigenbasis. A classical kernel chooses an output label ν , and the common covariant sector channel maps the conditional state into the ν -sector before the m_n -system output is reconstructed. For known p , the label kernel samples the exact target law $P_{m_n, p}$ independently of μ ; the spectrum-independent kernel instead randomly rounds $\gamma_n \mu$. Their label factors are therefore 1 and F_{cl} , while the common sector factor is F_{orb} , giving the exact asymptotic fidelities F_{orb} and F_{univ} for the known- and unknown-spectrum cloners, respectively. The details are developed in [Section 2](#).

In (8), the set K may inform the channel design, but the actual $p \in K$ and $g \in \text{U}(d)$ remain unknown. Unitary covariance removes the dependence on g , and the least favorable spectrum gives the infimum on the right. The channel $\mathcal{C}_n^{\text{univ}}$ is independent of K and attains the bound uniformly over K .²

For example, a natural admissible class keeps the smallest eigenvalue and every adjacent spectral gap uniformly positive. For $0 < \kappa < 2/[d(d+1)]$, set

$$\Delta_{d, \kappa} = \{p \in \mathbf{A}_d : p_d \geq \kappa, p_i - p_{i+1} \geq \kappa \ (i < d)\}.$$

This is a nonempty compact subset of Δ_d° and satisfies $\Delta_{d, \kappa} = \overline{\text{int}_{\mathbf{A}_d} \Delta_{d, \kappa}}^{\mathbf{A}_d}$, so (8) applies with $K = \Delta_{d, \kappa}$.³

²The condition $K = \overline{\text{int}_{\mathbf{A}_d} K}^{\mathbf{A}_d}$ allows the local converse at interior points to extend to all of K by continuity. Some thickness condition is essential: a singleton permits a channel tailored to its spectrum and has value F_{orb} , while other lower-dimensional spectral families can carry different local penalties.

³To verify nonempty relative interior, choose $\kappa' \in (\kappa, 2/[d(d+1)])$ and set $p_i = 1/d + ((d+1)/2 - i)\kappa'$. At the endpoint $\kappa = 2/[d(d+1)]$, the constraints force a singleton, so the compact-set theorem no longer applies.

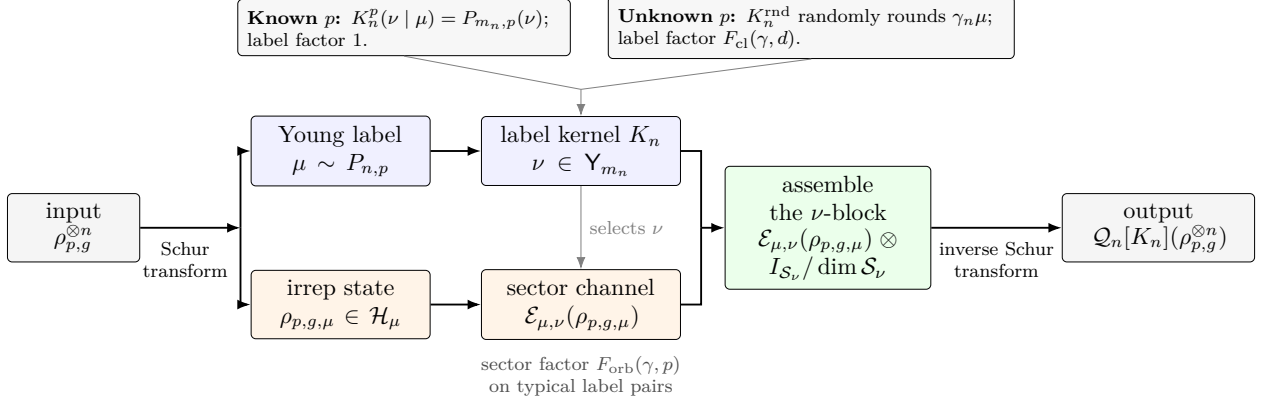


Figure 1: Cloner construction. The Schur transform separates the Young label μ from the conditional irrep state. A classical kernel chooses the output label ν : the known-spectrum kernel samples the target label law, whereas the unknown-spectrum kernel randomly dilates μ . Both constructions then apply the same Cartan-sector channel and restore a maximally mixed Specht factor. The label factor (1 or F_{cl}) and the common sector factor F_{orb} multiply to the asymptotic fidelities F_{orb} and F_{univ} of Theorems 1.1 and 1.2.

1.2 Remarks

Dependence on spectral mixedness The product formula makes precise one sense in which a more mixed state is easier to clone. For fixed $\gamma > 1$,

$$f'_\gamma(q) = \frac{(\gamma - 1)^2}{2\sqrt{q(\gamma - 1 + q)}(\sqrt{\gamma(\gamma - 1 + q)} + \sqrt{q})^2} > 0, \quad 0 < q < 1.$$

Consequently, if $p, p' \in \Delta_d^\circ$ satisfy

$$\frac{p'_j}{p'_i} \geq \frac{p_j}{p_i} \quad \text{for every } i < j, \quad (10)$$

that is, if every logarithmic spectral gap contracts in passing from p to p' , then

$$F_{\text{orb}}(\gamma, p') \geq F_{\text{orb}}(\gamma, p), \quad F_{\text{univ}}(\gamma, p') \geq F_{\text{univ}}(\gamma, p), \quad (11)$$

with strict inequality if any ratio is strictly larger. Condition (10) implies that p' is majorized by p but is stronger than majorization: for $d \geq 3$, majorization alone need not increase the cloning fidelity, whereas for $d = 2$ the single eigenvalue ratio does give majorization monotonicity. For example, at $\gamma = 2$, the simple spectra

$$p = (0.98, 0.011, 0.009), \quad p' = (0.94, 0.051, 0.009)$$

satisfy $p' \prec p$: the largest component decreases while the sum of the two largest components stays fixed. Nevertheless,

$$F_{\text{orb}}(2, p) \simeq 0.531279 > 0.520443 \simeq F_{\text{orb}}(2, p').$$

The same ordering holds for F_{univ} , whose classical factor is spectrum independent.

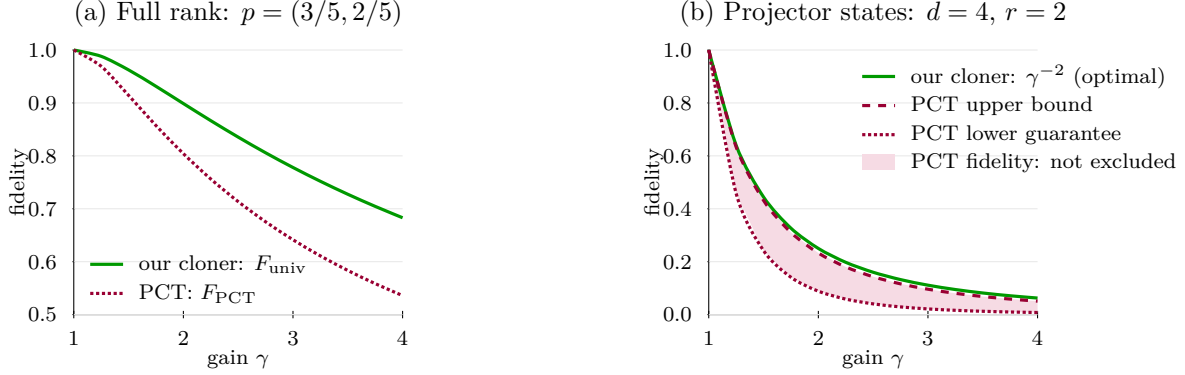


Figure 2: Asymptotic fidelities and bounds as functions of the gain $\gamma = \lim m_n/n$. (a) For the simple qubit spectrum $p = (3/5, 2/5)$, with neither protocol supplied the spectrum: F_{univ} is the exact asymptotic fidelity of our spectrum-independent cloner (Theorem 1.2) and F_{PCT} is the exact asymptotic fidelity of PCT (Theorem A.1). (b) For projector states with $d = 4$ and $r = 2$, with both protocols supplied r but not P : the solid curve is the optimal value γ^{-2} , attained by our projector-state cloner (Theorem 1.3); the dashed curve is the upper bound $\gamma^{-2}(\sqrt{2\gamma-1}/\gamma)^{1/2}$ on PCT from Proposition A.4, and the dotted curve is the published PCT lower guarantee $\gamma^{-7/2}$. The shaded region is the interval not excluded by these two bounds; it is not a confidence band and does not assert that the PCT fidelity converges. Since the dashed curve lies strictly below the solid one for every $\gamma > 1$, our projector cloner is strictly better.

Comparison with PCT The quantitative comparison with the purify–clone–trace protocol is developed in Section 5. At every fixed $\gamma > 1$, our spectrum-independent cloner has strictly larger asymptotic fidelity than PCT at every simple full-rank spectrum, and our projector-state cloner is strictly better than PCT for $1 < r < d$. Figure 2 previews both separations; Section 5 gives the exact formulas, their sample-complexity interpretation, and the order-of-limits caveat.

Limitations Apart from flat projector states, our theorems require a simple full-rank spectrum. We do not treat nonflat rank-deficient spectra, positive degeneracies, or neighborhoods in which the rank changes. Our estimates are also not uniform as $\gamma \downarrow 1$, and we do not determine the exact finite-copy optimum. The simple-spectrum restriction leads to noncommuting limits. For qubits,

$$\lim_{\substack{p \rightarrow (1/2, 1/2) \\ p_1 > p_2}} F_{\text{orb}}(\gamma, p) = \frac{2\sqrt{\gamma}}{1+\gamma} < 1,$$

whereas the known-spectrum family at $p = (1/2, 1/2)$ contains only $I_2/2$ and can be cloned perfectly. At any fixed sample size, preparing $(I_2/2)^{\otimes m_n}$ gives fidelity tending uniformly to one as $p \rightarrow (1/2, 1/2)$. Thus taking the degeneracy limit before the large-sample limit gives a different value: the orbit collapses at the degeneracy. Expected extensions to degenerate and rank-deficient spectra are discussed in Section 6.1.

1.3 Earlier and related work

Pure-state cloning. Early universal qubit cloners [BH96, GM97, BDE+98] were followed by Werner’s solution of global universal cloning in arbitrary dimension [Wer98] and Keyl–Werner’s optimality result for individual-clone tests [KW99]. State-dependent global cloning was developed

in [BDE+98, CB99]. For individual clones, the many-output limit is closely tied to estimation [MP95, BEM98, BA06, CD06]. For the joint output, the relation is subtler: global asymptotic cloning has been analyzed for broad pure-state families, and an optimal measure-and-prepare output need not consist of independent copies of the estimated state [CY14]. The criterion also matters for coherent-state and phase-covariant cloning [CKNWW05, KC22]: in particular, Cerf et al. prove Gaussian optimality for joint fidelity, while non-Gaussian operations can improve single-clone fidelity. Reviews include [SIGA05, Chi11, FWJ+14].

Mixed-state cloning, broadcasting, and PCT. The no-broadcasting theorem excludes exact cloning of a noncommuting family of density matrices [BCFJS96]; an entropic proof appears in [KH08]. Approximate mixed-state results include global-fidelity, relative-error, and state-dependent bounds for prescribed finite ensembles [Ras03a, Ras03b, Ras03c]; cloners for symmetric-subspace inputs and identical mixed qubits under one-site shrinking-factor criteria [Fan03, FLS07]; a restricted mixed-qubit construction under averaged Hilbert–Schmidt loss [Kaz10]; and entropic limitations on simultaneous approximate cloning and broadcasting [LW17]. Cirac–Ekert–Macchiavello solved optimal mixed-qubit purification and a cloning/estimation problem under local fidelity [CEM99]. Multi-input broadcasting and superbroadcasting likewise optimize marginals and permit correlations [DMP05, BDMP05, BDMP06, DF07]; these criteria do not determine fidelity with the joint product target.

Random purification followed by pure-state cloning gives a global-fidelity guarantee for bounded-rank mixed states [TWZ25, LTHC26]. Concurrent work shows that PCT attains the optimal copy complexity up to constants for squared fidelity $1 - \epsilon$, uniformly over states of rank at most r [FGSS26, JSO26]. These worst-case high-fidelity results complement our exact fixed-gain constants on specified spectral families; Section 5 compares them and distinguishes the orders of limits.

Gaussian cloning and the LAN. Guță–Matsumoto solved cloning of displaced thermal oscillator states with known temperature under joint norm-distance loss, reducing it to amplification and proving Gaussian optimality by stochastic ordering [GM06]. Bowles–Guță–Adesso used LAN for asymptotically optimal mixed-qubit purification and dilution under global trace-norm loss [BGA11]; related methods solve qudit channel reversal [BGA12]. These norm-distance optima do not by themselves determine our sharp fidelity constants in arbitrary fixed dimension, the separate spectral and orbital costs, or the explicit finite-dimensional Schur–Cartan cloners.

The relevant LAN framework was developed in [GJ07, GK06, KG09, YFG13]. Our fidelity amplification bound is attained by the least-noise amplifier [Cav82, GM06]. Low-rank LAN has a different mode structure [LN24]; we treat the boundary directly only for flat projector states.

1.4 Organization

Section 2 constructs the Schur–Cartan cloners and proves achievability from uniform label and sector estimates, established in Appendices B and C. Section 3 combines two-way LAN with the unrestricted Gaussian fidelity converse of Appendix D to prove full-rank optimality. Independently, Section 4 proves the projector-state theorem by exact compression and a finite-dimensional converse. Section 5 compares these results with PCT and its high-fidelity copy complexity; the required partial-trace and Fock-limit analysis is in Appendix A. The conclusion records the remaining problems and the conjectured boundary extensions.

Glossary of notation

We use $U(d)$ for the unitary group, $\mathcal{T}_1(\mathcal{H})$ for trace-class operators. The dimension d is fixed, $\gamma_n = m_n/n \rightarrow \gamma > 1$, N denotes a generic sample size, and $r_d = \binom{d}{2}$. Fidelity uses the unsquared convention, $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$. Unless stated otherwise, limits are as $n \rightarrow \infty$ with d, γ fixed; $o(1)$ is uniform only over the parameters specified in its statement.

Symbols	Meaning
$A_d, \Delta_d^\circ; K \in \Delta_d^\circ$	The affine hyperplane $\sum_i p_i = 1$, its strictly ordered positive chamber, and a compact subset of that chamber. Relative interiors and closures are in A_d .
$\rho_p, \rho_{p,g}; P \in \text{Gr}(r, d)$	$\rho_p = \text{diag}(p)$, $\rho_{p,g} = g\rho_p g^*$; a rank- r orthogonal projector, giving the state P/r , $1 \leq r < d$.
$q_{ij}, f_\gamma(q_{ij})$	$q_{ij} = p_j/p_i$ and its one-mode fidelity factor (4).
$F_{\text{orb}}, F_{\text{cl}}, F_{\text{univ}}$	Orbit optimum, classical dilation factor, and their product $F_{\text{univ}} = F_{\text{cl}}F_{\text{orb}}$, attained uniformly on compact spectral sets; see (5) and the minimax statement in Theorem 1.2.
$Y_N; \mathcal{H}_\lambda, \mathcal{S}_\lambda$	Partitions $\lambda \vdash N$ with at most d rows; the associated unitary and symmetric-group representation spaces.
$P_{N,p}, \rho_{p,\lambda}; \mathbb{T}_N(p)$	Young-label law of $\rho_p^{\otimes N}$, conditional irreducible state, and typical-label set.
$K_n(\nu \mid \mu); \mathcal{Q}_n[K_n]$	A Markov kernel on Young labels and its covariant channel lift. The spectral set K is distinct from the kernel K_n .
$\mathcal{C}_{\mu,\nu}, \mathcal{E}_{\mu,\nu}$	Cartan-sector channel for compatible labels and its extension to all label pairs by a fallback channel.
$\mathcal{M}_n; \mathcal{C}_n^p, \mathcal{C}_n^{\text{univ}}$	A general cloner; our known- and unknown-spectrum cloners.
$\mathbf{H}; h, z$	Spectral tangent hyperplane $\sum_i h_i = 0$; local spectral and orbital coordinates $h \in \mathbf{H}$, $z \in \mathbb{C}^{r_d}$.
$Y_p(z), g_{n,z}^{(p)}; \Omega_L$	Skew-Hermitian generator at the fixed base spectrum p , $g_{n,z}^{(p)} = e^{Y_p(z)/\sqrt{n}}$; bounded local parameter box.
$\mathcal{F}, \mathcal{F}_{\text{orb}}; \hat{N}$	One-mode Fock space, $\mathcal{F}_{\text{orb}} = \mathcal{F}^{\otimes r_d}$, and number operator; distinct from fidelity F and sample size N .
$\tau_p, \tau_p^{(\gamma)}$	Thermal product in the input and ideal target, and noisy amplifier output at zero displacement.
$\Phi_z^{(p)}, \Psi_z^{(p,\gamma)}; \Theta^{\text{in}}, \Theta^{\text{tar}}$	Displaced input and ideal target built from τ_p ; their counterparts including the classical Gaussian factor.

Symbols confined to one proof are defined there.

2 Cloner construction and fidelity analysis

Both cloning protocols use the same finite-dimensional architecture. We first recall the Schur–Weyl decomposition and lift an arbitrary Markov kernel on Young labels to a covariant quantum channel. We then choose the exact target-law kernel for known spectrum and the randomized dilation kernel for unknown spectrum, and combine their label estimates with the common sector analysis. The fidelities of the constructed cloners are evaluated in Section 2.6. We now develop the components summarized in Fig. 1.

2.1 Schur–Weyl decomposition

Let

$$\mathbf{Y}_N = \{\lambda \vdash N : \ell(\lambda) \leq d\}.$$

Schur–Weyl duality gives

$$(\mathbb{C}^d)^{\otimes N} \cong \bigoplus_{\lambda \in \mathbf{Y}_N} \mathcal{H}_\lambda \otimes \mathcal{S}_\lambda, \quad (12)$$

where $\mathcal{H}_\lambda$ is the irreducible $U(d)$ -module of highest weight λ and $\mathcal{S}_\lambda$ is the corresponding Specht module. For $\rho_p = \text{diag}(p_1, \dots, p_d)$,

$$\rho_p^{\otimes N} = \bigoplus_{\lambda \in \mathbf{Y}_N} P_{N,p}(\lambda) \rho_{p,\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}, \quad P_{N,p}(\lambda) = \dim \mathcal{S}_\lambda s_\lambda(p), \quad \rho_{p,\lambda} = \frac{\pi_\lambda(\rho_p)}{s_\lambda(p)}. \quad (13)$$

Here π_λ is the irreducible polynomial representation of highest weight λ , acting on $\mathcal{H}_\lambda$, and $s_\lambda(p) = \text{Tr } \pi_\lambda(\rho_p)$ is the corresponding Schur polynomial. For $\rho_{p,g} = g\rho_p g^*$, replace $\rho_{p,\lambda}$ by $\rho_{p,g,\lambda} = U_\lambda(g)\rho_{p,\lambda}U_\lambda(g)^*$, where $U_\lambda(g) = \pi_\lambda(g)$. The Young law $P_{N,p}$ depends on p , but not on g .

Define

$$\mathbf{T}_N(p) = \left\{ \lambda \in \mathbf{Y}_N : \left\| \frac{\lambda}{N} - p \right\|_\infty \leq N^{-1/3} \right\}. \quad (14)$$

Lemma 2.1 (Uniform Young concentration). *There is a constant $c_d > 0$, depending only on d , such that*

$$\sup_{p \in \Delta_d^\circ} P_{N,p}(\mathbf{T}_N(p)^c) \leq e^{-c_d N^{1/3}} \quad (15)$$

for all sufficiently large N .

Proof. Write $q = \lambda/N$. The Weyl dimension formula gives $\dim \mathcal{H}_\lambda \leq (N+1)^{d(d-1)/2}$. Every weight ξ of $\mathcal{H}_\lambda$ has $\lambda - \xi$ equal to a nonnegative integral combination of positive roots. Since $p_j/p_i < 1$ for $i < j$, this gives $p^\xi \leq p^\lambda$. Hence $s_\lambda(p) \leq \dim \mathcal{H}_\lambda p^\lambda$. The standard bounds $\dim \mathcal{S}_\lambda \leq \binom{N}{\lambda_1, \dots, \lambda_d}$ and $\binom{N}{\lambda_1, \dots, \lambda_d} p^\lambda \leq \exp[-ND(q\|p)]$ give

$$P_{N,p}(\lambda) \leq (N+1)^{d(d-1)/2} \exp[-ND(q\|p)], \quad (16)$$

which is the finite-sample estimate underlying the Keyl–Werner spectrum large-deviation theorem [KW01].

There are at most $(N+1)^{d-1}$ partitions of N with at most d rows. If $\lambda \notin \mathbf{T}_N(p)$, then $\|q - p\|_1 > N^{-1/3}$, and Pinsker’s inequality gives $D(q\|p) \geq \frac{1}{2}N^{-2/3}$. Therefore

$$P_{N,p}(\mathbf{T}_N(p)^c) \leq (N+1)^{d(d-1)/2+d-1} e^{-N^{1/3}/2}.$$

For all sufficiently large N , the polynomial factor is at most $e^{N^{1/3}/4}$. Hence the asserted bound holds with $c_d = 1/4$. $\square$

2.2 Cartan-sector channel and lifting of a label kernel

Let λ, θ be dominant weights, with unit highest-weight vectors v_λ, v_θ . The Cartan component $\mathcal{H}_{\lambda+\theta}$ occurs with multiplicity one in $\mathcal{H}_\lambda \otimes \mathcal{H}_\theta$. Let

$$V_{\lambda,\theta} : \mathcal{H}_{\lambda+\theta} \longrightarrow \mathcal{H}_\lambda \otimes \mathcal{H}_\theta, \quad V_{\lambda,\theta} v_{\lambda+\theta} = v_\lambda \otimes v_\theta, \quad (17)$$

be the normalized intertwining isometry. For $\mu \in \mathbf{Y}_n$ and $\nu \in \mathbf{Y}_m$, write $\nu - \mu \vdash m - n$ when the coordinatewise difference $\nu - \mu$ is a partition of $m - n$, and call such a pair (μ, ν) *compatible*. If this holds, put $\omega = \nu - \mu$ and define

$$\mathcal{C}_{\mu,\nu}(X) = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V_{\mu,\omega}^* (X \otimes I_{\mathcal{H}_\omega}) V_{\mu,\omega}. \quad (18)$$

The partial trace $\text{Tr}_{\mathcal{H}_\omega}(V_{\mu,\omega} V_{\mu,\omega}^*)$ commutes with the irreducible $\text{U}(d)$ -action on $\mathcal{H}_\mu$; hence Schur's lemma and a trace comparison give $\text{Tr}_{\mathcal{H}_\omega}(V_{\mu,\omega} V_{\mu,\omega}^*) = \frac{\dim \mathcal{H}_\nu}{\dim \mathcal{H}_\mu} I_{\mathcal{H}_\mu}$. Thus (18) is completely positive by tensoring with $I_{\mathcal{H}_\omega}$, applying $Y \mapsto V_{\mu,\omega}^* Y V_{\mu,\omega}$, and positive rescaling; the identity above makes it trace preserving, while equivariance of $V_{\mu,\omega}$ gives $\text{U}(d)$ -covariance.

When $\mu = (n, 0, \dots, 0)$ and $\nu = (m, 0, \dots, 0)$, the three irreducible spaces in (18) are symmetric powers, and the formula is Werner's universal pure-state n -to- m cloner [Wer98]. Thus the Cartan-sector channel extends Werner's construction from symmetric tensors to arbitrary compatible Schur sectors.

For every pair $(\mu, \nu) \in \mathbf{Y}_n \times \mathbf{Y}_m$, define a covariant channel $\mathcal{E}_{\mu,\nu} : \mathcal{T}_1(\mathcal{H}_\mu) \rightarrow \mathcal{T}_1(\mathcal{H}_\nu)$ by

$$\mathcal{E}_{\mu,\nu}(X) = \begin{cases} \mathcal{C}_{\mu,\nu}(X), & \nu - \mu \vdash m - n, \\ \text{Tr}(X) I_{\mathcal{H}_\nu} / \dim \mathcal{H}_\nu, & \text{otherwise.} \end{cases} \quad (19)$$

Let $K_n : \mathbf{Y}_n \rightsquigarrow \mathbf{Y}_{m_n}$, with transition probabilities $K_n(\nu \mid \mu)$, be any Markov kernel. Its Schur–Cartan lifting is the following channel, denoted by $\mathcal{Q}_n[K_n]$: apply the Schur transform, measure μ , discard the input Specht factor, sample ν from $K_n(\cdot \mid \mu)$, apply $\mathcal{E}_{\mu,\nu}$, adjoin $I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu$, and apply the inverse Schur transform. Each step is completely positive and trace preserving, so their composition is a channel. Because the label measurement and sampling are invariant and every map $\mathcal{E}_{\mu,\nu}$ is covariant, $\mathcal{Q}_n[K_n]$ is $\text{U}(d)$ -covariant.

Proposition 2.2 (Fidelity of a lifted label kernel). *Fix $p \in \Delta_d^\circ$ and $g \in \text{U}(d)$. Let*

$$\Pi_{n,p}(\mu, \nu) = P_{n,p}(\mu) K_n(\nu \mid \mu), \quad Q_{m_n,p}(\nu) = \sum_{\mu} \Pi_{n,p}(\mu, \nu), \quad (20)$$

and, for $\nu \in \mathbf{Y}_{m_n}$, let

$$A_{n,p,\nu} = \sum_{\mu \in \mathbf{Y}_n} \Pi_{n,p}(\mu, \nu) \mathcal{E}_{\mu,\nu}(\rho_{p,\mu}) \quad (21)$$

be the unnormalized irreducible output block, so that $\text{Tr } A_{n,p,\nu} = Q_{m_n,p}(\nu)$.

(i) *The fidelity of the lifted channel is given exactly by*

$$F(\mathcal{Q}_n[K_n](\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = \sum_{\nu \in \mathbf{Y}_{m_n}} \sqrt{P_{m_n,p}(\nu)} F(A_{n,p,\nu}, \rho_{p,\nu}). \quad (22)$$

(ii) *Suppose $G \subset \mathbf{Y}_n \times \mathbf{Y}_{m_n}$ satisfies $\nu - \mu \vdash m_n - n$ for every $(\mu, \nu) \in G$, and*

$$F(\mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}) \geq b \quad ((\mu, \nu) \in G)$$

for some $b \in [0, 1]$. Then

$$F(\mathcal{Q}_n[K_n](\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) \geq b \left(F(Q_{m_n,p}, P_{m_n,p}) - \sqrt{\Pi_{n,p}(G^c)} \right). \quad (23)$$

Proof. By covariance of $\mathcal{Q}_n[K_n]$ and unitary invariance of fidelity, we may take $g = I$. Under the output Schur decomposition, the actual and target states are

$$\bigoplus_{\nu \in \mathbf{Y}_{m_n}} A_{n,p,\nu} \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}, \quad \bigoplus_{\nu \in \mathbf{Y}_{m_n}} P_{m_n,p}(\nu) \rho_{p,\nu} \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}.$$

Fidelity is additive over orthogonal direct sums, multiplicative over tensor products, and satisfies $F(A, cB) = \sqrt{c} F(A, B)$ for $c \geq 0$. Since $F(I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu, I_{\mathcal{S}_\nu} / \dim \mathcal{S}_\nu) = 1$, this proves (22).

For (ii), set

$$r(\nu) = \sum_{\mu: (\mu, \nu) \notin G} \frac{\Pi_{n,p}(\mu, \nu)}{Q_{m_n,p}(\nu)} \quad \text{if } Q_{m_n,p}(\nu) > 0, \quad r(\nu) = 0 \quad \text{otherwise.}$$

Homogeneity and concavity of fidelity in its first argument give, for $Q_{m_n,p}(\nu) > 0$,

$$F(A_{n,p,\nu}, \rho_{p,\nu}) \geq \sqrt{Q_{m_n,p}(\nu)} \sum_{\mu} \frac{\Pi_{n,p}(\mu, \nu)}{Q_{m_n,p}(\nu)} F(\mathcal{E}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}) \geq b \sqrt{Q_{m_n,p}(\nu)} (1 - r(\nu)),$$

where the last step keeps only the pairs in G , on which $\mathcal{E}_{\mu,\nu} = \mathcal{C}_{\mu,\nu}$. Substituting into (22) yields

$$F(\mathcal{Q}_n[K_n](\rho_p^{\otimes n}), \rho_p^{\otimes m_n}) \geq b \left(F(Q_{m_n,p}, P_{m_n,p}) - \sum_{\nu} \sqrt{Q_{m_n,p}(\nu) P_{m_n,p}(\nu)} r(\nu) \right).$$

Since $0 \leq r \leq 1$ and $\sum_{\nu} Q_{m_n,p}(\nu) r(\nu) = \Pi_{n,p}(G^c)$, Cauchy–Schwarz yields

$$\sum_{\nu} \sqrt{Q_{m_n,p}(\nu) P_{m_n,p}(\nu)} r(\nu) \leq \sqrt{\Pi_{n,p}(G^c)} \left(\sum_{\nu} P_{m_n,p}(\nu) r(\nu) \right)^{1/2} \leq \sqrt{\Pi_{n,p}(G^c)}.$$

This proves (23). $\square$

We now specialize the lifted channel by choosing its output-label kernel. The two protocols differ only in this choice: the known-spectrum construction samples the exact target law, whereas the unknown-spectrum construction uses a randomized dilation of the measured input label.

2.3 Known spectrum: exact target-label sampling

Fix $p \in \Delta_d^\circ$. For every n such that $m_n > n$, define

$$K_n^p(\nu \mid \mu) = P_{m_n,p}(\nu). \quad (24)$$

Thus ν is independent of μ and has the exact target law. Define the amplification branch by

$$\mathcal{C}_n^p := \mathcal{Q}_n[K_n^p]. \quad (25)$$

Pairs with $\nu - \mu \nmid m_n - n$ use $\mathcal{E}_{\mu,\nu}(X) = \text{Tr}(X) I_{\mathcal{H}_\nu} / \dim \mathcal{H}_\nu$ without changing the sampled label ν . Together with the common convention in Section 1.1, this defines the full $U(d)$ -covariant sequence $(\mathcal{C}_n^p)_n$, whose output Young law on $\rho_{p,g}^{\otimes n}$ is $P_{m_n,p}$ for every n and g .

2.4 Unknown spectrum: randomized rounding

For $N \in \mathbb{N}$, let

$$\mathbb{L}_N = \left\{ x \in \mathbb{Z}^d : \sum_i x_i = N \right\}.$$

For every n such that $m_n > n$, set $\nu_n^{\text{fb}} = (m_n, 0, \dots, 0)$. Given $\mu \in \mathbb{Y}_n$, draw independent variables $U_1, \dots, U_{d-1}$, each uniform on $[-1/2, 1/2)$, and set

$$\tilde{\nu}_i = \left\lfloor \gamma_n(\mu_i + U_i) + \frac{1}{2} \right\rfloor \quad (1 \leq i < d), \quad \tilde{\nu}_d = m_n - \sum_{i < d} \tilde{\nu}_i. \quad (26)$$

This is a spectrum-independent lattice version of the dilation $\mu \mapsto \gamma_n \mu$: it sends typical labels near $n\mu$ to labels near $m_n \mu$, with $\tilde{\nu} - \mu$ typically a partition. The uniform shifts randomize the rounding error, avoiding lattice bias in the Gaussian limit. Then $\tilde{\nu} \in \mathbb{L}_{m_n}$. Let $D_n(\tilde{\nu} \mid \mu)$ denote its distribution. Define $K_n^{\text{rnd}} : \mathbb{Y}_n \rightsquigarrow \mathbb{Y}_{m_n}$ by

$$K_n^{\text{rnd}}(\nu \mid \mu) = \begin{cases} D_n(\nu \mid \mu), & \nu \neq \nu_n^{\text{fb}} \text{ and } \nu - \mu \vdash m_n - n, \\ 1 - \sum_{\substack{\lambda \in \mathbb{Y}_{m_n}, \\ \lambda \neq \nu_n^{\text{fb}}, \\ \lambda - \mu \vdash m_n - n}} D_n(\lambda \mid \mu), & \nu = \nu_n^{\text{fb}}, \\ 0, & \text{otherwise.} \end{cases} \quad (27)$$

Under (27), every rounded label outside $\mathbb{Y}_{m_n}$, and every $\tilde{\nu} \in \mathbb{Y}_{m_n}$ for which $\tilde{\nu} - \mu \not\vdash m_n - n$, is replaced by ν_n^{fb} , which also retains any mass rounded onto it. Define the amplification branch by

$$\mathcal{C}_n^{\text{univ}} := \mathcal{Q}_n[K_n^{\text{rnd}}]. \quad (28)$$

The rounding variables, (27), and the maps $\mathcal{E}_{\mu, \nu}$ are spectrum independent, while $\mathcal{Q}_n[K_n^{\text{rnd}}]$ is $\text{U}(d)$ -covariant. Together with the common convention in Section 1.1, this defines the full spectrum-independent, $\text{U}(d)$ -covariant sequence $(\mathcal{C}_n^{\text{univ}})_n$.

2.5 Sector and label asymptotics

The following proposition and its mixture-stability consequence provide the representation-theoretic asymptotic estimate used in the cloning protocols.

Proposition 2.3 (Uniform Cartan-sector fidelity). *Let $K \in \Delta_d^\circ$ and let $\delta_n \geq 0$ satisfy $\delta_n \rightarrow 0$. For all sufficiently large n , every $p \in K$, $\mu \in \mathbb{Y}_n$, and $\nu \in \mathbb{Y}_{m_n}$ satisfying*

$$\left\| \frac{\mu}{n} - p \right\|_\infty \leq \delta_n, \quad \left\| \frac{\nu}{m_n} - p \right\|_\infty \leq \delta_n$$

also satisfies $\nu - \mu \vdash m_n - n$. Moreover,

$$F(\mathcal{C}_{\mu, \nu}(\rho_{p, \mu}), \rho_{p, \nu}) = F_{\text{orb}}(\gamma, p) + o(1) \quad (29)$$

uniformly over all such p, μ, ν .

Lemma 2.4 (Uniform stability under sector mixing). *Let $K \Subset \Delta_d^\circ$ and let $\delta_n \geq 0$ satisfy $\delta_n \rightarrow 0$. For $p \in K$ and $\nu \in \mathbf{Y}_{m_n}$ satisfying*

$$\left\| \frac{\nu}{m_n} - p \right\|_\infty \leq \delta_n,$$

let $(a_\mu)_\mu$ be any probability distribution supported on labels $\mu \in \mathbf{Y}_n$ satisfying

$$\left\| \frac{\mu}{n} - p \right\|_\infty \leq \delta_n.$$

For all sufficiently large n , every pair in the support is compatible, and

$$F\left(\sum_{\mu} a_{\mu} \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}\right) = F_{\text{orb}}(\gamma, p) + o(1), \quad (30)$$

where the $o(1)$ is uniform over p, ν , and all probability distributions $(a_\mu)_\mu$ satisfying the preceding conditions.

The mechanism is already visible on states obtained by a fixed number of lowering operations. Writing $\omega = \nu - \mu$, the Cartan intertwiner satisfies

$$V_{\mu,\omega} F_{\alpha}^{(\nu)} = (F_{\alpha}^{(\mu)} \otimes I + I \otimes F_{\alpha}^{(\omega)}) V_{\mu,\omega},$$

where $F_{\alpha}^{(\lambda)}$ is the lowering operator for $\alpha = e_i - e_j$ on $\mathcal{H}_{\lambda}$. After normalization, each excitation splits binomially between the two factors, with fraction

$$s_{\alpha,n} = \frac{\mu_i - \mu_j}{\nu_i - \nu_j} = \frac{1}{\gamma} + o(1)$$

entering the μ -factor. Thus the conditional thermal parameter $q_{ij} = p_j/p_i$ is transformed into

$$1 - s_{\alpha,n} + s_{\alpha,n} q_{ij} \longrightarrow 1 - \frac{1 - q_{ij}}{\gamma} =: q_{ij}^{(\gamma)}. \quad (31)$$

The low-weight matrices of the sector output and ideal sector therefore approach those of $\tau_p^{(\gamma)}$ and τ_p , whose fidelity is $F_{\text{orb}}(\gamma, p)$. This is a statement about conditional states for typical label pairs, not convergence of the varying channels $\mathcal{C}_{\mu,\nu}$ to the Gaussian amplifier. The exact PBW calculation and uniform thermal-tail argument that complete the proof are given in Appendix B.

The second estimate concerns only K_n^{rnd} .

Proposition 2.5 (Randomized Young-label dilation). *There is a constant $c_{\text{rnd}} < \infty$, depending only on d and the fixed sequence (γ_n) , such that, for every compact $K \Subset \Delta_d^\circ$ and all sufficiently large n , every $p \in K$ and $\mu \in \mathbf{T}_n(p)$ satisfy*

$$\text{supp } K_n^{\text{rnd}}(\cdot \mid \mu) \subseteq \{\nu \in \mathbf{Y}_{m_n} : \nu - \mu \vdash m_n - n, \quad \|\nu - \gamma_n \mu\|_\infty \leq c_{\text{rnd}}\}. \quad (32)$$

For $m_n > n$, let

$$Q_{m_n,p}(\nu) := \sum_{\mu \in \mathbf{Y}_n} P_{n,p}(\mu) K_n^{\text{rnd}}(\nu \mid \mu), \quad \nu \in \mathbf{Y}_{m_n},$$

be the induced output-label law. Then, for every compact $K \Subset \Delta_d^\circ$,

$$F(Q_{m_n,p}, P_{m_n,p}) = F_{\text{cl}}(\gamma, d) + o(1) \quad \text{uniformly for } p \in K. \quad (33)$$

The uniform dither in (26) makes the label kernel a discrete realization of continuous dilation. After centering at Np and scaling by $\sqrt{N}$, embed a label law as a piecewise-constant density on the fundamental cells of the label lattice. Before fallback replacement, the intersection-volume identity (163) identifies the embedded rounded law with the target-cell average of the $\sqrt{\gamma_n}$ -dilation of the embedded input law. Write

$$\Sigma_p := (\text{diag}(p) - pp^\top) \Big|_{\{x \in \mathbb{R}^d: \sum_i x_i = 0\}}. \quad (34)$$

The Young local limit is $\mathbf{N}_{0,\Sigma_p}$, so randomized rounding changes the limit to $\mathbf{N}_{0,\gamma\Sigma_p}$, whereas the ideal m_n -label law still converges to $\mathbf{N}_{0,\Sigma_p}$. Their fidelity (Hellinger affinity) is

$$F(\mathbf{N}_{0,\gamma\Sigma_p}, \mathbf{N}_{0,\Sigma_p}) = \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{(d-1)/2} = F_{\text{cl}}(\gamma, d). \quad (35)$$

The support check and fallback rule affect only vanishing probability on typical labels.

The proofs of Proposition 2.3 and Lemma 2.4 are given in Appendix B; the proof of Proposition 2.5 is given in Appendix C.

2.6 Fidelity of the constructed cloners

We now combine the label and sector estimates with Proposition 2.2 to evaluate the fidelities of both cloners.

Known-spectrum achievability in Theorem 1.1. By the covariance established in Section 2.3, take $g = I$. For K_n^p ,

$$\Pi_{n,p}(\mu, \nu) = P_{n,p}(\mu)P_{m_n,p}(\nu), \quad Q_{m_n,p} = P_{m_n,p}.$$

Let $G_{n,p} = \mathbb{T}_n(p) \times \mathbb{T}_{m_n}(p)$. Young concentration, Lemma 2.1, gives $\Pi_{n,p}(G_{n,p}^c) = o(1)$. Apply Proposition 2.3 with $K = \{p\}$ and

$$\delta_n = n^{-1/3} + m_n^{-1/3}.$$

It follows that every pair in $G_{n,p}$ is compatible for all sufficiently large n , and there is an $\eta_n \rightarrow 0$ such that, uniformly on $G_{n,p}$,

$$F(\mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \rho_{p,\nu}) \geq F_{\text{orb}}(\gamma, p) - \eta_n.$$

Applying part (ii) of Proposition 2.2 and using $F(Q_{m_n,p}, P_{m_n,p}) = 1$ yields

$$F(\mathcal{C}_n^p(\rho_p^{\otimes n}), \rho_p^{\otimes m_n}) \geq (F_{\text{orb}}(\gamma, p) - \eta_n)(1 - \sqrt{\Pi_{n,p}(G_{n,p}^c)}) = F_{\text{orb}}(\gamma, p) - o(1).$$

□

Exact fidelity of $\mathcal{C}_n^{\text{univ}}$. By covariance, take $g = I$. Fix $K \in \Delta_d^\circ$, and use $\Pi_{n,p}$, $Q_{m_n,p}$, and $A_{n,p,\nu}$ from Proposition 2.2 with $K_n = K_n^{\text{rnd}}$. For brevity write $P = P_{m_n,p}$, $Q = Q_{m_n,p}$, and $A_\nu = A_{n,p,\nu}$. Part (i) of that proposition gives

$$F_{n,p}^{\text{out}} := F(\mathcal{C}_n^{\text{univ}}(\rho_p^{\otimes n}), \rho_p^{\otimes m_n}) = \sum_\nu \sqrt{P(\nu)} F(A_\nu, \rho_{p,\nu}). \quad (36)$$

Define the typical-input part and its label mass by

$$A_\nu^{\text{good}} = \sum_{\mu \in \mathbb{T}_n(p)} \Pi_{n,p}(\mu, \nu) \mathcal{E}_{\mu,\nu}(\rho_{p,\mu}), \quad Q^{\text{good}}(\nu) = \text{Tr } A_\nu^{\text{good}}.$$

Put $A_\nu^{\text{bad}} = A_\nu - A_\nu^{\text{good}} \succeq 0$ and $Q^{\text{bad}}(\nu) = \text{Tr } A_\nu^{\text{bad}}$. Trace preservation gives

$$Q^{\text{good}} + Q^{\text{bad}} = Q, \quad \sum_\nu Q^{\text{bad}}(\nu) = P_{n,p}(\mathbb{T}_n(p)^c) =: \epsilon_{n,p}. \quad (37)$$

For every typical input label, [Proposition 2.5](#) makes all output labels in its support compatible and gives

$$\left\| \frac{\nu}{m_n} - p \right\|_\infty \leq n^{-1/3} + \frac{c_{\text{rnd}}}{m_n} =: \delta_n.$$

Thus $\mathcal{E}_{\mu,\nu} = \mathcal{C}_{\mu,\nu}$ on these pairs, and both input and output normalized labels are within $\delta_n \rightarrow 0$ of p . Whenever $Q^{\text{good}}(\nu) > 0$, the state $A_\nu^{\text{good}}/Q^{\text{good}}(\nu)$ is therefore a mixture covered by [Lemma 2.4](#). Its uniform conclusion and homogeneity of fidelity give a sequence $\eta_n \rightarrow 0$, independent of $p \in K$ and ν , such that

$$\left| F(A_\nu^{\text{good}}, \rho_{p,\nu}) - F_{\text{orb}}(\gamma, p) \sqrt{Q^{\text{good}}(\nu)} \right| \leq \eta_n \sqrt{Q^{\text{good}}(\nu)}. \quad (38)$$

If $Q^{\text{good}}(\nu) = 0$, positivity implies $A_\nu^{\text{good}} = 0$, so the same estimate holds. Multiplying by $\sqrt{P(\nu)}$ and summing shows that

$$\left| \sum_\nu \sqrt{P(\nu)} F(A_\nu^{\text{good}}, \rho_{p,\nu}) - F_{\text{orb}}(\gamma, p) F(Q^{\text{good}}, P) \right| \leq \eta_n,$$

because Cauchy–Schwarz gives $F(Q^{\text{good}}, P) \leq \sqrt{\sum_\nu Q^{\text{good}}(\nu)} \leq 1$.

All blocks have trace at most one, and $\|A_\nu - A_\nu^{\text{good}}\|_1 = Q^{\text{bad}}(\nu)$. Consequently, [\(56\)](#) and Cauchy–Schwarz give

$$\left| F_{n,p}^{\text{out}} - \sum_\nu \sqrt{P(\nu)} F(A_\nu^{\text{good}}, \rho_{p,\nu}) \right| \leq \sum_\nu \sqrt{P(\nu) Q^{\text{bad}}(\nu)} \leq \sqrt{\epsilon_{n,p}}.$$

The same continuity estimate applies to the subnormalized label law Q^{good} : since $\|Q - Q^{\text{good}}\|_{\ell^1} = \epsilon_{n,p}$,

$$\left| F(Q^{\text{good}}, P) - F(Q, P) \right| \leq \sqrt{\epsilon_{n,p}}.$$

Combining these estimates and using $0 \leq F_{\text{orb}}(\gamma, p) \leq 1$ yields

$$\left| F_{n,p}^{\text{out}} - F_{\text{orb}}(\gamma, p) F(Q_{m_n,p}, P_{m_n,p}) \right| \leq \eta_n + 2\sqrt{\epsilon_{n,p}}. \quad (39)$$

Uniform Young concentration gives $\sup_{p \in K} \epsilon_{n,p} \rightarrow 0$, while [\(33\)](#) gives $F(Q_{m_n,p}, P_{m_n,p}) = F_{\text{cl}}(\gamma, d) + o(1)$ uniformly on K . Hence

$$\sup_{p \in K} \left| F\left(\mathcal{C}_n^{\text{univ}}(\rho_p^{\otimes n}), \rho_p^{\otimes m_n}\right) - F_{\text{cl}}(\gamma, d) F_{\text{orb}}(\gamma, p) \right| \rightarrow 0.$$

Covariance makes the fidelity independent of g , completing the proof. $\square$

3 Optimality from LAN and Gaussian converses

We prove the matching minimax upper bounds by restricting arbitrary cloners to local neighborhoods. Two-way LAN transports each cloner to an exact channel between Gaussian models, where dilation and amplification bounds apply. After defining the local coordinates, we establish these bounds, state LAN, and complete the minimax proofs. [Figure 3](#) summarizes the transport; [Lemma 3.5](#) proves the common fidelity estimate.

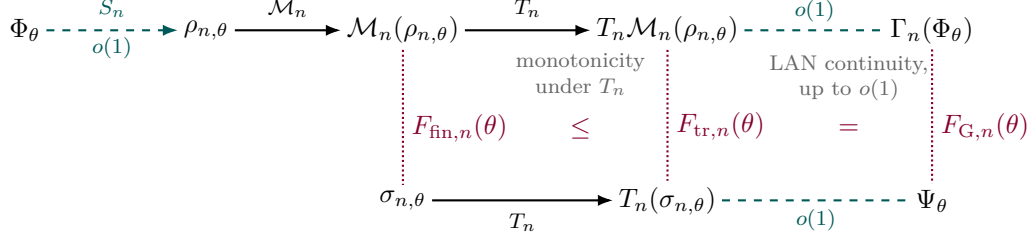


Figure 3: LAN transfer for $\Gamma_n = T_n \circ \mathcal{M}_n \circ S_n$ on a fixed compact local parameter set C . The top and bottom rows track output and target; dotted segments indicate fidelity pairs, solid arrows exact channels, and dashed links uniform $o(1)$ trace-norm errors. Monotonicity and continuity give $F_{\text{fin},n}(\theta) \leq F_{\text{tr},n}(\theta) = F_{\text{G},n}(\theta) + o(1)$ uniformly on C , as proved in [Lemma 3.5](#). The LAN maps need not be inverses, and Γ_n need not be Gaussian or converge.

3.1 Local coordinates

Fix a base spectrum $p \in \Delta_d^\circ$. Let

$$\mathbf{H} = \left\{ h \in \mathbb{R}^d : \sum_i h_i = 0 \right\}, \quad Y_p(z) = \sum_{j < k} \frac{z_{jk} E_{kj} - \bar{z}_{jk} E_{jk}}{\sqrt{p_j - p_k}}, \quad z \in \mathbb{C}^{r_d},$$

where E_{jk} are the standard matrix units, and define

$$g_{n,z}^{(p)} = \exp\left(\frac{Y_p(z)}{\sqrt{n}}\right).$$

In $\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}$, the coordinates h and z describe local changes of eigenvalues and eigenbasis, respectively. The denominators in Y_p use the fixed base spectrum p , including when the local spectrum is $p + h/\sqrt{n}$. Since $Y_p(z)^* = -Y_p(z)$, one has $g_{n,z}^{(p)} \in \text{U}(d)$. For $L > 0$, set

$$\Omega_L = \{(h, z) \in \mathbf{H} \times \mathbb{C}^{r_d} : \|h\|_\infty \leq L, |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for all } i < j\}.$$

For every fixed $p \in \Delta_d^\circ$ and $L > 0$, the vector $p + h/\sqrt{n}$ belongs to Δ_d° for every $(h, z) \in \Omega_L$ once n is sufficiently large. Under LAN, h becomes a $(d-1)$ -dimensional classical Gaussian shift, while z becomes a collection of displaced oscillator coordinates. The Gaussian theorem below optimizes the two corresponding fidelity losses jointly.

3.2 Gaussian amplification

Let $\mathcal{F}$ be the one-mode Fock space with number basis $\{|k\rangle : k \geq 0\}$, creation and annihilation operators a^*, a , and number operator $\hat{N} = a^*a$. We use the Weyl convention

$$D(z) = \exp(za^* - \bar{z}a). \quad (40)$$

For $0 < q < 1$, set

$$\tau_q = (1-q) \sum_{k \geq 0} q^k |k\rangle\langle k|.$$

Two thermal states commute, and hence

$$F(\tau_q, \tau_{q'}) = \frac{\sqrt{(1-q)(1-q')}}{1 - \sqrt{qq'}}. \quad (41)$$

For $G \geq 1$, put $q^{(G)} = 1 - (1 - q)/G$. The quantum-limited phase-insensitive amplifier $\mathcal{A}_G$ satisfies

$$\mathcal{A}_G(D(z)\tau_q D(z)^*) = D(\sqrt{G}z)\tau_{q^{(G)}}D(\sqrt{G}z)^*. \quad (42)$$

Equivalently, $\mathcal{A}_G(X) = \text{Tr}_b[S_r(X \otimes |0\rangle\langle 0|_b)S_r^*]$, where S_r is a two-mode squeezing unitary with $\cosh^2 r = G$. Squeezing supplies the gain $\sqrt{G}$, while tracing out the vacuum idler produces the minimum added noise encoded by $q^{(G)}$.

For $p \in \Delta_d^\circ$, define on $\mathcal{F}_{\text{orb}} = \mathcal{F}^{\otimes r_d}$

$$\tau_p = \bigotimes_{i < j} \tau_{q_{ij}}, \quad \tau_p^{(\gamma)} = \bigotimes_{i < j} \tau_{q_{ij}^{(\gamma)}}.$$

With $D(z) = \bigotimes_{i < j} D(z_{ij})$, the three relevant states are

$$\begin{aligned} \text{input:} \quad & \Phi_z^{(p)} = D(z)\tau_p D(z)^*, \\ \text{ideal gain-}\sqrt{\gamma} \text{ target:} \quad & \Psi_z^{(p,\gamma)} = D(\sqrt{\gamma}z)\tau_p D(\sqrt{\gamma}z)^*, \\ \text{amplifier output:} \quad & \mathcal{A}_\gamma^{\otimes r_d}(\Phi_z^{(p)}) = D(\sqrt{\gamma}z)\tau_p^{(\gamma)} D(\sqrt{\gamma}z)^*. \end{aligned} \quad (43)$$

Thus the ideal target retains the thermal state τ_p ; only the actual amplifier output contains the additional noise $\tau_p^{(\gamma)}$. By (41),

$$F(\tau_{q^{(\gamma)}}, \tau_q) = f_\gamma(q). \quad (44)$$

Thus, for every $z \in \mathbb{C}^{r_d}$, the product-amplifier output and the corresponding ideal state satisfy

$$F(\mathcal{A}_\gamma^{\otimes r_d}(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}) = F_{\text{orb}}(\gamma, p). \quad (45)$$

Equip $\mathbf{H}$ with its induced Euclidean measure and recall

$$\Sigma_p = (\text{diag}(p) - pp^\top)|_{\mathbf{H}} > 0.$$

Write $\mathbf{N}_{h, \Sigma_p}$ for the Gaussian law on $\mathbf{H}$ with mean h and covariance Σ_p . A classical-quantum state has the form

$$R(x) = r(x)\rho_x, \quad r(x) \geq 0, \quad \int_{\mathbf{H}} r(x) dx = 1,$$

where x is a classical variable, $r(x)$ is its density, and ρ_x is a quantum state on $\mathcal{F}_{\text{orb}}$. A channel Λ is completely positive and preserves $\int \text{Tr } R(x) dx$. Here complete positivity means positivity of $\text{id}_{M_k} \otimes \Lambda$ for every $k \geq 1$, with the matrix factor acting on the quantum trace-class fibers; the classical algebra is abelian. If $S(x) = s(x)\sigma_x$, then

$$F(R, S) = \int_{\mathbf{H}} F(R(x), S(x)) dx = \int_{\mathbf{H}} \sqrt{r(x)s(x)} F(\rho_x, \sigma_x) dx.$$

With $\|R\|_1 := \int_{\mathbf{H}} \|R(x)\|_1 dx$, the usual monotonicity and joint-concavity properties remain valid, and the first equality above is the continuous direct-sum identity for fidelity.

Set

$$\Theta_{h,z}^{\text{in}} = \mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)}, \quad \Theta_{h,z}^{\text{tar}} = \mathbf{N}_{\sqrt{\gamma}h, \Sigma_p} \otimes \Psi_z^{(p,\gamma)}. \quad (46)$$

Let

$$\mathcal{Z}_L = \{z \in \mathbb{C}^{r_d} : |\Re z_{ij}|, |\Im z_{ij}| \leq L \text{ for all } i < j\}.$$

Theorem 3.1 (Gaussian amplification optimum). *For every $p \in \Delta_d^\circ$,*

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|Z_L|} \int_{Z_L} F\left(\Lambda(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}\right) dz = F_{\text{orb}}(\gamma, p), \quad (47)$$

and

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Omega_L|} \int_{\Omega_L} F\left(\Lambda(\Theta_{h,z}^{\text{in}}), \Theta_{h,z}^{\text{tar}}\right) dh dz = F_{\text{univ}}(\gamma, p). \quad (48)$$

The first supremum is over all quantum channels on $\mathcal{F}_{\text{orb}}$; the second is over all classical-quantum channels. No Gaussianity, covariance, or product-structure assumption is imposed on the competing channels in either supremum.

The achievability mechanism is explicit. The product amplifier $\mathcal{A}_\gamma^{\otimes r_d}$ attains $F_{\text{orb}}(\gamma, p)$ in the orbital model. In the full model, tensor it with the classical pushforward $x \mapsto \sqrt{\gamma}x$. The resulting Gaussian affinity is $F_{\text{cl}}(\gamma, d)$, so multiplicativity gives $F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p) = F_{\text{univ}}(\gamma, p)$.

For the converse, average a near-optimal channel over translates of the expanding flat-prior boxes. A weighted-fidelity witness reduces the problem to an oscillator moment. The corresponding weighted oscillator maps have a subsequential limit, tested against compact observables, that is completely positive and displacement covariant. Its output trace on normalized inputs is bounded by 1 in the orbital model and, by a direct Gaussian translation estimate, by $F_{\text{cl}}(\gamma, d)$ in the full model. The least-noise inequality then gives the respective bounds $F_{\text{orb}}(\gamma, p)$ and $F_{\text{univ}}(\gamma, p)$. The compactness and witness arguments are given in [Proposition D.5](#) and [Appendix D](#).

3.3 Local asymptotic normality

Here LAN means that, uniformly for (h, z) in compact sets, the state families $\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n}$ and $\mathcal{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)}$ are asymptotically interconvertible by channels with vanishing trace-norm error. The Gaussian and oscillator factors represent spectral and eigenbasis fluctuations, respectively; cloning scales (h, z) by $\sqrt{\gamma}$.

We fix the oscillator normalization directly from the local tangent and the collective fluctuation operators. Put $\Delta_{ij} = p_i - p_j$. The chart in [Section 3.1](#) satisfies

$$\frac{[Y_p(z), \rho_p]_{ji}}{\sqrt{\Delta_{ij}}} = z_{ij}, \quad i < j. \quad (49)$$

For $a_{ij,n} = (n\Delta_{ij})^{-1/2} \sum_{s=1}^n E_{ij}^{(s)}$, where $E_{ij}^{(s)}$ acts on the s -th tensor factor, the expectations at $\rho_p^{\otimes n}$ obey

$$\langle [a_{ij,n}, a_{ij,n}^*] \rangle = 1, \quad \langle a_{ij,n}^* a_{ij,n} \rangle = \frac{p_j}{\Delta_{ij}} = \frac{q_{ij}}{1 - q_{ij}}.$$

Under the local state, (49) gives $\langle a_{ij,n} \rangle = z_{ij} + o(1)$, uniformly on compact parameter sets. Thus the limiting mode has mean z_{ij} and thermal parameter q_{ij} , agreeing with $D(z_{ij})\tau_{q_{ij}}D(z_{ij})^*$ in (40). This is also the rotation-to-displacement normalization in the constructive proof of [\[KG09, Lemma 6.4 and Section 7.3\]](#). We use their two-way LAN theorem with oscillator coordinates expressed in this convention.

Theorem 3.2 (Finite-dimensional LAN and target-sample scaling). *Fix $p \in \Delta_d^\circ$. There are channels*

$$T_N^{\text{full}} : \mathcal{T}_1((\mathbb{C}^d)^{\otimes N}) \longrightarrow L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}})), \quad S_N^{\text{full}} : L^1(\mathbf{H}; \mathcal{T}_1(\mathcal{F}_{\text{orb}})) \longrightarrow \mathcal{T}_1((\mathbb{C}^d)^{\otimes N})$$

such that, for every compact $C \subset \mathbf{H} \times \mathbb{C}^{r_d}$,

$$\sup_{(h,z) \in C} \left\| T_n^{\text{full}} \left(\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n} \right) - \mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)} \right\|_1 \longrightarrow 0, \quad (50)$$

$$\sup_{(h,z) \in C} \left\| S_n^{\text{full}} \left(\mathbf{N}_{h, \Sigma_p} \otimes \Phi_z^{(p)} \right) - \rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes n} \right\|_1 \longrightarrow 0, \quad (51)$$

and

$$\sup_{(h,z) \in C} \left\| T_{m_n}^{\text{full}} \left(\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}}^{\otimes m_n} \right) - \mathbf{N}_{\sqrt{\gamma}h, \Sigma_p} \otimes \Psi_z^{(p, \gamma)} \right\|_1 \longrightarrow 0. \quad (52)$$

Proof. The forward and reverse assertions (50)–(51) are [KG09, Theorem 4.3], after identifying their $(d-1)$ -dimensional spectral coordinate with $\mathbf{H}$ and using the oscillator normalization (49). For the classical factor, let

$$\iota : \mathbb{R}^{d-1} \longrightarrow \mathbf{H}, \quad \iota(u_1, \dots, u_{d-1}) = \left(u_1, \dots, u_{d-1}, -\sum_{i < d} u_i \right),$$

and let A be the matrix of ι . In the coordinates of [KG09, Eq. (3.3)], the classical covariance is the $(d-1) \times (d-1)$ matrix

$$V(p)_{ij} = \delta_{ij} p_i - p_i p_j, \quad i, j < d.$$

A direct calculation gives

$$AV(p)A^\top = \text{diag}(p) - pp^\top.$$

Hence, when $h = \iota(u)$, the pushforward of their classical Gaussian is $\mathbf{N}_{h, \Sigma_p}$. The fluctuation normalization above identifies the oscillator state with $\Phi_z^{(p)}$. Their Schrödinger-picture maps are completely positive and trace preserving on the trace-class and classical-quantum L^1 spaces shown above.

The target-sample assertion (52) is a consequence of the first assertion. Indeed, the identity

$$\rho_{p+h/\sqrt{n}, g_{n,z}^{(p)}} = \rho_{p+\sqrt{\gamma}h/\sqrt{m_n}, g_{m_n, \sqrt{\gamma}n}^{(p)}}$$

follows from $\sqrt{\gamma n}/\sqrt{m_n} = 1/\sqrt{n}$ and the real linearity of Y_p , with the same base p at both sample sizes. Apply (50) at sample size m_n on one compact set containing all scaled parameters. Since $\gamma_n \rightarrow \gamma$, translation of Gaussian densities and Weyl conjugation are uniformly trace-norm continuous there, which gives (52). $\square$

Remark 3.3 (Qubit normalization check). For $d = 2$, put $\Delta = p_1 - p_2 > 0$, $q = p_2/p_1$, and take $z = t \in \mathbb{R}$. The exact qubit fidelity identity gives

$$F\left(\rho_p^{\otimes n}, \rho_{p, g_{n,t}^{(p)}}^{\otimes n}\right) = \left(1 - \Delta^2 \sin^2 \frac{t}{\sqrt{n\Delta}}\right)^{n/2} \longrightarrow e^{-\Delta t^2/2}.$$

This equals $F(\tau_q, D(t)\tau_q D(t)^*) = \exp[-(1-q)t^2/(2(1+q))]$, confirming the displacement scale in Theorem 3.2.

Corollary 3.4 (Fixed-spectrum LAN). *Fix $p \in \Delta_d^\circ$. There are channels*

$$T_N^{\text{orb}} : \mathcal{T}_1((\mathbb{C}^d)^{\otimes N}) \rightarrow \mathcal{T}_1(\mathcal{F}_{\text{orb}}), \quad S_N^{\text{orb}} : \mathcal{T}_1(\mathcal{F}_{\text{orb}}) \rightarrow \mathcal{T}_1((\mathbb{C}^d)^{\otimes N})$$

such that, uniformly for z in compact sets,

$$\|T_n^{\text{orb}}(\rho_{p, g_n, z}^{\otimes n}) - \Phi_z^{(p)}\|_1 \rightarrow 0, \quad (53)$$

$$\|S_n^{\text{orb}}(\Phi_z^{(p)}) - \rho_{p, g_n, z}^{\otimes n}\|_1 \rightarrow 0, \quad (54)$$

and

$$\|T_{m_n}^{\text{orb}}(\rho_{p, g_n, z}^{\otimes m_n}) - \Psi_z^{(p, \gamma)}\|_1 \rightarrow 0. \quad (55)$$

Proof. Restrict [Theorem 3.2](#) to $h = 0$. The classical Gaussian factor $\mathbf{N}_{0, \Sigma_p}$ is independent of z . Set $T_N^{\text{orb}}(X) := \int_{\mathbf{H}} T_N^{\text{full}}(X)(x) dx$ and $S_N^{\text{orb}}(\sigma) := S_N^{\text{full}}(\mathbf{N}_{0, \Sigma_p} \otimes \sigma)$. The bounds $\|\int_{\mathbf{H}} R(x) dx\|_1 \leq \int_{\mathbf{H}} \|R(x)\|_1 dx$ and $\|\mathbf{N}_{0, \Sigma_p} \otimes X\|_1 = \|X\|_1$ give the three assertions. $\square$

We will use the following continuity estimate here and in the appendices. If A, C, A', C' are positive trace-class operators with trace at most one, then

$$|F(A, C) - F(A', C')| \leq \|A - A'\|_1^{1/2} + \|C - C'\|_1^{1/2}. \quad (56)$$

Indeed, the Powers–Størmer inequality bounds $\|\sqrt{A} - \sqrt{A'}\|_2$ by $\|A - A'\|_1^{1/2}$, and similarly for C, C' . Adding and subtracting $\sqrt{A'}\sqrt{C}$, the reverse triangle inequality and Schatten Hölder give [\(56\)](#), since $\|\sqrt{C}\|_2, \|\sqrt{A'}\|_2 \leq 1$.

The following lemma formalizes this LAN fidelity transfer.

Lemma 3.5 (LAN fidelity transfer). *Let $\{\rho_{n, \theta}\}_{\theta \in C}$ and $\{\sigma_{n, \theta}\}_{\theta \in C}$ be input and target models on a compact parameter set C . Let $\{\Phi_\theta\}$ and $\{\Psi_\theta\}$ be limiting models, and suppose channels S_n, T_n satisfy*

$$\begin{aligned} \delta_n &:= \sup_{\theta \in C} \|S_n(\Phi_\theta) - \rho_{n, \theta}\|_1 \rightarrow 0, \\ \eta_n &:= \sup_{\theta \in C} \|T_n(\sigma_{n, \theta}) - \Psi_\theta\|_1 \rightarrow 0. \end{aligned}$$

For every channel $\mathcal{M}_n$, put $\Gamma_n = T_n \circ \mathcal{M}_n \circ S_n$. Then, for every $\theta \in C$,

$$F(\mathcal{M}_n(\rho_{n, \theta}), \sigma_{n, \theta}) \leq F(\Gamma_n(\Phi_\theta), \Psi_\theta) + \sqrt{\delta_n} + \sqrt{\eta_n}. \quad (57)$$

The error is uniform over both $\theta \in C$ and $\mathcal{M}_n$.

Proof. Trace-norm contractivity of $T_n \circ \mathcal{M}_n$ gives $\|\Gamma_n(\Phi_\theta) - T_n \mathcal{M}_n(\rho_{n, \theta})\|_1 \leq \delta_n$. Monotonicity under T_n , followed by [\(56\)](#) and the definition of η_n , therefore yields

$$\begin{aligned} F(\mathcal{M}_n(\rho_{n, \theta}), \sigma_{n, \theta}) &\leq F(T_n \mathcal{M}_n(\rho_{n, \theta}), T_n(\sigma_{n, \theta})) \\ &\leq F(\Gamma_n(\Phi_\theta), \Psi_\theta) + \sqrt{\delta_n} + \sqrt{\eta_n}. \end{aligned}$$

$\square$

Conditional sector limits versus LAN. The sector proof conditions on typical input and output Young labels: on fixed-height cutoffs, the target and Cartan-sector output approach τ_p and $\tau_p^{(\gamma)}$, giving $F_{\text{orb}}(\gamma, p)$; see [\(160\)](#) and [\(159\)](#). Full LAN retains the centered label fluctuation as $\mathbf{N}_{0, \Sigma_p}$, whereas orbital LAN marginalizes it. The same τ_p thus arises through conditioning in the sector argument and marginalization in orbital LAN.

3.4 Completion of the minimax proofs

We apply the transfer lemma to the orbital and full Gaussian models to complete [Theorems 1.1](#) and [1.2](#), respectively.

Known-spectrum converse and completion of [Theorem 1.1](#). For the known-spectrum problem, set $u_n(p) := \sup_{\mathcal{M}_n} \inf_{g \in \mathcal{U}(d)} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n})$. For every channel $\mathcal{M}_n$ and fixed L , set

$$\Gamma_n = T_{m_n}^{\text{orb}} \circ \mathcal{M}_n \circ S_n^{\text{orb}}.$$

The local orbit $\{g_{n,z}^{(p)} : z \in \mathbb{Z}_L\}$ is contained in the full unitary orbit. Apply [Lemma 3.5](#), average over $\mathbb{Z}_L$, and take the supremum over $\mathcal{M}_n$. Since the transfer error is independent of $\mathcal{M}_n$,

$$u_n(p) \leq \sup_{\Lambda} \frac{1}{|\mathbb{Z}_L|} \int_{\mathbb{Z}_L} F(\Lambda(\Phi_z^{(p)}), \Psi_z^{(p,\gamma)}) dz + o(1),$$

where the supremum is over all channels on the Gaussian model and the error tends to zero for fixed L . Taking $\limsup_n$, then $L \rightarrow \infty$, [\(47\)](#) gives

$$\limsup_{n \rightarrow \infty} u_n(p) \leq F_{\text{orb}}(\gamma, p).$$

Together with $\liminf_n u_n(p) \geq F_{\text{orb}}(\gamma, p)$, proved in [Section 2](#), this proves [Theorem 1.1](#). $\square$

Fixing p in advance gives the known-spectrum problem and therefore cannot reveal the cost of an unknown spectrum. To capture that cost, we consider $n^{-1/2}$ -neighborhoods of a fixed interior spectrum and let the local window grow after the large- n limit. Each fixed local window eventually lies in the prescribed spectral set around any interior point, so the full Gaussian bound applies directly to the compact-set converse.

Proof of part (b) of [Theorem 1.2](#). For the compact-set problem, set $w_n(K) := \sup_{\mathcal{M}_n} \inf_{\substack{p \in K \\ g \in \mathcal{U}(d)}} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n})$. Part (a) of [Theorem 1.2](#) gives

$$\liminf_{n \rightarrow \infty} w_n(K) \geq \inf_{p \in K} F_{\text{univ}}(\gamma, p).$$

For the reverse inequality, fix $p_0 \in \text{int}_{\mathbb{A}_d} K$. For every fixed L , all local spectra $p_0 + h/\sqrt{n}$, $(h, z) \in \Omega_L$, lie in K for sufficiently large n . For every channel $\mathcal{M}_n$, form

$$\Gamma_n = T_{m_n}^{\text{full}} \circ \mathcal{M}_n \circ S_n^{\text{full}},$$

using the LAN maps at p_0 . The worst-case fidelity of $\mathcal{M}_n$ on K is bounded by its average over this local window. The pointwise transfer bound, uniform over $\mathcal{M}_n$, therefore gives

$$w_n(K) \leq \sup_{\Lambda} \frac{1}{|\Omega_L|} \int_{\Omega_L} F(\Lambda(\Theta_{h,z}^{\text{in}}), \Theta_{h,z}^{\text{tar}}) dh dz + o(1),$$

where both Gaussian models are based at p_0 , the supremum is over all classical-quantum channels, and L is fixed as $n \rightarrow \infty$. Taking $\limsup_n$ and then $L \rightarrow \infty$, [\(48\)](#) yields

$$\limsup_{n \rightarrow \infty} w_n(K) \leq F_{\text{univ}}(\gamma, p_0).$$

Take the infimum over $\text{int}_{\mathbb{A}_d} K$. Its closure in $\mathbb{A}_d$ is K , and $F_{\text{univ}}(\gamma, \cdot)$ is continuous, so

$$\limsup_{n \rightarrow \infty} w_n(K) \leq \inf_{p \in K} F_{\text{univ}}(\gamma, p).$$

$\square$

4 Cloning projector states

A flat rank- r projector state has the form $\rho_P = P/r$, where P is a rank- r orthogonal projector. These projectors form the complex Grassmannian $\text{Gr}(r, d)$. When $r > 1$, the state ρ_P has a nonzero eigenvalue of multiplicity r , so the simple-spectrum theorems do not apply. We prove [Theorem 1.3](#) directly, without a general rank-deficient LAN theorem. The exact compression lemma below holds for any positive spectrum on its support, although we use it only for the flat spectrum. By itself, the lemma does not establish optimal cloning for other rank-deficient states.

4.1 Exact rank compression

In this section, a superscript (s) indicates the ambient dimension of a representation space or map. For a partition λ with at most r rows, let

$$J_\lambda : \mathcal{H}_\lambda^{(r)} \longrightarrow \mathcal{H}_\lambda^{(d)}$$

be the isometric $U(r)$ -intertwiner that maps the unit highest-weight vector to the unit highest-weight vector. Its range is the $U(r)$ -submodule generated by that vector, and we set $P_\lambda = J_\lambda J_\lambda^*$.

For a compatible pair of r -row partitions $\mu \vdash n$ and $\nu \vdash m$, write $\omega = \nu - \mu \vdash m - n$. In ambient dimension $s \in \{r, d\}$, the Cartan-sector channel is

$$\mathcal{C}_{\mu, \nu}^{(s)}(X) = \frac{\dim \mathcal{H}_\mu^{(s)}}{\dim \mathcal{H}_\nu^{(s)}} (V_{\mu, \omega}^{(s)})^* (X \otimes I_{\mathcal{H}_\omega^{(s)}}) V_{\mu, \omega}^{(s)}.$$

For a probability vector $x \in \mathbb{R}^s$, including one with zero entries, write $\pi_\lambda^{(s)}$ for the polynomial $U(s)$ -representation, $s_\lambda^{(s)}$ for its Schur polynomial, and

$$\rho_{x, \lambda}^{(s)} = \frac{\pi_\lambda^{(s)}(\text{diag } x)}{s_\lambda^{(s)}(x)}$$

whenever the denominator is nonzero.

Lemma 4.1 (Exact rank-compression identity). *Let $p = (p_1, \dots, p_r)$ satisfy $p_i > 0$ and $\sum_i p_i = 1$, and put $\bar{p} = (p_1, \dots, p_r, 0, \dots, 0) \in \mathbb{R}^d$. For every r -row partition λ ,*

$$\rho_{\bar{p}, \lambda}^{(d)} = J_\lambda \rho_{p, \lambda}^{(r)} J_\lambda^*. \quad (58)$$

For every compatible r -row pair μ, ν and every trace-class operator X on $\mathcal{H}_\mu^{(r)}$,

$$P_\nu \mathcal{C}_{\mu, \nu}^{(d)}(J_\mu X J_\mu^*) P_\nu = a_{\mu, \nu}^{(r, d)} J_\nu \mathcal{C}_{\mu, \nu}^{(r)}(X) J_\nu^*, \quad (59)$$

where

$$a_{\mu, \nu}^{(r, d)} = \frac{\dim \mathcal{H}_\mu^{(d)} \dim \mathcal{H}_\nu^{(r)}}{\dim \mathcal{H}_\nu^{(d)} \dim \mathcal{H}_\mu^{(r)}} = \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{\mu_i + j - i}{\nu_i + j - i}. \quad (60)$$

Consequently, the fidelity for each pair of labels factorizes exactly:

$$F\left(\mathcal{C}_{\mu, \nu}^{(d)}(\rho_{\bar{p}, \mu}^{(d)}), \rho_{\bar{p}, \nu}^{(d)}\right) = \sqrt{a_{\mu, \nu}^{(r, d)}} F\left(\mathcal{C}_{\mu, \nu}^{(r)}(\rho_{p, \mu}^{(r)}), \rho_{p, \nu}^{(r)}\right). \quad (61)$$

Proof. The singular specialization identity for Schur polynomials is

$$s_\lambda^{(d)}(p_1, \dots, p_r, 0, \dots, 0) = s_\lambda^{(r)}(p_1, \dots, p_r),$$

and the left side vanishes when λ has more than r rows. We claim that

$$\pi_\lambda^{(d)}(\iota \operatorname{diag}(p) \iota^*) = J_\lambda \pi_\lambda^{(r)}(\operatorname{diag} p) J_\lambda^*.$$

Indeed, $\pi_\lambda^{(d)}(\iota \operatorname{diag}(p) \iota^*)$ acts on each weight space of $\mathcal{H}_\lambda^{(d)}$ by the monomial $\bar{p}^\xi$, which vanishes unless the weight ξ is supported in the first r coordinates. By the Gelfand–Tsetlin branching rule, the weight spaces so supported span exactly the $U(r)$ -submodule generated by the highest-weight vector, that is, the range of J_λ ; on that submodule the block-diagonal action reduces to the r -dimensional polynomial representation, proving the claim. Normalizing by the Schur-polynomial identity proves (58).

The two maps

$$V_{\mu,\omega}^{(d)} J_\nu, \quad (J_\mu \otimes J_\omega) V_{\mu,\omega}^{(r)}$$

are $U(r)$ -intertwiners from $\mathcal{H}_\nu^{(r)}$ to $\mathcal{H}_\mu^{(d)} \otimes \mathcal{H}_\omega^{(d)}$. Both send the unit highest-weight vector to $v_\mu \otimes v_\omega$, so they agree on the submodule generated by that vector:

$$V_{\mu,\omega}^{(d)} J_\nu = (J_\mu \otimes J_\omega) V_{\mu,\omega}^{(r)}. \quad (62)$$

Inserting (62) into the two sector-channel formulas proves (59), with the dimension ratio in (60). The Weyl dimension formula and the vanishing of all rows after r give the product in the same equation. Factors involving two positive rows cancel, while factors involving two zero rows equal one.

If $A, B \geq 0$ and $PBP = B$, then

$$F(A, B) = \operatorname{Tr} \sqrt{\sqrt{B} A \sqrt{B}} = F(PAP, B).$$

Apply this with $P = P_\nu$, then use (58), (59), invariance of fidelity under the isometry J_ν , and $F(aA, B) = \sqrt{a} F(A, B)$. This proves (61). Taking traces in (59) also gives

$$\operatorname{Tr} [P_\nu \mathcal{C}_{\mu,\nu}^{(d)} (J_\mu X J_\mu^*) P_\nu] = a_{\mu,\nu}^{(r,d)} \operatorname{Tr} X.$$

In particular, compatibility forces $a_{\mu,\nu}^{(r,d)} \leq 1$. □

4.2 Exact sector calculation

Let

$$D_s^\lambda = \dim \mathcal{H}_\lambda^{(s)}, \quad R_\lambda = \frac{D_d^\lambda}{D_r^\lambda} = \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{\lambda_i + j - i}{j - i}. \quad (63)$$

For the standard projector P_0 onto the first r coordinates, the conditional state in the λ -sector is

$$\rho_{P_0,\lambda} = J_\lambda \frac{I_{\mathcal{H}_\lambda^{(r)}}}{D_r^\lambda} J_\lambda^*. \quad (64)$$

The r -dimensional Cartan channel is $U(r)$ -covariant and therefore maps the maximally mixed state on the irreducible input module exactly to the maximally mixed state on the irreducible output module. The compression identity (61) consequently becomes, for every compatible pair,

$$F(\mathcal{C}_{\mu,\nu}^{(d)}(\rho_{P_0,\mu}), \rho_{P_0,\nu}) = \sqrt{\frac{R_\mu}{R_\nu}}. \quad (65)$$

For $\mu_i/n \rightarrow 1/r$ and $\nu_i/m_n \rightarrow 1/r$, this tends to the right-hand side of (9).

4.3 A compatible exact-marginal coupling of flat Schur laws

Let

$$\mathbf{P}_N^{(r)}(\lambda) = \frac{\dim \mathcal{S}_\lambda D_r^\lambda}{r^N}, \quad \lambda \vdash N, \quad \ell(\lambda) \leq r, \quad (66)$$

be the Schur–Weyl law of the maximally mixed state on $\mathbb{C}^r$. The complete Schur decomposition is

$$(P/r)^{\otimes N} = \bigoplus_{\ell(\lambda) \leq r} \mathbf{P}_N^{(r)}(\lambda) \frac{\Pi_{\lambda,P}}{D_r^\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}. \quad (67)$$

Here $\Pi_{\lambda,P}$ is the rank- D_r^λ support projection defined intrinsically by the range of P . Pad partitions with zero rows to length r , and define the flat typical set

$$\mathbf{T}_N^{\text{flat}} = \left\{ \lambda : \max_{1 \leq i \leq r} \left| \frac{\lambda_i}{N} - \frac{1}{r} \right| \leq N^{-1/3} \right\}. \quad (68)$$

The finite Young large-deviation estimate (16), applied in effective dimension r , gives $\mathbf{P}_N^{(r)}((\mathbf{T}_N^{\text{flat}})^c) \rightarrow 0$. The Schur–Weyl/random-word central limit theorem [Kup02, Joh01] states that

$$X_N := \frac{\lambda - (N/r)\mathbf{1}_r}{\sqrt{N}} \Rightarrow X, \quad (69)$$

where X is a fixed scaling of the ordered eigenvalue vector of a traceless $r \times r$ GUE matrix. In particular, $\Pr(X_1 > \dots > X_r) = 1$.

Both X_n and X_{m_n} converge to the same law. Their Prokhorov distance therefore tends to zero, so Strassen’s coupling theorem [Str65] supplies a coupling $\Pi_n(\mu, \nu)$ of the exact marginals $\mathbf{P}_n^{(r)}$ and $\mathbf{P}_{m_n}^{(r)}$ such that

$$\left\| \frac{\mu - (n/r)\mathbf{1}_r}{\sqrt{n}} - \frac{\nu - (m_n/r)\mathbf{1}_r}{\sqrt{m_n}} \right\|_\infty \rightarrow 0 \quad \text{in probability.} \quad (70)$$

Disintegrate this finite coupling to obtain a label kernel $K_n(\nu \mid \mu)$.

Put $\omega = \nu - \mu$. Its leading term is $(m_n - n)/r$ in every coordinate, while its fluctuations are of order $\sqrt{n}$. Hence $\omega_i > 0$ for every i with probability $1 - o(1)$. For adjacent rows,

$$\frac{\omega_i - \omega_{i+1}}{\sqrt{n}} = \sqrt{\gamma_n} \left[\frac{\nu_i - \nu_{i+1}}{\sqrt{m_n}} \right] - \left[\frac{\mu_i - \mu_{i+1}}{\sqrt{n}} \right] \Rightarrow (\sqrt{\gamma} - 1)(X_i - X_{i+1}) > 0.$$

Thus ω is a partition with probability $1 - o(1)$. Use the ambient Cartan-sector channel on this good event and, otherwise, the covariant replacement channel into the same sampled ν -block. The output-label marginal is exactly the target law. Let

$$G_n = \left\{ (\mu, \nu) : \nu - \mu \vdash m_n - n, \quad \mu \in \mathbf{T}_n^{\text{flat}}, \quad \nu \in \mathbf{T}_{m_n}^{\text{flat}} \right\}.$$

Then

$$\Pi_n(G_n^c) \leq \Pi_n\{\nu - \mu \not\vdash m_n - n\} + \sum_{N \in \{n, m_n\}} \mathbf{P}_N^{(r)}((\mathbf{T}_N^{\text{flat}})^c) \rightarrow 0.$$

Moreover, (63) gives, uniformly on G_n ,

$$\sqrt{\frac{R_\mu}{R_\nu}} = \gamma_n^{-r(d-r)/2} (1 + O_{r,d}(n^{-1/3})) \rightarrow \gamma^{-r(d-r)/2}. \quad (71)$$

The block formula (22), valid verbatim for the r -row label laws, and concavity of fidelity now give

$$F(\mathcal{Q}_n[K_n]((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) \geq \sum_{(\mu, \nu) \in G_n} \Pi_n(\mu, \nu) \sqrt{\frac{R_\mu}{R_\nu}} \longrightarrow \gamma^{-r(d-r)/2}. \quad (72)$$

On every compatible pair, $\nu_i \geq \mu_i$ and hence

$$0 \leq \sqrt{\frac{R_\mu}{R_\nu}} \leq 1.$$

Thus convergence in probability under the coupling also implies convergence of the displayed expectations. This proves achievability in [Theorem 1.3](#).

4.4 Direct finite-dimensional converse

For $N \in \{n, m_n\}$, let

$$\text{Tw}_N(X) = \frac{1}{N!} \sum_{\pi \in \mathfrak{S}_N} V_\pi X V_\pi^*$$

be the permutation twirl. Covariantize an arbitrary cloner over $U(d)$, then replace it by $\text{Tw}_{m_n} \circ \mathcal{M}_n \circ \text{Tw}_n$. The input twirl fixes every $(P/r)^{\otimes n}$, and the output twirl fixes every target. Unitary covariance of fidelity, its joint concavity under the $U(d)$ -average, and monotonicity under the output twirl show that these operations do not decrease the worst-case fidelity over projector states. From now on, $\mathcal{M}_n$ denotes this symmetrized channel; the bound proved below holds for it at every P , and hence bounds the worst-case fidelity of the original cloner.

Let $Q_\lambda^{(N)}$ be the Young-block projection, and define the Specht reduction and reinsertion maps

$$D_\lambda^{(N)}(X) = \text{Tr}_{\mathcal{S}_\lambda}(Q_\lambda^{(N)} X Q_\lambda^{(N)}), \quad I_\lambda^{(N)}(A) = Q_\lambda^{(N)} \left(A \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda} \right) Q_\lambda^{(N)}.$$

For the symmetrized channel set

$$\Phi_{\nu\mu} = D_\nu^{(m_n)} \circ \mathcal{M}_n \circ I_\mu^{(n)}.$$

Each $\Phi_{\nu\mu} : \mathcal{T}_1(\mathcal{H}_\mu^{(d)}) \rightarrow \mathcal{T}_1(\mathcal{H}_\nu^{(d)})$ is completely positive and $U(d)$ -equivariant, and trace preservation gives

$$\sum_\nu \Phi_{\nu\mu}^*(I_{\mathcal{H}_\nu^{(d)}}) = I_{\mathcal{H}_\mu^{(d)}}.$$

For fixed μ, ν , equivariance makes $\Phi_{\nu\mu}^*(I)$ commute with the irreducible $U(d)$ -action on $\mathcal{H}_\mu^{(d)}$. Schur's lemma therefore gives

$$\Phi_{\nu\mu}^*(I) = K(\nu \mid \mu) I, \quad K(\nu \mid \mu) \geq 0, \quad \sum_\nu K(\nu \mid \mu) = 1.$$

Thus K is a state-independent Markov kernel. Whenever $K(\nu \mid \mu) > 0$, the map

$$\mathcal{E}_{\mu, \nu} = \frac{\Phi_{\nu\mu}}{K(\nu \mid \mu)}$$

is a covariant channel; zero-probability branches may be chosen arbitrarily.

Only input labels $\ell(\mu) \leq r$ carry mass, and only target labels $\ell(\nu) \leq r$ contribute to fidelity. In the following sums we restrict μ, ν accordingly; the channel kernel still includes all output

branches. For $\ell(\lambda) \leq r$, let $\Pi_{\lambda,P}$ be the rank- D_r^λ support projection of the conditional λ -sector state. Covariance and irreducibility give

$$\mathcal{E}_{\mu,\nu}(I_{\mathcal{H}_\mu^{(d)}}) = \frac{D_d^\mu}{D_d^\nu} I_{\mathcal{H}_\nu^{(d)}}.$$

The unnormalized ν -block of the output is therefore

$$Y_{\nu,P} = \sum_{\mu} P_n^{(r)}(\mu) K(\nu | \mu) \mathcal{E}_{\mu,\nu} \left(\frac{\Pi_{\mu,P}}{D_r^\mu} \right),$$

whereas the target block is $P_{m_n}^{(r)}(\nu) \Pi_{\nu,P} / D_r^\nu$. Since $\Pi_{\mu,P} / D_r^\mu \leq I / D_r^\mu$, positivity gives

$$\text{Tr} \left[\Pi_{\nu,P} \mathcal{E}_{\mu,\nu} \left(\frac{\Pi_{\mu,P}}{D_r^\mu} \right) \right] \leq \frac{R_\mu}{R_\nu}. \quad (73)$$

For any positive, possibly subnormalized A ,

$$F \left(A, \frac{\Pi_{\nu,P}}{D_r^\nu} \right) \leq \sqrt{\text{Tr}(\Pi_{\nu,P} A)}, \quad (74)$$

Indeed, with $B = \Pi_{\nu,P} A \Pi_{\nu,P}$, the left side is $(D_r^\nu)^{-1/2} \text{Tr} \sqrt{B}$, and $\text{Tr} \sqrt{B} \leq \sqrt{\text{rank}(B) \text{Tr} B}$. Using the direct-sum fidelity formula, then (73)–(74), and finally Cauchy–Schwarz over ν , gives

$$F(\mathcal{M}_n((P/r)^{\otimes n}), (P/r)^{\otimes m_n}) \leq \left[\left(\sum_{\mu} P_n^{(r)}(\mu) R_\mu \right) \left(\sum_{\nu} \frac{P_{m_n}^{(r)}(\nu)}{R_\nu} \right) \right]^{1/2}. \quad (75)$$

Output mass in longer diagrams can only strengthen the inequality.

The finite Young large-deviation estimate (16), applied in effective dimension r , remains valid at the uniform spectrum: in its proof all weight monomials are then equal rather than strictly ordered. Together with (63), it gives

$$\frac{1}{N^{r(d-r)}} \sum_{\mu} P_n^{(r)}(\mu) R_\mu \longrightarrow \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{1}{r(j-i)}, \quad (76)$$

and

$$m_n^{r(d-r)} \sum_{\nu} \frac{P_{m_n}^{(r)}(\nu)}{R_\nu} \longrightarrow \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} r(j-i). \quad (77)$$

For completeness, use the flat typical set (68). At the uniform spectrum, the large-deviation estimate, the bound $(N+1)^{r-1}$ on the number of partitions, and Pinsker's inequality give

$$P_N^{(r)} \left((\mathcal{T}_N^{\text{flat}})^c \right) \leq (N+1)^{r(r-1)/2+r-1} e^{-N^{1/3}/2}.$$

Uniformly for $\lambda \in \mathcal{T}_N^{\text{flat}}$, (63) gives

$$\frac{R_\lambda}{N^{r(d-r)}} = \prod_{\substack{1 \leq i \leq r \\ r < j \leq d}} \frac{1}{r(j-i)} (1 + O_{r,d}(N^{-1/3})).$$

Taking reciprocals yields the corresponding expansion $N^{r(d-r)}/R_\lambda = \prod_{1 \leq i \leq r < j \leq d} r(j-i)(1 + O_{r,d}(N^{-1/3}))$. Globally, $R_\lambda/N^{r(d-r)}$ is at most $\prod_{1 \leq i \leq r < j \leq d} (1+j-i)/(j-i)$, while $N^{r(d-r)}/R_\lambda \leq N^{r(d-r)}$; these bounds follow from $\lambda_i \leq N$ and $R_\lambda \geq 1$, even when $\lambda_r = 0$. Multiplying the global bounds by the tail probability shows that both tail expectations vanish. The typical-set expansions therefore prove (76) and (77), with $N = n$ and $N = m_n$, respectively. Substitution into (75) yields

$$\limsup_{n \rightarrow \infty} F \leq \gamma^{-r(d-r)/2}, \quad (78)$$

which completes the converse and the proof of [Theorem 1.3](#).

5 Comparison with PCT

Let N be the number of inputs and M the number of *additional* outputs, so that

$$m = N + M, \quad \delta = \frac{M}{N}, \quad \gamma = 1 + \delta. \quad (79)$$

We use squared Uhlmann fidelity F^2 in the sample-complexity comparisons, matching the concurrent papers [\[FGSS26, JSO26\]](#). For $1 \leq r \leq d$ and sufficiently small $\epsilon > 0$, they prove that uniformly cloning states of rank at most r to squared fidelity $1 - \epsilon$ has optimal copy complexity

$$N = \Theta\left(\frac{Mrd}{\epsilon}\right). \quad (80)$$

Their lower bounds use flat projectors of rank $\ell = \Theta(r)$, possibly $\ell < r$; in particular, the rank bound $r = d$ does not restrict the hard family to I_d/d . Their upper bounds use purify–clone–trace (PCT). This optimal scaling does not determine exact fidelity at fixed gain. Below we take $N = n$, $N + M = m_n$, and $m_n/n \rightarrow \gamma > 1$ to compare the protocols as illustrated in [Fig. 2](#).

PCT applies random purification [\[TWZ25\]](#), Werner’s pure-state cloner in dimension dr [\[Wer98\]](#), and partial trace over the purifying registers [\[LTHC26, FGSS26, JSO26\]](#). Denote its output on $\rho^{\otimes N}$ by $\sigma_{N,N+M}^{\text{PCT}}(\rho)$. For every ρ of rank at most r , the published data-processing guarantee is

$$F\left(\sigma_{N,N+M}^{\text{PCT}}(\rho), \rho^{\otimes(N+M)}\right)^2 \geq \frac{\binom{N+dr-1}{dr-1}}{\binom{N+M+dr-1}{dr-1}} \rightarrow \gamma^{-(dr-1)}. \quad (81)$$

The ratio is exact for each purified component before averaging and discarding the purifying registers; it is only a lower bound for the final output. [Appendix A](#) analyzes that output to establish the strict protocol comparisons below.

Projector states. For $\rho = P/r$, $P \in \text{Gr}(r, d)$, and $1 \leq r < d$, the projector family has complex dimension $r(d-r)$. [Theorem 1.3](#) gives its exact limiting squared fidelity $\gamma^{-r(d-r)}$. For PCT with rank bound r and purification dimension dr , write $\sigma_{n,m}^{\text{PCT}}(P)$ for the output on $(P/r)^{\otimes n}$. The appendix gives

$$\limsup_{n \rightarrow \infty} F\left(\sigma_{n,m_n}^{\text{PCT}}(P), (P/r)^{\otimes m_n}\right)^2 \leq \gamma^{-r(d-r)} \left(\frac{\sqrt{2\gamma-1}}{\gamma}\right)^{r-1} < \gamma^{-r(d-r)} \quad (1 < r < d). \quad (82)$$

Thus our projector cloner strictly improves on PCT for every fixed $\gamma > 1$ and $1 < r < d$. For $r = 1$, both reduce to Werner’s optimal pure-state cloner; for $r = d$, the projector family is the known singleton I_d/d .

Inverting our limiting value at $1 - \epsilon$ gives

$$\frac{N}{M} = \frac{1}{(1 - \epsilon)^{-1/[r(d-r)]} - 1} = \frac{r(d-r)}{\epsilon} - \frac{r(d-r)+1}{2} + O(\epsilon). \quad (83)$$

The finite guarantee (81) certifies PCT squared fidelity at least $1 - \epsilon$ whenever

$$\frac{N}{M} \geq \frac{1}{(1 - \epsilon)^{-1/(rd-1)} - 1} = \frac{rd-1}{\epsilon} - \frac{rd}{2} + O(\epsilon), \quad (84)$$

because its binomial ratio is $\prod_{j=1}^{rd-1} (N+j)/(N+M+j) \geq [N/(N+M)]^{rd-1}$. The coefficient $rd-1$ describes this sufficient condition, while (82) compares the actual protocols.

Simple full-rank spectra. Fix $p_1 > \dots > p_d > 0$. PCT now uses rank bound $r = d$ and purification dimension d^2 ; neither it nor $\mathcal{C}_n^{\text{univ}}$ is supplied p . Write $\sigma_{n,m}^{\text{PCT}}(p)$ for its output on $\rho_p^{\otimes n}$. For every fixed $\delta > 0$ and $m_n/n \rightarrow 1 + \delta$, Theorems A.1 and 1.2 give

$$\lim_{n \rightarrow \infty} F\left(\sigma_{n,m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}\right)^2 = F_{\text{PCT}}(1 + \delta, p)^2 < F_{\text{univ}}(1 + \delta, p)^2 = \lim_{n \rightarrow \infty} F\left(\mathcal{C}_n^{\text{univ}}(\rho_p^{\otimes n}), \rho_p^{\otimes m_n}\right)^2. \quad (85)$$

Expanding these exact limiting fidelities as $\delta \downarrow 0$ gives

$$F_{\text{univ}}(1 + \delta, p)^2 = 1 - A_{\text{univ}}(p)\delta^2 + O(\delta^3), \quad (86)$$

$$F_{\text{PCT}}(1 + \delta, p)^2 = 1 - A_{\text{PCT}}(p)\delta^2 + O(\delta^3), \quad (87)$$

where the remainder constants may depend on p , and

$$A_{\text{univ}}(p) = \frac{d-1}{8} + \frac{1}{4} \sum_{i < j} \frac{p_i}{p_j}, \quad (88)$$

$$A_{\text{PCT}}(p) = \frac{d-1}{2} + \frac{1}{4} \sum_{i < j} \frac{(p_i + p_j)^2}{p_i p_j}. \quad (89)$$

In particular,

$$A_{\text{PCT}}(p) - A_{\text{univ}}(p) = \frac{3(d-1)}{8} + \frac{1}{4} \sum_{i < j} \left(2 + \frac{p_j}{p_i}\right) > 0. \quad (90)$$

First taking $n \rightarrow \infty$ at fixed $\delta > 0$, then inverting the limiting fidelities at $1 - \epsilon$ and sending $\epsilon \downarrow 0$, gives the respective thresholds

$$\frac{N}{M} = (1 + o(1)) \sqrt{\frac{A_{\text{univ}}(p)}{\epsilon}}, \quad (91)$$

$$\frac{N}{M} = (1 + o(1)) \sqrt{\frac{A_{\text{PCT}}(p)}{\epsilon}}, \quad (92)$$

with both $o(1)$ terms taken as $\epsilon \downarrow 0$. Our cloner therefore has the strictly smaller leading constant $\sqrt{A_{\text{univ}}(p)} < \sqrt{A_{\text{PCT}}(p)}$. This $1/\sqrt{\epsilon}$ law is confined to a fixed interior spectrum: both coefficients diverge as an eigenvalue approaches zero, so it does not contradict the uniform $1/\epsilon$ scaling.

Uniform model at fixed γ . For the minimax fidelity over all d -dimensional states,

$$w_n = \sup_{\mathcal{M}_n} \inf_{\substack{\rho \geq 0 \\ \text{Tr } \rho = 1}} F(\mathcal{M}_n(\rho^{\otimes n}), \rho^{\otimes m_n}), \quad (93)$$

the PCT guarantee gives $\liminf_n w_n \geq \gamma^{-(d^2-1)/2}$. Restricting to the compact simple-spectrum chambers $\Delta_{d,\kappa}$, applying (8), and then letting $\kappa \downarrow 0$ gives

$$\limsup_{n \rightarrow \infty} w_n \leq \inf_{p \in \Delta_d^\circ} F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d) \gamma^{-d(d-1)/4}. \quad (94)$$

At large gain, these lower and upper bounds scale respectively as $\gamma^{-(d^2-1)/2}$ and $\gamma^{-(d^2-1)/4}$, up to constants, leaving a factor of two between the polynomial exponents. The exact uniform minimax value and whether PCT attains it remain open.

Order of limits. The strict protocol comparisons hold at every fixed $\gamma > 1$. The small-error thresholds derived from our limiting fidelities use the iterated limit $N, M \rightarrow \infty$ with $M/N \rightarrow \delta > 0$, followed by $\delta \downarrow 0$. Our estimates are not uniform when $\gamma_n \downarrow 1$, so these thresholds do not provide finite-sample bounds in a joint high-fidelity regime or improve the uniform $1/\epsilon$ scaling up to constants. The published sufficient condition (84) is a finite-sample guarantee and does not require this iterated limit.

6 Conclusion

In the regime $m_n/n \rightarrow \gamma > 1$, we determined the optimal asymptotic cloning fidelity for known simple full-rank spectra, and the exact asymptotic fidelity of a single spectrum-independent cloner for unknown simple full-rank spectra, uniformly on compact spectral sets. On the compact spectral families specified in part (b) of [Theorem 1.2](#), this cloner is minimax optimal. At every fixed simple full-rank spectrum its limiting fidelity is F_{univ} , strictly above the exact PCT fidelity. For flat projector states, our projector-state cloner attains the exact projector-state minimax value, while PCT is strictly suboptimal for $1 < r < d$.

Several boundary regimes remain open. These include the positive-degeneracy and nonflat rank-deficient cases discussed below, and neighborhoods in which the rank changes. Estimates uniform for gains $\gamma_n = m_n/n \downarrow 1$ would be needed to turn these formulas into finite-sample bounds in the joint high-fidelity limit. The exact minimax value over all states at fixed γ , as well as the finite-copy optimum, also remains to be determined.

6.1 Conjectures for degenerate and rank-deficient spectra

We record two expected extensions of the proved formulas. Suppose a rank- r state ρ has distinct positive eigenvalues $u_1 > \dots > u_s > 0$, with multiplicities $r_1, \dots, r_s$, where $\sum_a r_a = r$ and $\sum_a r_a u_a = 1$. We conjecture that the known-spectrum optimum on its unitary orbit, defined as in (6), is

$$F_{\text{orb}}(\gamma, \rho) = \gamma^{-r(d-r)/2} \prod_{a < b} f_\gamma\left(\frac{u_b}{u_a}\right)^{r_a r_b}. \quad (95)$$

Only pairs of distinct positive eigenspaces contribute to the product: rotations within a degenerate eigenspace are stabilizer directions, while the $r(d-r)$ support-kernel directions give the first factor. The simple full-rank case $r = d$, $s = d$, and $r_a = 1$ reduces to the orbital formula in (5); the flat

projector case $s = 1$ reduces to $\gamma^{-r(d-r)/2}$, proved in [Theorem 1.3](#). In particular, $r = 1$ recovers Werner’s pure-state value $\gamma^{-(d-1)/2}$. The exact compression identity in [Lemma 4.1](#) establishes this support–kernel factorization for our sector channel, but does not by itself prove the conjecture for the remaining spectra.

For the local unknown-state problem, consider the smooth manifold of rank- r states

$$\mathcal{D}_r^\circ := \{\sigma \succeq 0 : \text{Tr } \sigma = 1, \text{ rank } \sigma = r\}. \quad (96)$$

If P is the support projection of ρ , its tangent space is

$$T_\rho \mathcal{D}_r^\circ = \{X = X^* : \text{Tr } X = 0, (I - P)X(I - P) = 0\}. \quad (97)$$

Equip this manifold with the Riemannian metric induced by the Hilbert–Schmidt inner product and denote its exponential map at ρ by Exp_ρ . For fixed L and all sufficiently large n , set $\rho_{n,X} = \text{Exp}_\rho(X/\sqrt{n})$ and define

$$v_{n,L}^{(r)}(\rho) := \sup_{\mathcal{M}_n} \inf_{\substack{X \in T_\rho \mathcal{D}_r^\circ \\ \|X\|_2 \leq L}} F\left(\mathcal{M}_n(\rho_{n,X}^{\otimes n}), \rho_{n,X}^{\otimes m_n}\right). \quad (98)$$

Here ρ is fixed before the supremum, so the channel sequence may depend on the center ρ , but not on the unknown local displacement X . We conjecture that the $\limsup$ and $\liminf$ of $v_{n,L}^{(r)}(\rho)$, first as $n \rightarrow \infty$ and then as $L \rightarrow \infty$, agree. Writing their common value as $F_{\text{univ}}^{(r)}(\gamma, \rho)$, the conjectured formula is

$$F_{\text{univ}}^{(r)}(\gamma, \rho) = \left(\frac{2\sqrt{\gamma}}{1+\gamma}\right)^{(\sum_{a=1}^s r_a^2 - 1)/2} F_{\text{orb}}(\gamma, \rho). \quad (99)$$

Only the positive-eigenvalue blocks enter the exponent; the support–kernel directions are already included in $F_{\text{orb}}(\gamma, \rho)$ through the factor $\gamma^{-r(d-r)/2}$. The quantity $\sum_{a=1}^s r_a^2 - 1$ is the real dimension of the traceless Hermitian commutant on the support. The exponent is suggested by the Gaussian matrix fluctuations that resolve a degenerate positive block [[Kup02](#), [Joh01](#)]. For simple positive spectra (every $r_a = 1$), the fixed-rank LAN theorem of [[LN24](#), Theorem 3.3] has exactly $r - 1$ classical Gaussian parameters, $r(r - 1)/2$ shifted thermal modes, and $r(d - r)$ shifted pure modes, matching the exponent $(r - 1)/2$ here and the factors in (95). When also $r = d$, the formula reduces to the local full-rank value in (5). At a flat projector state, $s = 1$, [Theorem 1.3](#) proves only the known-spectrum orbital factor, not this local conjecture. This is a pointwise local minimax conjecture, not a claim that one rank-aware channel works uniformly over the rank- r manifold, and we make no claim for neighborhoods in which the rank can change.

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Generative AI usage

The original idea for the cloner construction is due to the authors. We used OpenAI’s ChatGPT (Sol 5.6) and Anthropic’s Claude (Fable 5) to assist with the analysis: the evaluation of the cloning fidelity and the converse argument benefited from their suggestions. The same tools assisted with language editing, manuscript organization, and figure preparation. All AI-assisted material was reviewed and revised by the authors, who take full responsibility for the accuracy and content of the manuscript.

A Strict comparison with purify–clone–trace

We compute the exact asymptotic PCT fidelity at simple full-rank spectra and derive a strict upper bound for flat projector states with $1 < r < d$, proving the comparisons stated in [Section 5](#).

A.1 Simple full-rank spectra

Fix $p \in \Delta_d^\circ$, so that $\rho_p = \text{diag}(p_1, \dots, p_d)$ with $p_1 > \dots > p_d > 0$, and recall $q_{ij} = p_j/p_i$ and the fidelity convention [\(3\)](#). Put

$$h_q(x) = \frac{\sqrt{(1-q)(1-x)}}{1 - \sqrt{qx}}, \quad (100)$$

and define

$$q_{ij}^{\text{PCT}} = \frac{\gamma - 1 + \gamma q_{ij}}{\gamma + (\gamma - 1)q_{ij}}. \quad (101)$$

Define the asymptotic PCT fidelity by

$$F_{\text{PCT}}(\gamma, p) = \left(\frac{\sqrt{2\gamma - 1}}{\gamma} \right)^{(d-1)/2} \prod_{i < j} h_{q_{ij}}(q_{ij}^{\text{PCT}}). \quad (102)$$

Let $\sigma_{n, m_n}^{\text{PCT}}(p)$ denote the output on the cloned d -dimensional systems. PCT applies random purification in $\mathbb{C}^d \otimes \mathbb{C}^d$, Werner's pure-state $n \rightarrow m_n$ cloner in dimension d^2 , and then discards the purifying registers.

Theorem A.1 (Exact full-rank fidelities and comparison). *For fixed d , p , and $\gamma > 1$ as above,*

$$\lim_{n \rightarrow \infty} F(\sigma_{n, m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}) = F_{\text{PCT}}(\gamma, p) < F_{\text{univ}}(\gamma, p) = \lim_{n \rightarrow \infty} F(\mathcal{C}_n^{\text{univ}}(\rho_p^{\otimes n}), \rho_p^{\otimes m_n}), \quad (103)$$

with F_{univ} and f_γ as in [\(5\)](#) and [\(4\)](#).

The final equality in [\(103\)](#) is part (a) of [Theorem 1.2](#). It remains to prove the PCT limit and the strict inequality.

We first identify the limiting channel obtained by composing partial trace with interior LAN, and then feed into that channel the Fock-space limit of Werner's cloner.

A.1.1 Partial trace followed by interior LAN

Let $H = E = \mathbb{C}^d$, set

$$S = \text{diag}(\sqrt{p_1}, \dots, \sqrt{p_d}), \quad |\psi_p\rangle = \sum_{i=1}^d \sqrt{p_i} |ii\rangle,$$

and identify a vector in $H \otimes E$ with its coefficient matrix:

$$|Z\rangle = \sum_{i,j} Z_{ij} |ij\rangle.$$

The one-particle purification-tangent space is

$$K_p := \psi_p^\perp = K_p^{\text{diag}} \oplus \bigoplus_{i < j} \text{span}\{|ij\rangle, |ji\rangle\},$$

where

$$K_p^{\text{diag}} = \left\{ \sum_i z_i |ii\rangle : \sum_i \sqrt{p_i} z_i = 0 \right\}.$$

Let $\Gamma_s(K_p)$ be the symmetric Fock space over K_p , and let $b(u)$ denote the annihilation operator in the one-particle direction $u \in K_p$. We abbreviate $b_{ij} = b(|ij\rangle)$ for $i \neq j$.

For a matrix tangent $Z \in K_p$, the differential of partial trace at the canonical purification is

$$D_p(Z) := \left. \frac{d}{dt} \right|_{t=0} \text{Tr}_E |\psi_p + tZ\rangle\langle\psi_p + tZ| = ZS + SZ^*. \quad (104)$$

Its diagonal and positive-CCR orbital coordinates are

$$h_i(Z) = 2\sqrt{p_i} \Re Z_{ii}. \quad (105)$$

$$\beta_{ij}(Z) = \frac{D_p(Z)_{ji}}{\sqrt{p_i - p_j}} = \frac{\sqrt{p_i} Z_{ji} + \sqrt{p_j} \overline{Z_{ij}}}{\sqrt{p_i - p_j}}, \quad i < j. \quad (106)$$

Notice that $\sum_i h_i(Z) = 0$. Set

$$\mathbf{H}_0 = \left\{ h \in \mathbb{R}^d : \sum_i h_i = 0 \right\}, \quad \bar{\Sigma}_p = \text{diag}(p) - pp^\top, \quad \Sigma_p = \bar{\Sigma}_p|_{\mathbf{H}_0}.$$

The explicit limiting channel. We now construct a classical-quantum Gaussian channel

$$\Lambda_p : \mathcal{T}_1(\Gamma_s(K_p)) \longrightarrow L^1\left(\mathbf{H}_0; \mathcal{T}_1(\mathcal{F}^{\otimes \binom{d}{2}})\right).$$

For the diagonal block, define

$$v_i = \sqrt{p_i} |ii\rangle - p_i |\psi_p\rangle \in K_p^{\text{diag}}, \quad X_i = b(v_i) + b(v_i)^*. \quad (107)$$

These fields commute. Indeed,

$$\langle v_i, v_j \rangle = p_i \delta_{ij} - p_i p_j = (\bar{\Sigma}_p)_{ij} \quad (108)$$

is real, so $[X_i, X_j] = 0$. Moreover $\sum_i X_i = 0$. The diagonal part of Λ_p is the joint spectral measurement of $(X_1, \dots, X_d)$, regarded as a classical variable in $\mathbf{H}_0$.

For each $i < j$, put $\Delta_{ij} = p_i - p_j$ and define

$$a_{ij} = \frac{\sqrt{p_i} b_{ji} + \sqrt{p_j} b_{ij}^*}{\sqrt{\Delta_{ij}}}. \quad (109)$$

$$c_{ij} = \frac{\sqrt{p_i} b_{ij} + \sqrt{p_j} b_{ji}^*}{\sqrt{\Delta_{ij}}}. \quad (110)$$

A direct CCR calculation gives

$$[a_{ij}, a_{ij}^*] = [c_{ij}, c_{ij}^*] = 1, \quad [a_{ij}, c_{ij}] = [a_{ij}, c_{ij}^*] = 0.$$

Thus $(b_{ij}, b_{ji}) \mapsto (a_{ij}, c_{ij})$ is a two-mode squeezing transformation. The (i, j) -part of Λ_p applies this transformation and discards the complementary mode c_{ij} , retaining a_{ij} . Different unordered pairs use disjoint input modes and commute with the diagonal fields, so these blocks form a product channel.

Let $|Z\rangle$ now denote the Fock coherent vector of amplitude $Z \in K_p$. The preceding construction gives

$$\Lambda_p(|Z\rangle\langle Z|) = \mathbf{N}_{h(Z), \Sigma_p} \otimes \bigotimes_{i < j} D(\beta_{ij}(Z)) \tau_{q_{ij}} D(\beta_{ij}(Z))^*, \quad (111)$$

where

$$\tau_q = (1 - q) \sum_{r \geq 0} q^r |r\rangle\langle r|.$$

To verify the diagonal part, use $\langle X_i \rangle_Z = 2\Re\langle v_i, Z \rangle = h_i(Z)$ and (108). For the off-diagonal part, $\langle a_{ij} \rangle_Z = \beta_{ij}(Z)$, while on the input vacuum

$$\langle a_{ij}^* a_{ij} \rangle = \frac{p_j}{p_i - p_j}.$$

The corresponding geometric parameter is therefore $p_j/p_i = q_{ij}$.

Strong convergence of the finite channels. Let R_M project onto Fock number vectors of total occupation at most M , and let J_M map such a number vector to the normalized symmetric occupation vector with all remaining particles in ψ_p . Define the trace-preserving occupation embedding

$$\mathcal{J}_M(X) = J_M R_M X R_M J_M^* + \text{Tr}[(I - R_M)X] |\psi_p\rangle\langle\psi_p|^{\otimes M}. \quad (112)$$

Let T_M^p be an interior classical-quantum LAN extraction channel centered at ρ_p , with its orbital normalization and phase chosen so that its output annihilator is the LAN limit of the positive-CCR fluctuation

$$\frac{1}{\sqrt{M(p_i - p_j)}} \sum_{\ell=1}^M E_{ij}^{(\ell)}, \quad i < j.$$

Define

$$\Lambda_{p,M} = T_M^p \circ \text{Tr}_{E^M} \circ \mathcal{J}_M. \quad (113)$$

Lemma A.2 (Interior partial-trace/LAN limits). *Let S_M^p be the reverse channel of the interior LAN theorem centered at ρ_p , expressed in the (h, β) coordinates used by T_M^p . For every trace-class operator X on $\Gamma_s(K_p)$,*

$$\|\Lambda_{p,M}(X) - \Lambda_p(X)\|_1 \longrightarrow 0. \quad (114)$$

$$\|(\text{Tr}_{E^M} \circ \mathcal{J}_M)(X) - (S_M^p \circ \Lambda_p)(X)\|_1 \longrightarrow 0. \quad (115)$$

Proof. First take a coherent amplitude $Z \in K_p$. Put

$$|\psi_{Z,M}\rangle = \frac{|\psi_p\rangle + M^{-1/2}|Z\rangle}{\sqrt{1 + \|Z\|^2/M}}.$$

We first prove that, uniformly for Z in bounded subsets of K_p ,

$$\left\| \mathcal{J}_M(|Z\rangle\langle Z|) - |\psi_{Z,M}\rangle\langle\psi_{Z,M}|^{\otimes M} \right\|_1 \longrightarrow 0. \quad (116)$$

Indeed, the truncation term in (112) carries the Poisson tail weight

$$\text{Tr}[(I - R_M)|Z\rangle\langle Z|] = \sum_{j > M} e^{-\|Z\|^2} \frac{\|Z\|^{2j}}{j!},$$

which vanishes uniformly for $\|Z\| \leq L$. Up to this term both states in (116) are pure, so it suffices that their overlap tends to one. Expanding both vectors in the occupation basis and collecting the amplitudes with the multinomial theorem gives the positive overlap

$$\langle Z | \psi_{Z,M}^{\otimes M} \rangle = e^{-\|Z\|^2/2} \left(1 + \frac{\|Z\|^2}{M}\right)^{-M/2} \sum_{j=0}^M \frac{\|Z\|^{2j}}{j!} \sqrt{\frac{M!}{(M-j)! M^j}}.$$

Every square root lies in $[0, 1]$ and tends to one at each fixed j , so dominated convergence makes the sum tend to $e^{\|Z\|^2}$, while the prefactors tend to $e^{-\|Z\|^2}$; the convergence is uniform on $\|Z\| \leq L$, and the trace distance of two pure states is $2\sqrt{1 - |\langle \cdot, \cdot \rangle|^2}$. After the purifying systems are traced out, the second state in (116) becomes the product state $\rho_{Z,M}^{\otimes M}$, where

$$\rho_{Z,M} = \rho_p + M^{-1/2}(ZS + SZ^*) + O_Z(M^{-1}). \quad (117)$$

To make the local chart explicit, define the anti-Hermitian matrix

$$G(\beta) = Y_p(\beta) = \sum_{i < j} \frac{\beta_{ij} E_{ji} - \overline{\beta_{ij}} E_{ij}}{\sqrt{p_i - p_j}}. \quad (118)$$

It satisfies $[G(\beta), \rho_p]_{ji} = \sqrt{p_i - p_j} \beta_{ij}$, so (106) is precisely the off-diagonal coordinate of $D_p(Z)$. To obtain exact local coordinates, consider

$$(u, w) \mapsto e^{Y_p(w)} \text{diag}(p + u) e^{-Y_p(w)}, \quad u \in \mathbf{H}_0, \quad w \in \mathbb{C}^{(d)}.$$

Its derivative at $(0, 0)$ is $(u, w) \mapsto \text{diag}(u) + [Y_p(w), \rho_p]$, a real linear isomorphism onto the traceless Hermitian matrices: its diagonal is u and its (j, i) -entry is $\sqrt{p_i - p_j} w_{ij}$. The inverse function theorem therefore gives a smooth local chart. Applying its inverse to (117) yields

$$\rho_{Z,M} = g_{M,z_M}^{(p)} \text{diag}(p + M^{-1/2} h_M) (g_{M,z_M}^{(p)})^*, \quad (h_M, z_M) = (h(Z), \beta(Z)) + O_L(M^{-1/2}),$$

uniformly for $\|Z\| \leq L$. Thus $\rho_{Z,M}$ lies in the interior-LAN local chart, with bounded coordinates $\theta_M = (h_M, z_M)$ converging to $\theta := (h(Z), \beta(Z))$. Write Θ_η for the interior Gaussian state at coordinate η ; by (111), $\Lambda_p(|Z\rangle\langle Z|) = \Theta_\theta$. The forward and reverse LAN estimates (50)–(51), which are uniform on bounded local parameter sets, together with trace-norm continuity of $\eta \mapsto \Theta_\eta$, give

$$\|T_M^p(\rho_{Z,M}^{\otimes M}) - \Theta_\theta\|_1 \leq \|T_M^p(\rho_{Z,M}^{\otimes M}) - \Theta_{\theta_M}\|_1 + \|\Theta_{\theta_M} - \Theta_\theta\|_1 \longrightarrow 0,$$

$$\|S_M^p(\Theta_\theta) - \rho_{Z,M}^{\otimes M}\|_1 \leq \|\Theta_\theta - \Theta_{\theta_M}\|_1 + \|S_M^p(\Theta_{\theta_M}) - \rho_{Z,M}^{\otimes M}\|_1 \longrightarrow 0,$$

uniformly for Z in a fixed bounded set. Combined with (116) and contractivity of $T_M^p \circ \text{Tr}_{E^M}$ and of Tr_{E^M} , these are (114) and (115) for $X = |Z\rangle\langle Z|$.

It remains to extend both convergences to arbitrary trace-class inputs. The complex linear span of coherent projectors is trace-norm dense in the trace class. Indeed, a bounded operator annihilating that span has identically zero Husimi symbol; analyticity then makes every number-basis matrix element zero, and Hahn–Banach gives density. All four maps $\Lambda_{p,M}$, Λ_p , $\text{Tr}_{E^M} \circ \mathcal{J}_M$, and $S_M^p \circ \Lambda_p$ are trace-norm contractions. Hence, if Y is a finite linear combination of coherent projectors, then for either pair (A_M, B_M) of maps compared above,

$$\|A_M X - B_M X\|_1 \leq 2\|X - Y\|_1 + \|A_M Y - B_M Y\|_1.$$

First let $M \rightarrow \infty$, then let $Y \rightarrow X$, to obtain the claims. $\square$

Action on the thermal tangent state. Let τ_ϑ be a one-mode thermal state with mean photon number

$$\bar{n}_{\text{th}} = \frac{\vartheta}{1 - \vartheta}.$$

Because the product thermal state is invariant under passive changes of one-particle basis, it factorizes according to the diagonal/off-diagonal decomposition of K_p . The joint fields in (107) have covariance $(2\bar{n}_{\text{th}} + 1)\Sigma_p$. For each $i < j$, the retained mode has mean photon number

$$\bar{n}_{ij}^{\text{out}} = \frac{p_i \bar{n}_{\text{th}} + p_j (\bar{n}_{\text{th}} + 1)}{p_i - p_j}. \quad (119)$$

The input is centered Gaussian and gauge invariant, and $\langle a_{ij}^2 \rangle = 0$, so the retained one-mode Gaussian state is thermal, rather than merely having the displayed mean photon number. Consequently,

$$\frac{\bar{n}_{ij}^{\text{out}}}{\bar{n}_{ij}^{\text{out}} + 1} = \frac{\bar{n}_{\text{th}} + (\bar{n}_{\text{th}} + 1)q_{ij}}{\bar{n}_{\text{th}} + 1 + \bar{n}_{\text{th}}q_{ij}}. \quad (120)$$

Taking

$$\vartheta = \frac{\gamma - 1}{\gamma}, \quad \bar{n}_{\text{th}} = \gamma - 1,$$

turns (120) into (101). We have proved

$$\Lambda_p\left(\tau_\vartheta^{\otimes(d^2-1)}\right) = \mathbf{N}_{0,(2\gamma-1)\Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}}^{\text{PCT}}. \quad (121)$$

A.1.2 The Fock limit of Werner cloning

We record explicitly the arbitrary-dimensional occupation calculation used below. Let $D \geq 2$, fix a unit vector $\psi \in \mathbb{C}^D$, and choose an orthonormal basis $e_1, \dots, e_{D-1}$ of $\psi^\perp$. For $\mathbf{k} = (k_1, \dots, k_{D-1})$, let $|\mathbf{k}\rangle_m$ be the normalized symmetric occupation vector with k_j particles in e_j and the remaining $m - |\mathbf{k}|$ particles in ψ .

Lemma A.3 (Werner occupation law). *Let $\mathcal{W}_{n,m}^{(D)}$ be Werner's universal pure-state cloner in dimension D . Its output on $|\psi\rangle\langle\psi|^{\otimes n}$ is diagonal in the occupation basis, with eigenvalue*

$$\pi_{n,m}^{(D)}(\mathbf{k}) = \frac{\binom{m-|\mathbf{k}|}{n}}{\binom{m+D-1}{n+D-1}}, \quad |\mathbf{k}| \leq m - n. \quad (122)$$

If $m_n/n \rightarrow \gamma > 1$, then under the occupation-Fock embedding

$$\mathcal{W}_{n,m_n}^{(D)}(|\psi\rangle\langle\psi|^{\otimes n}) \longrightarrow \tau_{(\gamma-1)/\gamma}^{\otimes(D-1)} \quad (123)$$

in trace norm.

Proof. Werner's channel [Wer98] is

$$\mathcal{W}_{n,m}^{(D)}(X) = \frac{\mathbf{d}_D(n)}{\mathbf{d}_D(m)} P_m(X \otimes I^{\otimes(m-n)}) P_m, \quad \mathbf{d}_D(N) = \dim \text{Sym}^N(\mathbb{C}^D) = \binom{N+D-1}{D-1},$$

where P_m projects onto the symmetric subspace. On an occupation vector with total transverse occupation $t = |\mathbf{k}|$, the expectation of $P_m(|\psi\rangle\langle\psi|^{\otimes n} \otimes I) P_m$ is $\binom{m-t}{n} / \binom{m}{n}$. Multiplying by $\mathbf{d}_D(n)/\mathbf{d}_D(m)$

gives (122). Irreducibility of $\text{Sym}^t(\mathbb{C}^{D-1})$ under the stabilizer $U(D-1)$ makes the operator scalar on each fixed- t subspace, while the stabilizer's central phase eliminates cross- t blocks. Thus these are all of its eigenvalues. Their normalization also follows from Vandermonde's identity.

For fixed $\mathbf{k}$,

$$\pi_{n,m_n}^{(D)}(\mathbf{k}) \longrightarrow \gamma^{-(D-1)} \left(\frac{\gamma-1}{\gamma} \right)^{|\mathbf{k}|}.$$

The right side is the product of $D-1$ normalized geometric laws. Since both sides are probability masses on the countable occupation lattice, the discrete Scheffé lemma upgrades pointwise convergence to ℓ^1 convergence, which is exactly the trace-norm convergence (123). $\square$

A.1.3 Proof of the exact limit and strict comparison

Proof of Theorem A.1: upper bound and strict inequality. Use the canonical purification ψ_p . Every other minimal purification has the form $(I \otimes U)\psi_p$, with $U \in U(d)$. Covariance of Werner's cloner and invariance of partial trace make the reduced Werner output independent of U . Thus the Haar average in random purification disappears exactly after the purifying registers are discarded:

$$\sigma_{n,m}^{\text{PCT}}(p) = \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(d^2)}(|\psi_p\rangle\langle\psi_p|^{\otimes n}). \quad (124)$$

Pull the Werner output in (124) back to $\Gamma_s(K_p)$ using the occupation embedding, and call the resulting Fock state $\omega_{n,m}$. Lemma A.3, with $D = d^2$, gives

$$\left\| \omega_{n,m_n} - \tau_{(\gamma-1)/\gamma}^{\otimes(d^2-1)} \right\|_1 \longrightarrow 0. \quad (125)$$

The definition of $\Lambda_{p,m}$ and (124) give the exact identity

$$T_{m_n}^p(\sigma_{n,m_n}^{\text{PCT}}(p)) = \Lambda_{p,m_n}(\omega_{n,m_n}).$$

By contractivity, (114), and (125), the pulled-back output converges in trace norm:

$$\left\| \Lambda_{p,m_n}(\omega_{n,m_n}) - \Lambda_p\left(\tau_{(\gamma-1)/\gamma}^{\otimes(d^2-1)}\right) \right\|_1 \longrightarrow 0.$$

Using (121), the LAN limit of the PCT output is therefore

$$\Theta_p^{\text{PCT}} = \mathbf{N}_{0,(2\gamma-1)\Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}}^{\text{PCT}}. \quad (126)$$

Ordinary interior LAN applied to the ideal target gives

$$T_{m_n}^p(\rho_p^{\otimes m_n}) \longrightarrow \Theta_p := \mathbf{N}_{0,\Sigma_p} \otimes \bigotimes_{i < j} \tau_{q_{ij}} \quad (127)$$

in trace norm.

Fidelity is nondecreasing under $T_{m_n}^p$ and continuous under trace-norm convergence. Hence

$$\limsup_{n \rightarrow \infty} F\left(\sigma_{n,m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}\right) \leq F(\Theta_p^{\text{PCT}}, \Theta_p). \quad (128)$$

The fidelity on the right factorizes. In $d-1$ real dimensions,

$$F\left(\mathbf{N}_{0,(2\gamma-1)\Sigma_p}, \mathbf{N}_{0,\Sigma_p}\right) = \left(\frac{2\sqrt{2\gamma-1}}{1+(2\gamma-1)} \right)^{(d-1)/2} = \left(\frac{\sqrt{2\gamma-1}}{\gamma} \right)^{(d-1)/2}. \quad (129)$$

For each thermal pair,

$$F(\tau_{q_{ij}^{\text{PCT}}}, \tau_{q_{ij}}) = h_{q_{ij}}(q_{ij}^{\text{PCT}}).$$

Equations (128) and (129) prove $\limsup_n F \leq F_{\text{PCT}}(\gamma, p)$.

It remains to prove strictness. The optimal one-mode amplifier has thermal parameter

$$q_{ij}^{\text{opt}} = \frac{\gamma - 1 + q_{ij}}{\gamma}, \quad f_\gamma(q_{ij}) = h_{q_{ij}}(q_{ij}^{\text{opt}}).$$

Moreover,

$$q_{ij}^{\text{opt}} - q_{ij} = \frac{(\gamma - 1)(1 - q_{ij})}{\gamma} > 0.$$

Direct calculation gives

$$q_{ij}^{\text{PCT}} - q_{ij}^{\text{opt}} = \frac{q_{ij}(\gamma - 1)(1 - q_{ij})}{\gamma[\gamma + (\gamma - 1)q_{ij}]} > 0.$$

For fixed $q \in (0, 1)$,

$$\frac{d}{dx} \log h_q(x) = \frac{\sqrt{q} - \sqrt{x}}{2\sqrt{x}(1-x)(1 - \sqrt{qx})} < 0, \quad x > q.$$

Thus every quantum factor in F_{PCT} is strictly smaller than the corresponding $f_\gamma(q_{ij})$. The classical factors obey

$$\frac{\sqrt{2\gamma - 1}}{\gamma} < \frac{2\sqrt{\gamma}}{1 + \gamma}, \quad \gamma > 1,$$

because, after squaring and clearing positive denominators, the difference is proportional to $(\gamma - 1)^2(2\gamma + 1) > 0$. Multiplying the strict modewise inequalities proves $F_{\text{PCT}} < F_{\text{univ}}$. $\square$

Proof of Theorem A.1: matching lower bound. Apply (115) to the fixed state $X = \tau_{(\gamma-1)/\gamma}^{\otimes(d^2-1)}$, and combine it with (124), (125), and (121): since $\sigma_{n,m_n}^{\text{PCT}}(p) = (\text{Tr}_{E^{m_n}} \circ \mathcal{J}_{m_n})(\omega_{n,m_n})$ and $\|\omega_{n,m_n} - \tau_{(\gamma-1)/\gamma}^{\otimes(d^2-1)}\|_1 \rightarrow 0$,

$$\|\sigma_{n,m_n}^{\text{PCT}}(p) - S_{m_n}^p(\Theta_p^{\text{PCT}})\|_1 \rightarrow 0,$$

with Θ_p^{PCT} as in (126). The reverse LAN approximation at the center likewise gives $\|S_{m_n}^p(\Theta_p) - \rho_p^{\otimes m_n}\|_1 \rightarrow 0$, with Θ_p as in (127). Monotonicity of fidelity under $S_{m_n}^p$ and its trace-norm continuity therefore yield

$$F(\sigma_{n,m_n}^{\text{PCT}}(p), \rho_p^{\otimes m_n}) \geq F(S_{m_n}^p(\Theta_p^{\text{PCT}}), S_{m_n}^p(\Theta_p)) - o(1) \geq F(\Theta_p^{\text{PCT}}, \Theta_p) - o(1) = F_{\text{PCT}}(\gamma, p) - o(1),$$

which is the required lower bound. Together with (128), this completes the proof of Theorem A.1. $\square$

A.2 Projector states

For projector states, we compare the fidelity attained by PCT itself with the optimum in Theorem 1.3, rather than relying only on its published data-processing lower bound.

For $s \geq 1$, let $\text{d}_s(N) = \dim \text{Sym}^N(\mathbb{C}^s) = \binom{N+s-1}{s-1}$ be as in the proof of Lemma A.3, and let $\Pi_{\text{sym},N}^{(s)}$ be the projection onto this symmetric subspace. For $m \geq n$ and an input X supported on $\text{Sym}^n(\mathbb{C}^s)$, Werner's pure-state n -to- m cloner in dimension s is

$$\mathcal{W}_{n,m}^{(s)}(X) = \frac{\text{d}_s(n)}{\text{d}_s(m)} \Pi_{\text{sym},m}^{(s)}(X \otimes I_s^{\otimes(m-n)}) \Pi_{\text{sym},m}^{(s)}. \quad (130)$$

Fix $1 \leq r < d$ and $P \in \text{Gr}(r, d)$, choose an isometry $V_P : \mathbb{C}^r \rightarrow \mathbb{C}^d$ with $V_P V_P^* = P$, and put

$$|\Phi_P\rangle = \frac{1}{\sqrt{r}} \sum_{i=1}^r V_P |i\rangle \otimes |i\rangle, \quad |\Phi_r\rangle = \frac{1}{\sqrt{r}} \sum_{i=1}^r |ii\rangle.$$

Let $\sigma_{n,m}^{\text{PCT}}(P)$ denote the system output of PCT on $(P/r)^{\otimes n}$, using a purifying register $\mathbb{C}^r$.

Proposition A.4 (PCT is strictly suboptimal for non-pure projector states). *Define*

$$s_{n,m}^{(r,d)} = \frac{\mathbf{d}_{rd}(n)/\mathbf{d}_{rd}(m)}{\mathbf{d}_{r^2}(n)/\mathbf{d}_{r^2}(m)} = \prod_{j=r^2}^{rd-1} \frac{n+j}{m+j}. \quad (131)$$

There is an exact finite-sample factorization

$$F(\sigma_{n,m}^{\text{PCT}}(P), (P/r)^{\otimes m}) = \sqrt{s_{n,m}^{(r,d)}} F(\zeta_{n,m}^{(r)}, (I_r/r)^{\otimes m}), \quad (132)$$

where

$$\zeta_{n,m}^{(r)} = \text{Tr}_{E^m} \mathcal{W}_{n,m}^{(r^2)}(|\Phi_r\rangle\langle\Phi_r|^{\otimes n}). \quad (133)$$

If $m_n/n \rightarrow \gamma > 1$, then

$$\limsup_{n \rightarrow \infty} F(\sigma_{n,m_n}^{\text{PCT}}(P), (P/r)^{\otimes m_n}) \leq \gamma^{-r(d-r)/2} \left(\frac{\sqrt{2\gamma-1}}{\gamma} \right)^{(r-1)/2}. \quad (134)$$

Consequently, for every $1 < r < d$ and fixed $\gamma > 1$, PCT is strictly below the optimal projector-state value (9). For $r = 1$, the internal factor in (132) is one and PCT reduces to Werner's optimal pure-state cloner.

Proof. Every purification in the random-purification output has the form $(I \otimes U)\Phi_P$, with $U \in \text{U}(r)$. Unitary covariance of $\mathcal{W}_{n,m}^{(rd)}$ and invariance of the partial trace make the reduced output independent of U . Thus the Haar average disappears exactly after the purifying registers are discarded.

Let $Q = P \otimes I_r$ on $\mathbb{C}^d \otimes \mathbb{C}^r$. Its rank is r^2 , its range contains Φ_P , and $Q^{\otimes m}$ commutes with the symmetric projection. Compressing (130) therefore gives

$$\begin{aligned} Q^{\otimes m} \mathcal{W}_{n,m}^{(rd)}(|\Phi_P\rangle\langle\Phi_P|^{\otimes n}) Q^{\otimes m} \\ = s_{n,m}^{(r,d)} (V_P \otimes I_r)^{\otimes m} \mathcal{W}_{n,m}^{(r^2)}(|\Phi_r\rangle\langle\Phi_r|^{\otimes n}) (V_P^* \otimes I_r)^{\otimes m}. \end{aligned} \quad (135)$$

Taking the partial trace over E^m yields

$$P^{\otimes m} \sigma_{n,m}^{\text{PCT}}(P) P^{\otimes m} = s_{n,m}^{(r,d)} V_P^{\otimes m} \zeta_{n,m}^{(r)} (V_P^*)^{\otimes m}. \quad (136)$$

The target is supported on $P^{\otimes m}$. Hence fidelity is unchanged by compressing the first argument to that support, and its homogeneity proves (132).

It remains to bound the internal fidelity. Assume $r > 1$, put

$$\mathbf{H}_r = \left\{ h \in \mathbb{R}^r : \sum_{i=1}^r h_i = 0 \right\}, \quad u_r = \frac{1}{r} \mathbf{1}_r, \quad \Sigma_r = (\text{diag}(u_r) - u_r u_r^\top) \Big|_{\mathbf{H}_r},$$

and measure every system output in the computational basis. If $C_m = (C_{m,1}, \dots, C_{m,r})$ is the resulting count vector, set

$$Z_m = \frac{C_m - m u_r}{\sqrt{m}} \in \mathbf{H}_r.$$

Under the target $(I_r/r)^{\otimes m}$, the multinomial central limit theorem gives

$$Z_m \Longrightarrow \mathbf{N}_{0, \Sigma_r}. \quad (137)$$

We next show that under $\zeta_{n, m_n}^{(r)}$,

$$Z_{m_n} \Longrightarrow \mathbf{N}_{0, (2\gamma-1)\Sigma_r}. \quad (138)$$

For this purpose put $R = r^2$ and use the symmetric occupation basis about the condensate Φ_r . [Lemma A.3](#), applied with $D = R$ and $\psi = \Phi_r$, shows that under the corresponding embedding into the $(R - 1)$ -mode Fock space the purified Werner output converges in trace norm to

$$\tau_{\vartheta}^{\otimes (R-1)}, \quad \tau_{\vartheta} = (1 - \vartheta) \sum_{j \geq 0} \vartheta^j |j\rangle\langle j|, \quad \vartheta = \frac{\gamma - 1}{\gamma}, \quad (139)$$

whose mean occupation in every mode is $\bar{n}_{\text{th}} = \gamma - 1$.

To read off the count fluctuations, fix $h \in \mathbf{H}_r$ and set

$$H_h = \text{diag}(h) \otimes I_r, \quad v_h = H_h \Phi_r \in \Phi_r^\perp.$$

Then

$$\langle v_g, v_h \rangle = \frac{1}{r} \sum_{i=1}^r g_i h_i = g^\top \Sigma_r h. \quad (140)$$

Let J_m be the condensate occupation embedding, $\mathcal{N}$ the Fock number operator, $b(v)$ the annihilation operator in direction v , $\text{d}\Gamma(A)$ the second quantization of A , and $P_\perp = I - |\Phi_r\rangle\langle\Phi_r|$. The standard condensate fluctuation identity gives, on each finite-particle sector,

$$J_m^* \left[\frac{1}{\sqrt{m}} \sum_{\ell=1}^m H_h^{(\ell)} \right] J_m = b^*(v_h) \sqrt{1 - \frac{\mathcal{N}}{m}} + \sqrt{1 - \frac{\mathcal{N}}{m}} b(v_h) + \frac{1}{\sqrt{m}} \text{d}\Gamma(P_\perp H_h P_\perp) \longrightarrow b^*(v_h) + b(v_h). \quad (141)$$

The fields on the right commute for $h \in \mathbf{H}_r$, because the inner products in [\(140\)](#) are real. Put

$$P_{\leq m} = \mathbf{1}_{\{\mathcal{N} \leq m\}}, \quad f_m(\mathcal{N}) = \sqrt{(1 - \mathcal{N}/m)_+},$$

and define on the full Fock space

$$A_m(h) = b^*(v_h) f_m(\mathcal{N}) + f_m(\mathcal{N}) b(v_h) + \frac{1}{\sqrt{m}} P_{\leq m} \text{d}\Gamma(P_\perp H_h P_\perp) P_{\leq m}.$$

This is a bounded self-adjoint operator and agrees with the pulled-back count field on the range of J_m . Let

$$\mathcal{D}_{\text{fin}} = \bigcup_{L \geq 0} \mathbf{1}_{\{\mathcal{N} \leq L\}} \Gamma_s(\Phi_r^\perp).$$

The finite-particle vectors are analytic for $\phi(h) = b^*(v_h) + b(v_h)$, so $\phi(h)$ is essentially self-adjoint on $\mathcal{D}_{\text{fin}}$. If ψ has at most L particles and $m > L$, then

$$\|(A_m(h) - \phi(h))\psi\| = O_{L,h}(m^{-1/2}) \|\psi\| \longrightarrow 0.$$

Thus $A_m(h) \rightarrow \overline{\phi(h)}$ on a core for the limiting self-adjoint operator. The standard core criterion [\[RS80, Theorem VIII.25\]](#) gives strong-resolvent convergence and hence $e^{itA_m(h)} \rightarrow e^{it\overline{\phi(h)}}$ strongly

for every $t \in \mathbb{R}$. Together with the trace-norm state convergence in (139), this gives convergence of the characteristic functions for every h and t . Explicitly, if $\varrho_m \rightarrow \varrho$ in trace norm and $U_m \rightarrow U$ strongly with all operators unitary, then

$$|\mathrm{Tr}(\varrho_m U_m) - \mathrm{Tr}(\varrho U)| \leq \|\varrho_m - \varrho\|_1 + |\mathrm{Tr}(\varrho(U_m - U))| \rightarrow 0.$$

In the gauge-invariant thermal state of mean $\bar{n}_{\mathrm{th}}$,

$$\mathrm{Tr}\left[\tau_{\vartheta}^{\otimes(R-1)} e^{it(b^*(v_h) + b(v_h))}\right] = \exp\left(-\frac{t^2}{2}(2\bar{n}_{\mathrm{th}} + 1)\|v_h\|^2\right).$$

Since measuring the system counts after tracing out E^m is the same as measuring H_h before that trace, the Cramér–Wold theorem, $2\bar{n}_{\mathrm{th}} + 1 = 2\gamma - 1$, and (140) prove (138).

Weak convergence suffices after an arbitrarily small common classical smoothing. For $t > 0$, add to Z_m an independent $\mathbf{N}_{0,t\Sigma_r}$ random vector on $\mathbf{H}_r$. This is a classical channel, so fidelity monotonicity bounds the internal quantum fidelity by that of the two smoothed count laws. Convolution with the fixed nondegenerate Gaussian has densities that converge pointwise under weak convergence, and hence in L^1 by Scheffé’s lemma. The limiting laws have covariances $(1+t)\Sigma_r$ and $(2\gamma - 1 + t)\Sigma_r$. The L^1 convergence of both densities implies convergence of their fidelity, and the limiting Gaussian fidelity is

$$\left(\frac{\sqrt{(1+t)(2\gamma - 1 + t)}}{\gamma + t}\right)^{(r-1)/2}.$$

It follows, after $t \downarrow 0$, that

$$\limsup_{n \rightarrow \infty} F\left(\zeta_{n,m_n}^{(r)}, (I_r/r)^{\otimes m_n}\right) \leq \left(\frac{\sqrt{2\gamma - 1}}{\gamma}\right)^{(r-1)/2}. \quad (142)$$

Finally, (131) gives

$$s_{n,m_n}^{(r,d)} \rightarrow \gamma^{-r(d-r)}.$$

Combining this with (132) and (142) proves (134). Its right-hand side is strictly smaller than $\gamma^{-r(d-r)/2}$ when $1 < r < d$, because $2\gamma - 1 < \gamma^2$. The $r = 1$ assertion follows directly from (133). $\square$

Remark A.5 (The limit $\gamma \downarrow 1$). Writing $\gamma = 1 + \delta$, the additional factor detected by the count statistic has the expansion

$$\left(\frac{\sqrt{2\gamma - 1}}{\gamma}\right)^{(r-1)/2} = 1 - \frac{r-1}{4}\delta^2 + O(\delta^3).$$

Thus this statistic detects a second-order loss. The exact factorization (132) also bounds the total PCT infidelity from both sides in the iterated limit of Section 5. For fixed $\delta > 0$ and $m_n/n \rightarrow 1 + \delta$, the upper bound from the support factor implies

$$\liminf_{n \rightarrow \infty} \left[1 - F\left(\sigma_{n,m_n}^{\mathrm{PCT}}(P), (P/r)^{\otimes m_n}\right)^2\right] \geq 1 - (1 + \delta)^{-r(d-r)},$$

while (81) implies

$$\limsup_{n \rightarrow \infty} \left[1 - F\left(\sigma_{n,m_n}^{\mathrm{PCT}}(P), (P/r)^{\otimes m_n}\right)^2\right] \leq 1 - (1 + \delta)^{-(rd-1)}.$$

Thus any first-order coefficient, if it exists after first taking $n \rightarrow \infty$ with $m_n/n \rightarrow 1 + \delta$, lies between $r(d - r)$ and $rd - 1$. It can never improve on the optimal projector-state coefficient $r(d - r)$, and it equals that coefficient precisely when the internal squared infidelity in (132) is $o(\delta)$. The count statistic detects a second-order loss in the $r - 1$ commuting diagonal directions. Whether the total internal loss is $o(\delta)$ remains open; we expect it to be $O(\delta^2)$, but proving this requires a degenerate central-limit or LAN statement.

B Exact-weight PBW calculation of the sector fidelity

This appendix proves Proposition 2.3 and Lemma 2.4 by comparing the target sector $\rho_{p,\nu}$ with the Cartan-sector output $\mathcal{C}_{\mu,\nu}(\rho_{p,\mu})$ on finitely many weight spaces near their highest weights, then controlling the remaining target mass. Lemma B.1 shows that normalized lowering-operator vectors become asymptotically orthonormal on these spaces, while Lemma B.2 gives the uniform limit $p^\lambda/s_\lambda(p) \rightarrow \prod_{i < j} (1 - p_j/p_i)$. The Cartan inclusion $V_{\mu,\nu-\mu}$ then splits each lowering operation between $\mathcal{H}_\mu$ and $\mathcal{H}_{\nu-\mu}$ with asymptotically binomial weights, producing the thermal limits τ_p and $\tau_p^{(\gamma)}$ and hence the fidelity $F_{\text{orb}}(\gamma, p)$.

Because the calculations involve only weights of height bounded in terms of L , $O_L(\cdot)$ may depend on L, d, c, C in $cN \leq \Delta_\alpha^\lambda \leq CN$ for $\alpha \in \mathbb{R}_+$, but is uniform in N and all dominant λ satisfying these inequalities.

B.1 PBW vectors on complete weight spaces

Identify the positive roots of $\mathbb{U}(d)$ with

$$\mathbb{R}_+ = \{e_i - e_j : 1 \leq i < j \leq d\}.$$

For $\alpha = e_i - e_j$, set

$$F_\alpha = E_{ji}, \quad E_\alpha = E_{ij}, \quad H_\alpha = [E_\alpha, F_\alpha] = E_{ii} - E_{jj}, \quad \Delta_\alpha^\lambda = \lambda_i - \lambda_j.$$

Fix a total order on $\mathbb{R}_+$. For $\mathbf{r} \in \mathbb{Z}_{\geq 0}^{\mathbb{R}_+}$, write

$$F^\mathbf{r} = \prod_{\alpha \in \mathbb{R}_+}^{\prec} F_\alpha^{r_\alpha}, \quad |\mathbf{r}| = \sum_{\alpha} r_\alpha.$$

For a nonnegative simple-root combination η , let $\text{ht}(\eta)$ be its simple-root height and put

$$\mathbb{P}(\eta) = \left\{ \mathbf{r} : \sum_{\alpha \in \mathbb{R}_+} r_\alpha \alpha = \eta \right\}.$$

The set $\mathbb{P}(\eta)$ is finite, and the PBW theorem implies that $\{F^\mathbf{r} v_\lambda : \mathbf{r} \in \mathbb{P}(\eta)\}$ spans $\mathcal{H}_\lambda[\lambda - \eta]$ [FH91]. Define

$$\mathcal{K}_{\lambda,L} = \bigoplus_{\text{ht}(\eta) \leq L} \mathcal{H}_\lambda[\lambda - \eta]. \quad (143)$$

Let $P_{\lambda,L}$ denote the orthogonal projection onto $\mathcal{K}_{\lambda,L}$.

Lemma B.1 (PBW Gram matrices at fixed height). *Let $\lambda = \lambda^{(N)}$ range over dominant weights satisfying*

$$cN \leq \Delta_\alpha^\lambda \leq CN \quad (\alpha \in \mathbb{R}_+)$$

for fixed $0 < c < C < \infty$. Define

$$\tilde{e}_{\mathbf{r}}^{\lambda} = \prod_{\alpha \in \mathbf{R}_+}^{\prec} \frac{F_{\alpha}^{r_{\alpha}}}{\sqrt{r_{\alpha}!} (\Delta_{\alpha}^{\lambda})^{r_{\alpha}/2}} v_{\lambda}. \quad (144)$$

For every fixed L , uniformly over $\text{ht}(\eta) \leq L$ and $\mathbf{r}, \mathbf{s} \in \mathbf{P}(\eta)$,

$$\langle \tilde{e}_{\mathbf{r}}^{\lambda}, \tilde{e}_{\mathbf{s}}^{\lambda} \rangle = \delta_{\mathbf{r}, \mathbf{s}} + O_L(N^{-1/2}). \quad (145)$$

Consequently, for all sufficiently large N , the full Gram matrix $G_{\lambda, \eta}$ on every such exact-weight block is invertible. If

$$(e_{\mathbf{r}}^{\lambda})_{\mathbf{r} \in \mathbf{P}(\eta)} = (\tilde{e}_{\mathbf{r}}^{\lambda})_{\mathbf{r} \in \mathbf{P}(\eta)} G_{\lambda, \eta}^{-1/2}, \quad (146)$$

then these vectors form an orthonormal basis of $\mathcal{H}_{\lambda}[\lambda - \eta]$, and

$$e_{\mathbf{r}}^{\lambda} = \tilde{e}_{\mathbf{r}}^{\lambda} + O_L(N^{-1/2}) \quad (147)$$

uniformly for $\text{ht}(\eta) \leq L$ and $\mathbf{r} \in \mathbf{P}(\eta)$.

Proof. For $\mathbf{r} \in \mathbf{P}(\eta)$, every positive root has positive simple-root height, so $|\mathbf{r}| \leq \text{ht}(\eta)$. Put

$$\hat{F}_{\alpha} = \frac{F_{\alpha}}{\sqrt{\Delta_{\alpha}^{\lambda}}}, \quad \hat{E}_{\alpha} = \frac{E_{\alpha}}{\sqrt{\Delta_{\alpha}^{\lambda}}}.$$

Each unnormalized Gram entry $\langle v_{\lambda}, (\hat{F}^{\mathbf{r}})^* \hat{F}^{\mathbf{s}} v_{\lambda} \rangle$ is therefore a word of length at most $2L$. Commuting root operators cannot increase the word length, and each root has height at most $d - 1$. Hence every intermediate vector lies in

$$\mathcal{K}_{\lambda, R_L}, \quad R_L := 2L(d - 1).$$

For $0 \leq R \leq R_L$, the restrictions

$$\hat{E}_{\alpha} : \mathcal{K}_{\lambda, R} \rightarrow \mathcal{K}_{\lambda, R}, \quad \hat{F}_{\alpha} : \mathcal{K}_{\lambda, R} \rightarrow \mathcal{K}_{\lambda, R+d-1}$$

satisfy $\|\hat{E}_{\alpha}\| + \|\hat{F}_{\alpha}\| = O_L(1)$. To see this, decompose $\mathcal{H}_{\lambda}$ orthogonally into irreducible $\mathfrak{sl}_2(\alpha)$ -summands within joint eigenspaces of the Cartan subalgebra commuting with $\mathfrak{sl}_2(\alpha)$. Each string then consists of full Cartan weight vectors. If a summand meets the weight space of $\lambda - \eta$, where $\text{ht}(\eta) \leq R$, let t be its level there and M its highest H_{α} -eigenvalue. The top of this string, of weight $\lambda - \eta + t\alpha$, is again a weight of $\mathcal{H}_{\lambda}$; hence $\eta - t\alpha$ is a nonnegative simple-root combination and $t \text{ht}(\alpha) \leq \text{ht}(\eta) \leq R$. Moreover,

$$M = \Delta_{\alpha}^{\lambda} - \eta(H_{\alpha}) + 2t = \Delta_{\alpha}^{\lambda} + O_L(1).$$

At level t , the F_{α} - and E_{α} -coefficients are $\sqrt{(t+1)(M-t)}$ and $\sqrt{t(M-t+1)}$. After division by $\sqrt{\Delta_{\alpha}^{\lambda}}$, both are $O_L(1)$, proving the norm bound. All operator norms below are taken after restriction to $\mathcal{K}_{\lambda, R_L}$, with the codomain norm inherited from $\mathcal{H}_{\lambda}$.

The matrix-unit commutator formula now gives

$$[\hat{E}_{\alpha}, \hat{F}_{\beta}] = \delta_{\alpha\beta} I + O_L(N^{-1/2}). \quad (148)$$

For $\alpha = \beta$, the left side is $H_{\alpha}/\Delta_{\alpha}^{\lambda} = I + O_L(N^{-1})$ on the cutoff. For $\alpha \neq \beta$, the unnormalized commutator is zero or, up to sign, a root operator E_{γ} or F_{γ} . In the nonzero case, the preceding bound gives $\|[\hat{E}_{\alpha}, \hat{F}_{\beta}]\| = O_L(\sqrt{\Delta_{\gamma}^{\lambda}}/\sqrt{\Delta_{\alpha}^{\lambda}\Delta_{\beta}^{\lambda}}) = O_L(\sqrt{N}/N) = O_L(N^{-1/2})$.

Commute every $\widehat{E}_\alpha$ to the right in $\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle$. The scalar commutators in (148) contribute only when $\mathbf{r} = \mathbf{s}$, in which case they give $\prod_\alpha r_\alpha!$. Every other term vanishes or contains an $O_L(N^{-1/2})$ remainder. Only $O_L(1)$ terms occur, and all remaining operator factors have norm $O_L(1)$; consequently,

$$\langle v_\lambda, (\widehat{F}^{\mathbf{r}})^* \widehat{F}^{\mathbf{s}} v_\lambda \rangle = \delta_{\mathbf{r}, \mathbf{s}} \prod_\alpha r_\alpha! + O_L(N^{-1/2}).$$

Dividing by the factorials in (144) proves (145).

The number and sizes of the exact-weight blocks are bounded in terms of L , so the entrywise estimate gives $G_{\lambda, \eta} = I + O_L(N^{-1/2})$ in operator norm. Thus $G_{\lambda, \eta}$ is invertible for large N , and the PBW spanning vectors form a basis. Finally, functional calculus gives $G_{\lambda, \eta}^{-1/2} = I + O_L(N^{-1/2})$; hence (146) is an orthonormal basis and satisfies (147). $\square$

B.2 Character normalization and fixed-height matrices

For $a = (a_1, \dots, a_d)$, write $p^a = \prod_i p_i^{a_i}$.

Lemma B.2 (Uniform character normalization). *Let $K \Subset \Delta_d^\circ$, let $\delta_N \rightarrow 0$, and suppose $p \in K$, $\lambda \vdash N$, and $\|\lambda/N - p\|_\infty \leq \delta_N$. Then, uniformly over this family,*

$$\frac{p^\lambda}{s_\lambda(p)} = \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right) + o(1). \quad (149)$$

Proof. Write the numerator in the Weyl character formula as

$$\sum_{\sigma \in \mathfrak{S}_d} \text{sgn}(\sigma) A_\sigma, \quad A_\sigma = \prod_i p_i^{\lambda_{\sigma(i)} + d - \sigma(i)}.$$

For $\sigma \neq \text{id}$,

$$\frac{1}{N} \log \frac{A_{\text{id}}}{A_\sigma} = \sum_i \frac{\lambda_i}{N} \log p_i - \sum_i \frac{\lambda_{\sigma(i)}}{N} \log p_i + O_K(N^{-1}).$$

For $x \in \mathbb{R}^d$, put

$$D_\sigma(x, p) = \sum_i x_i \log p_i - \sum_i x_{\sigma(i)} \log p_i.$$

Strict rearrangement gives $D_\sigma(p, p) > 0$. Since K is compact, the permutations are finite, and the logarithms are uniformly bounded on K , the assumptions $\|\lambda/N - p\|_\infty \leq \delta_N$ and $\delta_N \rightarrow 0$ give $c_K > 0$ such that, for every $\sigma \neq \text{id}$ and all sufficiently large N ,

$$\frac{1}{N} \log \frac{A_{\text{id}}}{A_\sigma} \geq c_K, \quad A_\sigma \leq e^{-c_K N} A_{\text{id}}.$$

Consequently,

$$\sum_{\sigma \in \mathfrak{S}_d} \text{sgn}(\sigma) A_\sigma = A_{\text{id}} (1 + O_K(e^{-c_K N})), \quad A_{\text{id}} = p^\lambda \prod_i p_i^{d-i}.$$

Since

$$\det(p_i^{d-j})_{i,j=1}^d = \prod_i p_i^{d-i} \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right),$$

the Weyl character formula gives, uniformly,

$$s_\lambda(p) = p^\lambda \prod_{i < j} \left(1 - \frac{p_j}{p_i}\right)^{-1} (1 + O_K(e^{-c_K N})).$$

This proves (149). $\square$

For the remainder of the appendix, index the number basis of $\mathcal{F}_{\text{orb}}$ by $\mathbf{r} \in \mathbb{Z}_{\geq 0}^{\mathbf{R}_+}$, and write it as $\{|\mathbf{r}\rangle\}$. Define the finite-rank projection

$$P_L^{\mathcal{F}} = \sum_{\text{ht}(\sum_{\alpha} r_{\alpha} \alpha) \leq L} |\mathbf{r}\rangle\langle \mathbf{r}|. \quad (150)$$

For every fixed L and all sufficiently large N , each highest weight λ satisfying the gap bounds in [Lemma B.1](#) has the orthonormal basis $e_{\mathbf{r}}^{\lambda}$ of $\mathcal{K}_{\lambda,L}$ constructed there. Define

$$J_{\lambda,L} : \mathcal{K}_{\lambda,L} \longrightarrow P_L^{\mathcal{F}} \mathcal{F}_{\text{orb}}, \quad J_{\lambda,L} e_{\mathbf{r}}^{\lambda} = |\mathbf{r}\rangle. \quad (151)$$

By [Lemma B.1](#), the two spaces have bases indexed by the same $\mathbf{r}$, so $J_{\lambda,L}$ is unitary onto $P_L^{\mathcal{F}} \mathcal{F}_{\text{orb}}$. It is defined only on $\mathcal{K}_{\lambda,L}$, and it depends on λ and L .

Proof of Proposition 2.3 and Lemma 2.4. Fix K and δ_n as in the proposition, write $m = m_n$, and put $\omega = \nu - \mu$. Since $m/n = \gamma_n \rightarrow \gamma > 1$, we have $m/n - 1 \geq (\gamma - 1)/2$ for all sufficiently large n . Uniformly over the allowed parameters,

$$\left\| \frac{\omega}{m - n} - p \right\|_{\infty} \leq \frac{m/n + 1}{m/n - 1} \delta_n = o(1).$$

Every coordinate and every adjacent gap is uniformly positive on K . Hence $\omega \vdash m - n$ for all sufficiently large n . For $\alpha = e_i - e_j$, set

$$s_{\alpha,n} := \frac{\Delta_{\alpha}^{\mu}}{\Delta_{\alpha}^{\nu}} = \frac{n}{m} \frac{\mu_i/n - \mu_j/n}{\nu_i/m - \nu_j/m} = \frac{1}{\gamma} + o(1), \quad 1 - s_{\alpha,n} = \frac{\Delta_{\alpha}^{\omega}}{\Delta_{\alpha}^{\nu}}. \quad (152)$$

Consequently, there are constants $0 < c_K < C_K < \infty$, depending only on K and γ , such that, for all large n ,

$$c_K n \leq \Delta_{\alpha}^{\mu}, \Delta_{\alpha}^{\nu}, \Delta_{\alpha}^{\omega} \leq C_K n \quad (\alpha \in \mathbf{R}_+). \quad (153)$$

Let $V = V_{\mu,\omega} : \mathcal{H}_{\nu} \rightarrow \mathcal{H}_{\mu} \otimes \mathcal{H}_{\omega}$, and write $F_{\alpha}^{(\lambda)} = d\pi_{\lambda}(F_{\alpha})$ for the action of the root operator on $\mathcal{H}_{\lambda}$. Intertwining and the highest-weight normalization give

$$VF_{\alpha}^{(\nu)} = (F_{\alpha}^{(\mu)} \otimes I + I \otimes F_{\alpha}^{(\omega)})V.$$

Expanding each $(F_{\alpha}^{(\mu)} \otimes I + I \otimes F_{\alpha}^{(\omega)})^{r_{\alpha}}$ binomially gives the exact identity

$$V\tilde{e}_{\mathbf{r}}^{\nu} = \sum_{\mathbf{a} \leq \mathbf{r}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} \tilde{e}_{\mathbf{a}}^{\mu} \otimes \tilde{e}_{\mathbf{r}-\mathbf{a}}^{\omega}, \quad (154)$$

where

$$\beta_{\mathbf{r},\mathbf{a}}^{(n)} = \prod_{\alpha \in \mathbf{R}_+} \sqrt{\binom{r_{\alpha}}{a_{\alpha}} s_{\alpha,n}^{a_{\alpha}} (1 - s_{\alpha,n})^{r_{\alpha} - a_{\alpha}}}. \quad (155)$$

If $\text{ht}(\sum_{\alpha} r_{\alpha} \alpha) \leq L$, then (154) has $O_L(1)$ terms, all of height at most L , with $0 \leq \beta_{\mathbf{r},\mathbf{a}}^{(n)} \leq 1$. Since $m/n \rightarrow \gamma > 1$, the sizes $n, m, m - n$ are uniformly comparable, so (153) and [Lemma B.1](#) give $e_{\mathbf{b}}^{\lambda} = \tilde{e}_{\mathbf{b}}^{\lambda} + O_L(n^{-1/2})$ for $\lambda \in \{\mu, \nu, \omega\}$ and all relevant $\mathbf{b}$. Substituting these relations into (154) and using $\|Vx\| = \|x\|$ gives

$$Ve_{\mathbf{r}}^{\nu} = \sum_{\mathbf{a} \leq \mathbf{r}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} e_{\mathbf{a}}^{\mu} \otimes e_{\mathbf{r}-\mathbf{a}}^{\omega} + O_L(n^{-1/2}) \quad (156)$$

uniformly in norm for $\text{ht}(\sum_{\alpha} r_{\alpha} \alpha) \leq L$.

Put

$$\sigma_{\mu,\nu} = \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}), \quad c_{\lambda}(p) = \frac{p^{\lambda}}{s_{\lambda}(p)}, \quad q_{\alpha} = \frac{p_j}{p_i} \quad (\alpha = e_i - e_j).$$

For $\lambda \in \{\mu, \nu\}$, the weight of $e_{\mathbf{b}}^{\lambda}$ is $\lambda - \sum_{\alpha} b_{\alpha} \alpha$, and hence

$$\rho_{p,\lambda} e_{\mathbf{b}}^{\lambda} = c_{\lambda}(p) \prod_{\alpha} q_{\alpha}^{b_{\alpha}} e_{\mathbf{b}}^{\lambda}.$$

The sector channel gives the local formula

$$\sigma_{\mu,\nu} = \frac{\dim \mathcal{H}_{\mu}}{\dim \mathcal{H}_{\nu}} V^{*}(\rho_{p,\mu} \otimes I_{\mathcal{H}_{\omega}}) V.$$

All asymptotic estimates below are uniform over the allowed family. The character estimate, [Lemma B.2](#), gives

$$c_{\lambda}(p) = \prod_{\alpha} (1 - q_{\alpha}) + o(1) \quad (\lambda = \mu, \nu),$$

while the Weyl dimension formula, (153), and (152) give

$$\frac{\dim \mathcal{H}_{\mu}}{\dim \mathcal{H}_{\nu}} = \prod_{\alpha=e_i-e_j} \frac{\Delta_{\alpha}^{\mu} + j - i}{\Delta_{\alpha}^{\nu} + j - i} = \prod_{\alpha \in \mathbf{R}_{+}} (s_{\alpha,n} + O(n^{-1})) = \gamma^{-r_d} + o(1).$$

Thus, substituting (156) and the weight action above into this formula, and using $\|\rho_{p,\mu}\| \leq 1$, gives, for fixed L ,

$$\begin{aligned} \langle e_{\mathbf{r}}^{\nu}, \sigma_{\mu,\nu} e_{\mathbf{s}}^{\nu} \rangle &= \frac{\dim \mathcal{H}_{\mu}}{\dim \mathcal{H}_{\nu}} \sum_{\substack{\mathbf{a} \leq \mathbf{r} \\ \mathbf{b} \leq \mathbf{s}}} \beta_{\mathbf{r},\mathbf{a}}^{(n)} \beta_{\mathbf{s},\mathbf{b}}^{(n)} \langle e_{\mathbf{a}}^{\mu}, \rho_{p,\mu} e_{\mathbf{b}}^{\mu} \rangle \langle e_{\mathbf{r}-\mathbf{a}}^{\omega}, e_{\mathbf{s}-\mathbf{b}}^{\omega} \rangle + O_L(n^{-1/2}) \\ &= \delta_{\mathbf{r},\mathbf{s}} \frac{\dim \mathcal{H}_{\mu}}{\dim \mathcal{H}_{\nu}} c_{\mu}(p) \prod_{\alpha \in \mathbf{R}_{+}} (1 - s_{\alpha,n} + s_{\alpha,n} q_{\alpha})^{r_{\alpha}} + O_L(n^{-1/2}), \end{aligned} \quad (157)$$

$$\langle e_{\mathbf{r}}^{\nu}, \rho_{p,\nu} e_{\mathbf{s}}^{\nu} \rangle = \delta_{\mathbf{r},\mathbf{s}} c_{\nu}(p) \prod_{\alpha \in \mathbf{R}_{+}} q_{\alpha}^{r_{\alpha}}. \quad (158)$$

Recall the following notation from [Section 3.2](#):

$$q_{\alpha}^{(\gamma)} = 1 - \frac{1 - q_{\alpha}}{\gamma}, \quad \tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle \langle k|, \quad \tau_p = \bigotimes_{\alpha \in \mathbf{R}_{+}} \tau_{q_{\alpha}}, \quad \tau_p^{(\gamma)} = \bigotimes_{\alpha \in \mathbf{R}_{+}} \tau_{q_{\alpha}^{(\gamma)}}.$$

Since $s_{\alpha,n} = \gamma^{-1} + o(1)$, (157) and (158) now give

$$\begin{aligned} \langle e_{\mathbf{r}}^{\nu}, \sigma_{\mu,\nu} e_{\mathbf{s}}^{\nu} \rangle &= \delta_{\mathbf{r},\mathbf{s}} \prod_{\alpha} \left[(1 - q_{\alpha}^{(\gamma)}) (q_{\alpha}^{(\gamma)})^{r_{\alpha}} \right] + o(1), \\ \langle e_{\mathbf{r}}^{\nu}, \rho_{p,\nu} e_{\mathbf{s}}^{\nu} \rangle &= \delta_{\mathbf{r},\mathbf{s}} \prod_{\alpha} [(1 - q_{\alpha}) q_{\alpha}^{r_{\alpha}}] + o(1). \end{aligned}$$

These are precisely the number-basis matrix elements of $\tau_p^{(\gamma)}$ and τ_p . For fixed L , the cutoff space has fixed finite dimension, so the uniform entrywise limits imply convergence in trace norm:

$$\left\| J_{\nu,L} P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L} J_{\nu,L}^{*} - P_L^{\mathcal{F}} \tau_p^{(\gamma)} P_L^{\mathcal{F}} \right\|_1 \longrightarrow 0. \quad (159)$$

$$\left\| J_{\nu,L} P_{\nu,L} \rho_{p,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^{\mathcal{F}} \tau_p P_L^{\mathcal{F}} \right\|_1 \longrightarrow 0. \quad (160)$$

We now prove the fidelity assertions of [Proposition 2.3](#) and [Lemma 2.4](#) together. Let $(a_\mu)_\mu$ be a probability distribution supported on labels $\mu \in \mathbb{Y}_n$ with $\|\mu/n - p\|_\infty \leq \delta_n$; for the proposition, take a point mass. All such pairs (μ, ν) are compatible for large n , by the first part of the proof. Put

$$\bar{\sigma}_{p,\nu} = \sum_\mu a_\mu \sigma_{\mu,\nu}, \quad \sigma_{\mu,\nu} = \mathcal{C}_{\mu,\nu}(\rho_{p,\mu}).$$

Fix L . By (159) there is $\varepsilon_{n,L} \rightarrow 0$, for fixed L and uniformly over the allowed parameters, such that every μ in the support satisfies

$$\left\| J_{\nu,L} P_{\nu,L} \sigma_{\mu,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^{\mathcal{F}} \tau_p^{(\gamma)} P_L^{\mathcal{F}} \right\|_1 \leq \varepsilon_{n,L}.$$

The frame $J_{\nu,L} P_{\nu,L}$ depends only on ν and L , so the triangle inequality and $\sum_\mu a_\mu = 1$ give the same bound with $\sigma_{\mu,\nu}$ replaced by $\bar{\sigma}_{p,\nu}$. After increasing $\varepsilon_{n,L}$ if necessary, (160) gives

$$\left\| J_{\nu,L} P_{\nu,L} \rho_{p,\nu} P_{\nu,L} J_{\nu,L}^* - P_L^{\mathcal{F}} \tau_p P_L^{\mathcal{F}} \right\|_1 \leq \varepsilon_{n,L}.$$

Next we control the target mass above height L . Since K is compact in Δ_d° , there is $b_K < 1$ such that $q_\alpha \leq b_K$ for every $p \in K$ and every positive root α . The occupation numbers under τ_p are independent geometric variables with parameters at most b_K . If their total occupation is s , then their total root height is at most $(d-1)s$; moreover, the probability of any fixed occupation vector of total occupation s is at most b_K^s , and there are $\binom{s+r_d-1}{r_d-1}$ such vectors. Therefore

$$t_L := \sup_{p \in K} \text{Tr}[(I - P_L^{\mathcal{F}}) \tau_p] \leq \sum_{s > L/(d-1)} \binom{s+r_d-1}{r_d-1} b_K^s \longrightarrow 0 \quad (L \rightarrow \infty). \quad (161)$$

Taking traces in the target estimate above gives

$$\text{Tr}[(I - P_{\nu,L}) \rho_{p,\nu}] \leq t_L + \varepsilon_{n,L}. \quad (162)$$

Every $\sigma_{\mu,\nu}$, and hence $\bar{\sigma}_{p,\nu}$, commutes with the diagonal torus, by torus invariance of $\rho_{p,\mu}$ and covariance of $\mathcal{C}_{\mu,\nu}$; so does $\rho_{p,\nu}$. Consequently, $P_{\nu,L}$ reduces both states, and the direct-sum identity splits their fidelity into the sum of the cutoff-block and complementary-block fidelities. Schatten Hölder gives $F(A, C) \leq \sqrt{\text{Tr } A \text{ Tr } C}$, so by (162) and $\text{Tr}[(I - P_{\nu,L}) \bar{\sigma}_{p,\nu}] \leq 1$ the complementary fidelity is at most $\sqrt{t_L + \varepsilon_{n,L}}$. The same decomposition for $F(\tau_p^{(\gamma)}, \tau_p)$ with respect to $P_L^{\mathcal{F}}$ bounds its complementary part by $\sqrt{t_L}$. Finally, fidelity is invariant under the isometry $J_{\nu,L}$, so the continuity estimate (56) and the two $\varepsilon_{n,L}$ -bounds above show that the two cutoff fidelities differ by at most $2\sqrt{\varepsilon_{n,L}}$. Altogether,

$$\left| F(\bar{\sigma}_{p,\nu}, \rho_{p,\nu}) - F(\tau_p^{(\gamma)}, \tau_p) \right| \leq 2\sqrt{\varepsilon_{n,L}} + \sqrt{t_L + \varepsilon_{n,L}} + \sqrt{t_L}.$$

First let $n \rightarrow \infty$ at fixed L , then let $L \rightarrow \infty$; the convergence is uniform over $p \in K$, over the admissible labels, and over the weights (a_μ) . Finally,

$$F(\tau_p^{(\gamma)}, \tau_p) = \prod_{i < j} F(\tau_{q_{ij}^{(\gamma)}}, \tau_{q_{ij}}) = F_{\text{orb}}(\gamma, p),$$

where multiplicativity of fidelity follows from multiplicativity of the trace norm under tensor products, and the last equality uses (41) and (44). For a point mass this proves [Proposition 2.3](#); in general it proves [Lemma 2.4](#). $\square$

C Young local limit and randomized rounding

This appendix proves [Proposition 2.5](#). We embed Young-label laws as densities on lattice cells and prove their uniform Gaussian limit. Randomized rounding then becomes exactly dilation followed by cell averaging, which yields the affinity $F_{\text{cl}}(\gamma, d)$. The fallback rule changes only the vanishing mass of atypical input labels.

Define the half-open fundamental cell

$$\mathbf{C} = \left\{ \sum_{i=1}^{d-1} t_i(e_i - e_d) : -\frac{1}{2} \leq t_i < \frac{1}{2} \right\} \subset \mathbf{H}.$$

Volumes are taken with the induced $(d-1)$ -dimensional Lebesgue measure. The rounding rule (26) is equivalent, outside a null set of cell boundaries, to choosing the unique $\tilde{\nu} \in \mathbf{L}_{m_n}$ for which

$$\gamma_n(\mu + U) \in \tilde{\nu} + \mathbf{C}, \quad U = \sum_{i=1}^{d-1} U_i(e_i - e_d).$$

The first $d-1$ coordinates give the nearest-integer rule, and the last is fixed by the coordinate sum. Since U is uniform on $\mathbf{C}$,

$$D_n(\tilde{\nu} \mid \mu) = \frac{|\gamma_n(\mu + \mathbf{C}) \cap (\tilde{\nu} + \mathbf{C})|}{\gamma_n^{d-1} |\mathbf{C}|}. \quad (163)$$

All label laws below are regarded as functions on $\mathbf{L}_N$: if a law R is originally defined on $\mathbf{Y}_N$, set $R(\lambda) = 0$ for every $\lambda \in \mathbf{L}_N \setminus \mathbf{Y}_N$.

For $\lambda \in \mathbf{L}_N$, define the recentered and rescaled lattice cell

$$\mathbf{C}_{N,p}(\lambda) = \frac{\lambda - Np + \mathbf{C}}{\sqrt{N}}.$$

For any $R : \mathbf{L}_N \rightarrow \mathbb{R}$ satisfying $\sum_{\lambda \in \mathbf{L}_N} |R(\lambda)| < \infty$, define

$$(\mathcal{E}_{N,p}R)(x) = \frac{N^{(d-1)/2}}{|\mathbf{C}|} R(\lambda) \quad (x \in \mathbf{C}_{N,p}(\lambda)).$$

The cells $\mathbf{C}_{N,p}(\lambda)$, $\lambda \in \mathbf{L}_N$, tile $\mathbf{H}$, and their boundaries have zero induced Lebesgue measure. The map $\mathcal{E}_{N,p}$ preserves ℓ^1 -distance, $\|\mathcal{E}_{N,p}R - \mathcal{E}_{N,p}S\|_1 = \|R - S\|_{\ell^1}$, and, for nonnegative R, S , Hellinger affinity, $\int_{\mathbf{H}} \sqrt{(\mathcal{E}_{N,p}R)(\mathcal{E}_{N,p}S)} \, dx = \sum_{\lambda \in \mathbf{L}_N} \sqrt{R(\lambda)S(\lambda)}$. Indeed, each $\mathbf{C}_{N,p}(\lambda)$ has volume $|\mathbf{C}|N^{-(d-1)/2}$, which cancels the prefactor in both cellwise integrals.

Lemma C.1 (Uniform Young local limit). *For every compact $K \Subset \Delta_d^\circ$,*

$$\sup_{p \in K} \|\mathcal{E}_{N,p}P_{N,p} - \varphi_{0,\Sigma_p}\|_1 \longrightarrow 0. \quad (164)$$

Proof. Write $f_{N,p} = \mathcal{E}_{N,p}P_{N,p}$, and fix a ball $B_R = \{x \in \mathbf{H} : \|x\| \leq R\}$. If $\mathbf{C}_{N,p}(\lambda)$ meets B_R , then $x_\lambda = (\lambda - Np)/\sqrt{N}$ stays bounded uniformly in N, p . Since the smallest eigenvalue and the adjacent gaps of $p \in K$ are bounded below, every such λ is a Young diagram for large N . The Frobenius dimension formula and [Lemma B.2](#) give, uniformly on these cells,

$$P_{N,p}(\lambda) = \frac{N! \prod_i p_i^{\lambda_i}}{\prod_i \lambda_i!} \prod_{i < j} \frac{\lambda_i - \lambda_j + j - i}{N(p_i - p_j)} \prod_i \prod_{a=1}^{d-i} \frac{Np_i}{\lambda_i + a} (1 + o(1)).$$

Each factor in the two last products is $1 + O_{K,R}(N^{-1/2})$. Stirling's formula and a Taylor expansion of the first factor therefore yield

$$P_{N,p}(\lambda) = \frac{\exp\left[-\frac{1}{2} \sum_i x_{\lambda,i}^2 / p_i\right]}{(2\pi N)^{(d-1)/2} \sqrt{\prod_i p_i}} (1 + o(1)).$$

The normalization agrees with the Gaussian density on $\mathbf{H}$:

$$|\mathbf{C}| = \sqrt{d}, \quad \det_{\mathbf{H}} \Sigma_p = d \prod_i p_i, \quad \langle x, \Sigma_p^{-1} x \rangle = \sum_i \frac{x_i^2}{p_i} \quad (x \in \mathbf{H}).$$

Indeed, the Gram matrix of $(e_i - e_d)_{i < d}$ has determinant d . The ambient covariance $\text{diag}(p) - pp^\top$ has kernel $\text{span}\{(1, \dots, 1)\}$ and principal cofactors $\prod_i p_i$, giving its restricted determinant. Its inverse on $\mathbf{H}$ sends x to $\text{diag}(p)^{-1}x$ up to a multiple of $(1, \dots, 1)$, giving the quadratic identity. The cell diameter is $O(N^{-1/2})$, so uniform continuity of the Gaussian densities on bounded sets now gives

$$\sup_{p \in K} \sup_{x \in B_R} |f_{N,p}(x) - \varphi_{0,\Sigma_p}(x)| \longrightarrow 0.$$

Both densities have integral one. Consequently,

$$\|f_{N,p} - \varphi_{0,\Sigma_p}\|_1 \leq 2 \int_{B_R} |f_{N,p} - \varphi_{0,\Sigma_p}| \, dx + 2 \int_{B_R^c} \varphi_{0,\Sigma_p}(x) \, dx.$$

First let $N \rightarrow \infty$, then $R \rightarrow \infty$. The Gaussian tails vanish uniformly on compact K , proving (164). $\square$

Lemma C.2 (Continuous limit of randomized dilation). *Let $\tilde{Q}_{m_n,p} = P_{n,p} D_n$ denote the $\tilde{\nu}$ -marginal of the joint law obtained by composing the kernel D_n with $P_{n,p}$, before the replacement in (27). For every compact $K \subseteq \Delta_d^\circ$,*

$$\sup_{p \in K} \|\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma_{\Sigma_p}}\|_1 \longrightarrow 0. \quad (165)$$

$$\sup_{p \in K} |F(\tilde{Q}_{m_n,p}, P_{m_n,p}) - F_{\text{cl}}(\gamma, d)| \longrightarrow 0. \quad (166)$$

Proof. For $a > 0$ and $f \in L^1(\mathbf{H})$, set

$$(\text{dil}_a f)(y) = a^{-(d-1)/2} f(y/\sqrt{a}),$$

and define the target-cell averaging operator explicitly by

$$(\text{Av}_{m_n,p} f)(x) := \frac{m_n^{(d-1)/2}}{|\mathbf{C}|} \int_{\mathbf{C}_{m_n,p}(\nu)} f(y) \, dy, \quad x \in \mathbf{C}_{m_n,p}(\nu).$$

If $X = (\mu - np + U)/\sqrt{n}$, where $\mu \sim P_{n,p}$ and U is uniform on $\mathbf{C}$, then X has density $\mathcal{E}_{n,p} P_{n,p}$. Since $m_n = \gamma_n n$, the event $\sqrt{\gamma_n} X \in \mathbf{C}_{m_n,p}(\nu)$ is exactly the rounding event $\gamma_n(\mu + U) \in \nu + \mathbf{C}$. Thus the dilated density assigns mass $\tilde{Q}_{m_n,p}(\nu)$ to that target cell, and

$$\text{Av}_{m_n,p} \text{dil}_{\gamma_n}(\mathcal{E}_{n,p} P_{n,p}) = \mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p}. \quad (167)$$

Cell averaging is an L^1 -contraction and dilation an L^1 -isometry. Since $\text{dil}_{\gamma_n} \varphi_{0,\Sigma_p} = \varphi_{0,\gamma_n \Sigma_p}$, (167) yields

$$\begin{aligned} \|\mathcal{E}_{m_n,p} \tilde{Q}_{m_n,p} - \varphi_{0,\gamma \Sigma_p}\|_1 &\leq \|\mathcal{E}_{n,p} P_{n,p} - \varphi_{0,\Sigma_p}\|_1 \\ &\quad + \|\text{Av}_{m_n,p} \varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma_n \Sigma_p}\|_1 \\ &\quad + \|\varphi_{0,\gamma_n \Sigma_p} - \varphi_{0,\gamma \Sigma_p}\|_1. \end{aligned}$$

The first term tends uniformly to zero by Lemma C.1, and the last does so because $\gamma_n \rightarrow \gamma$. For the middle term, the cellwise L^1 Poincaré inequality gives, for smooth f with integrable gradient,

$$\|\text{Av}_{m_n,p} f - f\|_1 \leq C_d m_n^{-1/2} \|\nabla f\|_1.$$

Gaussian gradients have uniformly bounded L^1 -norm on the compact covariance family $\gamma_n \Sigma_p$, so this term is $O_K(m_n^{-1/2})$. This proves (165).

Applying Lemma C.1 at sample size m_n gives

$$\mathcal{E}_{m_n,p} P_{m_n,p} \longrightarrow \varphi_{0,\Sigma_p}$$

uniformly in L^1 . The embedding preserves Hellinger affinity, and (56) also holds for nonnegative L^1 densities. Therefore

$$F(\tilde{Q}_{m_n,p}, P_{m_n,p}) \longrightarrow F(\mathbf{N}_{0,\gamma \Sigma_p}, \mathbf{N}_{0,\Sigma_p}) = F_{\text{cl}}(\gamma, d)$$

uniformly on K . $\square$

Proof of Proposition 2.5. Write $e = \tilde{\nu} - \gamma_n \mu$. The nearest-integer rule gives $|e_i| \leq (\gamma_n + 1)/2$ for $i < d$, and $e_d = -\sum_{i < d} e_i$. Thus $\|e\|_\infty \leq c_{\text{rnd}}$, where

$$c_{\text{rnd}} = \frac{d-1}{2} \left(1 + \sup_n \gamma_n \right) < \infty.$$

Compactness of K gives $c_0 > 0$ such that, for all large n , every $p \in K$ and $\mu \in \mathbf{T}_n(p)$ satisfy

$$\mu_d \geq c_0 n, \quad \mu_i - \mu_{i+1} \geq c_0 n \quad (i < d).$$

For $\omega = \tilde{\nu} - \mu = (\gamma_n - 1)\mu + e$, these bounds imply

$$\omega_d \geq (\gamma_n - 1)c_0 n - c_{\text{rnd}}, \quad \omega_i - \omega_{i+1} \geq (\gamma_n - 1)c_0 n - 2c_{\text{rnd}} \quad (i < d).$$

Since $\gamma_n \rightarrow \gamma > 1$, both are positive for large n . Thus $\omega \vdash m_n - n$ and $\tilde{\nu} = \mu + \omega \in \mathbf{Y}_{m_n}$. No rounded label is changed for typical μ , proving (32).

Coupling the raw and final labels by the fallback rule therefore gives

$$\|Q_{m_n,p} - \tilde{Q}_{m_n,p}\|_{\ell^1} \leq 2P_{n,p}(\mathbf{T}_n(p)^c) = o(1)$$

uniformly on K , by Lemma 2.1. Combining (166) with (56) proves (33). $\square$

D Unified Gaussian flat-prior converse

This appendix proves the two converse inequalities in Theorem 3.1. The weighted-fidelity bound (Lemma D.1) reduces fidelity to an oscillator moment; compact-observable limits (Lemma D.2) and the least-noise inequality (Proposition D.4) bound that moment. Proposition D.5 combines these ingredients with a direct Gaussian translation estimate for the classical factor, giving $F_{\text{orb}}(\gamma, p)$ in the orbital model and $F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p)$ in the full model.

The classical parameter space has dimension $k = 0$ for the orbital problem and $k = d - 1$ for the full problem. All trace-class-valued integrals below are Bochner integrals in trace norm.

D.1 Weighted fidelity and compact limits

Lemma D.1 (Weighted fidelity bound). *Let R, T be positive trace-class operators, and let W be bounded, positive, and injective. Define*

$$\mathrm{Tr}(TW^{-1}) := \lim_{\varepsilon \downarrow 0} \mathrm{Tr}(T(W + \varepsilon I)^{-1}) \in [0, \infty].$$

If this quantity is finite, then

$$F(R, T)^2 \leq \mathrm{Tr}(RW) \mathrm{Tr}(TW^{-1}). \quad (168)$$

Let $\mathbf{E}$ be a finite-dimensional Euclidean space, let $R \in L^1(\mathbf{E}; \mathcal{T}_1(\mathcal{F}_{\mathrm{orb}}))$ be positive, let g be a probability density on $\mathbf{E}$, and let $\chi > 0$ be measurable. For positive R, S in this L^1 -space, set $F(R, S) = \int_{\mathbf{E}} F(R(y), S(y)) \, dy$, extending the hybrid-state convention in [Section 3](#). The notation gT means the density $y \mapsto g(y)T$. Whenever the right-hand side below is finite,

$$F(R, gT)^2 \leq \left(\int_{\mathbf{E}} \chi(y) \mathrm{Tr}[R(y)W] \, dy \right) \left(\int_{\mathbf{E}} \frac{g(y)}{\chi(y)} \, dy \right) \mathrm{Tr}(TW^{-1}). \quad (169)$$

Proof. For $W_\varepsilon = W + \varepsilon I$, the Schatten Hölder inequality gives

$$F(R, T) = \|(R^{1/2}W_\varepsilon^{1/2})(W_\varepsilon^{-1/2}T^{1/2})\|_1 \leq \sqrt{\mathrm{Tr}(RW_\varepsilon)} \sqrt{\mathrm{Tr}(TW_\varepsilon^{-1})}.$$

Squaring and letting $\varepsilon \downarrow 0$, using $\mathrm{Tr}(RW_\varepsilon) = \mathrm{Tr}(RW) + \varepsilon \mathrm{Tr} R$ and monotone convergence for the second trace, proves [\(168\)](#). For the hybrid state,

$$F(R, gT) = \int_{\mathbf{E}} \sqrt{g(y)} F(R(y), T) \, dy.$$

Apply the first inequality pointwise and then Cauchy–Schwarz to the factors $\sqrt{\chi(y) \mathrm{Tr}[R(y)W]}$ and $\sqrt{g(y)/\chi(y)}$. This gives [\(169\)](#). $\square$

Lemma D.2 (Compact-observable limits of completely positive maps). *Let $\mathcal{H}_0, \mathcal{H}_1$ be separable Hilbert spaces and let $\Gamma_j : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$ be completely positive maps with $\sup_j \|\Gamma_j\|_{1 \rightarrow 1} < \infty$. There are a subsequence and a completely positive map $\Gamma : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$ such that*

$$\mathrm{Tr}[\Gamma_j(X)K] \longrightarrow \mathrm{Tr}[\Gamma(X)K] \quad (170)$$

for every $X \in \mathcal{T}_1(\mathcal{H}_0)$ and every compact $K \in \mathcal{K}(\mathcal{H}_1)$. If $\mathrm{Tr} \Gamma_j(X) \leq c \mathrm{Tr} X$ for every $X \geq 0$, then $\mathrm{Tr} \Gamma(X) \leq c \mathrm{Tr} X$.

Suppose in addition that $U_g \in \mathrm{U}(\mathcal{H}_0)$ and $V_g \in \mathrm{U}(\mathcal{H}_1)$ are families of unitaries indexed by a set G , and that, for every fixed $g \in G$ and $X \in \mathcal{T}_1(\mathcal{H}_0)$,

$$\|\Gamma_j(U_g X U_g^*) - V_g \Gamma_j(X) V_g^*\|_1 \longrightarrow 0.$$

Then

$$\Gamma(U_g X U_g^*) = V_g \Gamma(X) V_g^* \quad (171)$$

for every $g \in G$ and $X \in \mathcal{T}_1(\mathcal{H}_0)$.

Proof. Let $M = \sup_j \|\Gamma_j\|_{1 \rightarrow 1}$, and let $\mathcal{D} \subset \mathcal{K}(\mathcal{H}_1)$ be a countable norm-dense set. The predual $\mathcal{T}_1(\mathcal{H}_0)$ is separable, so a diagonal weak-* compactness argument gives a subsequence for which $\Gamma_j^*(K)$ converges weak-* in $\mathcal{B}(\mathcal{H}_0)$ for every $K \in \mathcal{D}$. The bound $\|\Gamma_j^*(K)\| \leq M\|K\|$ extends both the limit and the convergence to every compact K , producing a bounded complex-linear map $\Gamma^* : \mathcal{K}(\mathcal{H}_1) \rightarrow \mathcal{B}(\mathcal{H}_0)$.

For every positive block matrix $A = [K_{ab}]$ of compact operators, $[\Gamma_j^*(K_{ab})] \geq 0$; its entrywise weak-* limit is $[\Gamma^*(K_{ab})] \geq 0$. Thus Γ^* is completely positive.

Since $\mathcal{K}(\mathcal{H}_1)^* = \mathcal{T}_1(\mathcal{H}_1)$, define $\Gamma(X)$ by

$$\mathrm{Tr}[\Gamma(X)K] = \mathrm{Tr}[X\Gamma^*(K)] \quad (K \in \mathcal{K}(\mathcal{H}_1)).$$

Then Γ is completely positive and (170) holds. If $P_N \uparrow I$ are finite-rank projections and $X \geq 0$, then

$$\mathrm{Tr} \Gamma(X) = \sup_N \lim_j \mathrm{Tr}[\Gamma_j(X)P_N] \leq c \mathrm{Tr} X.$$

Finally, for fixed g, X and compact K , the operator $V_g^* K V_g$ is compact. Passing to the limit gives

$$\mathrm{Tr}[\Gamma(U_g X U_g^*)K] = \mathrm{Tr}[V_g \Gamma(X) V_g^* K].$$

Compact operators separate trace-class operators, so the limiting map is covariant. $\square$

D.2 The least-noise oscillator moment

Fix $\gamma > 1$. For $s \geq 1$ and $q_1, \dots, q_s \in (0, 1)$, put

$$q_\ell^{(\gamma)} = 1 - \frac{1 - q_\ell}{\gamma}, \quad \tau = \bigotimes_{\ell=1}^s \tau_{q_\ell}, \quad \tau^{(\gamma)} = \bigotimes_{\ell=1}^s \tau_{q_\ell^{(\gamma)}}.$$

The proof below uses one structural input, the idler representation of covariant one-mode channels, from which we then derive the required stochastic-ordering lemma.

Fact D.I (Covariant-amplifier representation). *Every displacement-covariant one-mode channel of amplitude gain $\sqrt{\gamma}$ admits an idler realization*

$$a_{\mathrm{out}} = \sqrt{\gamma} a_{\mathrm{in}} + \sqrt{\gamma - 1} b^*, \tag{172}$$

in which the idler mode b is prepared in a state that is uniquely determined by the channel and need not be Gaussian. If the channel is also phase covariant, the idler state may be chosen number diagonal.

Proof. Covariance implies that $\Lambda^*(D(w))D(\sqrt{\gamma}w)^*$ commutes with every displacement. Irreducibility of the Weyl system therefore gives

$$\Lambda^*(D(w)) = f(w)D(\sqrt{\gamma}w)$$

for a scalar function f . In real phase-space coordinates the associated defect symplectic form is $(1 - \gamma)\Omega$, which is nondegenerate because $\gamma > 1$. The quantum Bochner criterion [Hol82] then identifies f , after the fixed rescaling and phase conjugation determined by the gain, with the characteristic function of a unique one-mode state. The exact dilation theorem for a nondegenerate defect form [LSW18, Lemma 7], specialized to $X = \sqrt{\gamma}I_2$, realizes the resulting channel by the displayed two-mode squeezer. The gain-2 case is also derived directly in [GM06, Appendix A]. If the channel is phase covariant, uniqueness makes the idler invariant under phase rotations; equivalently, phase averaging the idler leaves the channel unchanged. The idler is therefore number diagonal. $\square$

Lemma D.3 (Vacuum-idler stochastic order). *Fix $\gamma > 1$ and $q \in (0, 1)$, and let Λ be a completely positive trace-preserving one-mode map that is phase covariant and displacement covariant with amplitude gain $\sqrt{\gamma}$. Let $p_k = \langle k | \Lambda(\tau_q) | k \rangle$ be its output number distribution on the thermal input. Then*

$$\sum_{k=0}^m (1 - q^{(\gamma)})(q^{(\gamma)})^k \geq \sum_{k=0}^m p_k \quad (m \geq 0), \quad (173)$$

where $q^{(\gamma)} = 1 - (1 - q)/\gamma$. Thus the quantum-limited vacuum-idler output $\tau_{q^{(\gamma)}}$ is stochastically smallest. For the gain-2 amplifier this ordering is Lemma 2.3 of [GM06].

Proof. Write $\mathcal{A}_G$ for the quantum-limited amplifier of gain $G \geq 1$, namely the channel of Fact D.I with vacuum idler. The normal-ordered disentangling of the two-mode squeeze operator [BR97] gives, with $\cosh^2 s = G$ and $\lambda = \tanh^2 s = 1 - 1/G$, the joint state created from the signal vacuum and an idler number state $|l\rangle$:

$$U_G|0, l\rangle = (1 - \lambda)^{(l+1)/2} \sum_{k \geq 0} \sqrt{\binom{k+l}{l}} \lambda^{k/2} |k, k+l\rangle. \quad (174)$$

Its signal marginal is therefore the negative binomial law $\text{NB}(l+1, \lambda)$,

$$\Pr[k] = \binom{k+l}{l} \lambda^k (1 - \lambda)^{l+1} \quad (k \geq 0). \quad (175)$$

In particular $\mathcal{A}_G(|0\rangle\langle 0|) = \tau_{1-1/G}$, so the thermal input is itself an amplified vacuum:

$$\tau_q = \mathcal{A}_{G'}(|0\rangle\langle 0|), \quad G' = \frac{1}{1 - q}. \quad (176)$$

Consider the composite channel $\Lambda \circ \mathcal{A}_{G'}$. It is completely positive, trace preserving, phase covariant, and displacement covariant with total amplitude gain $\sqrt{\gamma G'}$. By Fact D.I, it admits an idler realization at gain $\sqrt{\gamma G'}$ with a number-diagonal idler state $\sum_l c_l |l\rangle\langle l|$. Evaluating the composite on the vacuum and using (175) at gain $G = \gamma G'$ shows that the number distribution of $\Lambda(\tau_q) = (\Lambda \circ \mathcal{A}_{G'}) (|0\rangle\langle 0|)$ is the mixture

$$\sum_l c_l \text{NB}(l+1, \lambda), \quad \lambda = 1 - \frac{1}{\gamma G'} = 1 - \frac{1 - q}{\gamma} = q^{(\gamma)}.$$

A negative binomial variable $\text{NB}(l+1, \lambda)$ is a sum of $l+1$ independent geometric variables of parameter λ , so its cumulative distribution function is dominated pointwise by that of $\text{NB}(1, \lambda)$, which is the geometric law of $\tau_{q^{(\gamma)}}$. Averaging these pointwise inequalities over the weights c_l proves the claim. $\square$

Proposition D.4 (Least-noise moment). *Let Γ be a completely positive trace-nonincreasing map on the s -mode trace class satisfying*

$$\Gamma(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\Gamma(X)D(\sqrt{\gamma}z)^* \quad (z \in \mathbb{C}^s).$$

If

$$W(\hat{\mathbf{N}}) = \prod_{\ell=1}^s w_\ell(\hat{N}_\ell),$$

where every w_ℓ is bounded, nonnegative, and nonincreasing on $\mathbb{Z}_{\geq 0}$, then

$$\text{Tr}[\Gamma(\tau)W(\hat{\mathbf{N}})] \leq \text{Tr}[\Gamma(\tau)] \text{Tr}[\tau^{(\gamma)}W(\hat{\mathbf{N}})]. \quad (177)$$

Proof. Averaging Γ over independent phase rotations preserves its trace bound and displacement covariance, and it does not change either side of (177). We may therefore assume that Γ is phase covariant in every mode.

Consider one mode. Write $\text{Tr} \Gamma(X) = \text{Tr}(AX)$ with $0 \leq A \leq I$. Displacement covariance implies $D(z)^* A D(z) = A$; hence irreducibility of the Weyl representation gives $A = cI$, where $c = \text{Tr} \Gamma(\tau_q)$. If $c > 0$, then $\Lambda = c^{-1}\Gamma$ is a completely positive trace-preserving, phase-covariant map of amplitude gain $\sqrt{\gamma}$. Lemma D.3 applied to Λ gives

$$\sum_{n=0}^m (1 - q^{(\gamma)})(q^{(\gamma)})^n \geq \sum_{n=0}^m \langle n | \Lambda(\tau_q) | n \rangle \quad (m \geq 0). \quad (178)$$

The distribution on the left is the vacuum-idler output, namely that of $\tau_{q^{(\gamma)}}$, and both output states are number diagonal. For a bounded nonincreasing w , summation by parts writes its expectation as a nonnegative linear combination of these cumulative probabilities. Both distributions in (178) have total mass one, so the common limiting constant cancels. Hence

$$\text{Tr}[\Lambda(\tau_q)w(\hat{N})] \leq \text{Tr}[\tau_{q^{(\gamma)}}w(\hat{N})].$$

Multiplication by c proves the one-mode case; the case $c = 0$ is immediate.

For the multimode case, the claim is immediate if some w_ℓ vanishes identically; otherwise, divide each w_ℓ by $w_\ell(0)$ and assume $0 \leq w_\ell \leq 1$. Put

$$W_{\geq \ell} = I_1 \otimes \cdots \otimes I_{\ell-1} \otimes \bigotimes_{j=\ell}^s w_j(\hat{N}_j), \quad W_{\geq s+1} = I,$$

and, for an operator X on mode ℓ , define

$$\Gamma_\ell(X) = \text{Tr}_{\neq \ell} \left[W_{\geq \ell+1}^{1/2} \Gamma(\tau_{q_1} \otimes \cdots \otimes \tau_{q_{\ell-1}} \otimes X \otimes \tau_{q_{\ell+1}} \otimes \cdots \otimes \tau_{q_s}) W_{\geq \ell+1}^{1/2} \right].$$

Because $0 \leq W_{\geq \ell+1} \leq I$ and this operator is the identity on mode ℓ , each Γ_ℓ is completely positive, trace nonincreasing, and displacement covariant in that mode. The phase-covariant reduction following (177) also makes Γ_ℓ phase covariant because the fixed thermal inputs and $W_{\geq \ell+1}$ are phase invariant. Moreover,

$$\text{Tr}[\Gamma_\ell(\tau_{q_\ell} w_\ell(\hat{N}_\ell))] = \text{Tr}[\Gamma(\tau) W_{\geq \ell}], \quad \text{Tr} \Gamma_\ell(\tau_{q_\ell}) = \text{Tr}[\Gamma(\tau) W_{\geq \ell+1}].$$

Applying the one-mode inequality to Γ_ℓ and iterating over $\ell = 1, \dots, s$ gives (177). $\square$

Specialize to $s = r_d$, $q_\ell = q_{ij}$, $\tau = \tau_p$, and $\tau^{(\gamma)} = \tau_p^{(\gamma)}$. For $\mathbf{k} = (k_{ij})_{i < j} \in \mathbb{Z}_{\geq 0}^{r_d}$, let $t_{\mathbf{k}}$ and $t_{\mathbf{k}}^{(\gamma)}$ be the corresponding number-basis eigenvalues, and define

$$W_\gamma = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle \langle \mathbf{k}|. \quad (179)$$

Equivalently,

$$W_\gamma = \prod_{i < j} w_{ij}(\hat{N}_{ij}), \quad w_{ij}(n) = \sqrt{\frac{1 - q_{ij}}{1 - q_{ij}^{(\gamma)}}} \left(\frac{q_{ij}}{q_{ij}^{(\gamma)}} \right)^{n/2}.$$

Here $\hat{N}_{ij}$ is the number operator of the mode indexed by $i < j$, and $|\mathbf{k}| := \sum_{i < j} k_{ij}$ is the total occupation number. Since $q_{ij} < q_{ij}^{(\gamma)}$, each $w_{ij}(n)$ is positive, bounded, and strictly decreasing in n ,

and the eigenvalues of W_γ tend to zero as $|\mathbf{k}| \rightarrow \infty$. Thus W_γ is bounded, injective, and compact. Its definition gives

$$\mathrm{Tr}(\tau_p^{(\gamma)} W_\gamma) = \mathrm{Tr}(\tau_p W_\gamma^{-1}) = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}} = F_{\mathrm{orb}}(\gamma, p), \quad (180)$$

where the inverse is understood as in [Lemma D.1](#).

D.3 One flat-prior converse for both Gaussian models

Proposition D.5 (Flat-prior converse inequality). *Let $k \in \{0, d-1\}$. For $k = 0$, let $\mathbf{E} = \{0\}$, omit the classical factor, and interpret h -integration as evaluation at 0. For $k = d-1$, let $\mathbf{E} = \mathbf{H}$ with its induced Euclidean measure. Let*

$$\mathbf{E}_L = \{h \in \mathbf{E} : \|h\|_\infty \leq L\}, \quad \Xi_L = \mathbf{E}_L \times \mathbf{Z}_L,$$

with $|\mathbf{E}_L| = 1$ when $k = 0$. Define

$$R_{0,z}^{(0)} = \Phi_z^{(p)}, \quad S_{0,z}^{(0)} = \Psi_z^{(p,\gamma)},$$

$$R_{h,z}^{(d-1)} = \Theta_{h,z}^{\mathrm{in}}, \quad S_{h,z}^{(d-1)} = \Theta_{h,z}^{\mathrm{tar}}, \quad c_k = \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{k/2}.$$

For $k = 0$, the supremum below is over quantum channels on $\mathcal{F}_{\mathrm{orb}}$; for $k = d-1$, it is over classical-quantum channels. Then

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|\Xi_L|} \int_{\Xi_L} F(\Lambda(R_{h,z}^{(k)}), S_{h,z}^{(k)}) \, dh \, dz \leq c_k F_{\mathrm{orb}}(\gamma, p). \quad (181)$$

Proof. For $k = 0$, set $f_1 = f_\gamma = \chi = 1$ on the one-point space. For $k = d-1$, let f_1 and f_γ be the centered Gaussian densities with covariances Σ_p and $\gamma \Sigma_p$, and set

$$\chi(y) = \sqrt{\frac{f_1(y)}{f_\gamma(y)}} = \gamma^{k/4} \exp \left[-\frac{\gamma-1}{4\gamma} \langle y, \Sigma_p^{-1} y \rangle \right].$$

This function is bounded and strictly positive. In both cases,

$$\int_{\mathbf{E}} \frac{f_1(y)}{\chi(y)} \, dy = \int_{\mathbf{E}} \sqrt{f_1(y) f_\gamma(y)} \, dy = c_k. \quad (182)$$

For $k = d-1$, completing the square also gives

$$H(r) := \int_{\mathbf{E}} f_1(h) \chi(r + \sqrt{\gamma} h) \, dh = c_k \exp \left[-\frac{\gamma-1}{2\gamma(\gamma+1)} \langle r, \Sigma_p^{-1} r \rangle \right] \leq c_k. \quad (183)$$

For $\xi = (h, z)$, define

$$(\mathcal{U}_\xi^{\mathrm{in}} R)(x) = D(z) R(x - h) D(z)^*, \quad (\mathcal{U}_\xi^{\mathrm{out}} R)(y) = D(\sqrt{\gamma} z) R(y - \sqrt{\gamma} h) D(\sqrt{\gamma} z)^*.$$

Then $R_\xi^{(k)} = \mathcal{U}_\xi^{\mathrm{in}}(f_1 \tau_p)$ and $S_\xi^{(k)} = \mathcal{U}_\xi^{\mathrm{out}}(f_1 \tau_p)$. For a channel Λ , let

$$\bar{\Lambda}_L = \frac{1}{|\Xi_L|} \int_{\Xi_L} \mathcal{U}_{-\eta}^{\mathrm{out}} \circ \Lambda \circ \mathcal{U}_\eta^{\mathrm{in}} \, d\eta.$$

Invariance of F under the isometries $\mathcal{U}_\xi^{\text{out}}$ and joint concavity, applied to the Bochner average, give

$$F_L(\Lambda) = \frac{1}{|\Xi_L|} \int_{\Xi_L} F(\Lambda(R_\xi^{(k)}), S_\xi^{(k)}) d\xi \leq F(\bar{\Lambda}_L(f_1\tau_p), f_1\tau_p). \quad (184)$$

Let $F_* = \limsup_{L \rightarrow \infty} \sup_\Lambda F_L(\Lambda)$. Choose $L_j \rightarrow \infty$ with $\sup_\Lambda F_{L_j}(\Lambda) \rightarrow F_*$, and channels $\tilde{\Lambda}_j$ satisfying $F_{L_j}(\tilde{\Lambda}_j) \geq \sup_\Lambda F_{L_j}(\Lambda) - j^{-1}$. Set

$$\Lambda_j = \overline{\tilde{\Lambda}_{jL_j}}, \quad R_j = \Lambda_j(f_1\tau_p).$$

Then

$$F_{L_j}(\tilde{\Lambda}_j) \rightarrow F_*, \quad F_{L_j}(\tilde{\Lambda}_j) \leq F(R_j, f_1\tau_p). \quad (185)$$

Translating the averaging box and cancelling its intersection with the original box gives

$$\|\Lambda_j(\mathcal{U}_\xi^{\text{in}} R) - \mathcal{U}_\xi^{\text{out}} \Lambda_j(R)\|_1 \leq b_j(\xi) \|R\|_1, \quad b_j(\xi) := \frac{|(\Xi_{L_j} + \xi) \triangle \Xi_{L_j}|}{|\Xi_{L_j}|}. \quad (186)$$

Here $0 \leq b_j \leq 2$, and $b_j(\xi) = O_\xi(L_j^{-1}) \rightarrow 0$ for fixed ξ , since Ξ_1 is a bounded polytope and $\Xi_L = L\Xi_1$.

Define

$$\Gamma_j^\chi(X) = \int_{\mathbb{E}} \chi(y) [\Lambda_j(f_1 X)](y) dy.$$

These maps are completely positive and satisfy $\|\Gamma_j^\chi\|_{1 \rightarrow 1} \leq \|\chi\|_\infty$. We first show the sharper trace bound

$$\text{Tr } \Gamma_j^\chi(X) \leq (c_k + e_j) \text{Tr } X \quad (X \geq 0), \quad e_j := \|\chi\|_\infty \int_{\mathbb{E}} f_1(h) b_j(h, 0) dh \rightarrow 0. \quad (187)$$

For $k = 0$, this is trace preservation, with $c_0 = 1$ and $e_j = 0$. For $k = d - 1$, take a nonnegative probability approximate identity u_ε on $\mathbb{E}$ and put $B_{j,\varepsilon} = \Lambda_j(u_\varepsilon X)$, whose total trace is $\text{Tr } X$. Expressing $f_1 * u_\varepsilon$ as an average of translates of u_ε , (186) gives

$$\left\| \Lambda_j((f_1 * u_\varepsilon)X) - \int_{\mathbb{E}} f_1(h) \mathcal{U}_{(h,0)}^{\text{out}} B_{j,\varepsilon} dh \right\|_1 \leq \text{Tr } X \int_{\mathbb{E}} f_1(h) b_j(h, 0) dh.$$

Pairing with χ , changing variables, and using (183) therefore yields

$$\int_{\mathbb{E}} \chi(y) \text{Tr}[\Lambda_j((f_1 * u_\varepsilon)X)(y)] dy \leq \int_{\mathbb{E}} H(r) \text{Tr } B_{j,\varepsilon}(r) dr + e_j \text{Tr } X \leq (c_k + e_j) \text{Tr } X.$$

Letting $\varepsilon \downarrow 0$ proves (187), since $f_1 * u_\varepsilon \rightarrow f_1$ in L^1 . Dominated convergence gives $e_j \rightarrow 0$.

Taking $\xi = (0, z)$ in (186), multiplying by χ , and integrating gives

$$\|\Gamma_j^\chi(D(z)X D(z)^*) - D(\sqrt{\gamma}z) \Gamma_j^\chi(X) D(\sqrt{\gamma}z)^*\|_1 \leq \|\chi\|_\infty b_j(0, z) \|X\|_1 \rightarrow 0.$$

Hence, after passing to a subsequence, Lemma D.2 gives a completely positive, displacement-covariant limit Γ^χ such that

$$\text{Tr}[\Gamma_j^\chi(X)K] \rightarrow \text{Tr}[\Gamma^\chi(X)K] \quad (188)$$

for trace-class X and compact K . For $X \geq 0$, testing against finite-rank projections and using (187) gives

$$\text{Tr } \Gamma^\chi(X) \leq c_k \text{Tr } X.$$

In particular, the limiting map is trace nonincreasing. With

$$M_j = \int_{\mathbb{E}} \chi(y) \operatorname{Tr}[R_j(y)W_\gamma] \, dy = \operatorname{Tr}[\Gamma_j^\chi(\tau_p)W_\gamma],$$

compactness of W_γ gives

$$M_j \longrightarrow \operatorname{Tr}[\Gamma^\chi(\tau_p)W_\gamma]. \quad (189)$$

Moreover, [Lemma D.1](#), (182), and (180) give

$$F(R_j, f_1\tau_p)^2 \leq M_j c_k F_{\text{orb}}(\gamma, p). \quad (190)$$

By [Proposition D.4](#),

$$\lim_j M_j = \operatorname{Tr}[\Gamma^\chi(\tau_p)W_\gamma] \leq \operatorname{Tr} \Gamma^\chi(\tau_p) F_{\text{orb}}(\gamma, p) \leq c_k F_{\text{orb}}(\gamma, p).$$

Combining (185) with (190) yields $F_* \leq c_k F_{\text{orb}}(\gamma, p)$, proving both cases. $\square$

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